Classical Field Theory

Lecture Notes

Fields, symmetries, variational principles, and physical laws.

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Introduction

Purpose and Character of These Notes

Classical field theory is often presented in an extremely compressed form. A few symbols are introduced, an action is written down, a variation is performed in one line, and a field equation appears. To an experienced reader, such a presentation may be efficient. To a student meeting the subject for the first time, it can make the theory look like a collection of formulas that arrive without a clear reason, operate by hidden rules, and disappear before their meaning has been examined.

These notes are designed to avoid that style of exposition. Their purpose is not merely to collect standard formulas from analytical mechanics, special relativity, electrodynamics, and gauge theory. Their purpose is to show how the main constructions of classical field theory are built, why they are introduced, what assumptions they use, how the calculations are carried out, and how the final expressions are interpreted physically and mathematically.

A formula will therefore not be treated as the end of an explanation. Whenever possible, it will be presented as one stage in a sequence:

1. identify the physical or mathematical problem;
2. determine what objects must be introduced;
3. state the assumptions and conventions;
4. derive the relevant relation;
5. test it on a simple model;
6. interpret the result;
7. examine limiting cases and possible failures.

The central principle of the course is that a student should not only recognize an equation, but should know what to do with it. For example, it is not enough to state the Euler–Lagrange field equations. A student should see how the variation of a field is defined, how derivatives of the variation arise, where integration by parts is used, why a boundary term disappears, how indices are managed, and how the resulting equation is applied to an explicit Lagrangian density. Similarly, it is not enough to write down Noether's theorem. The theorem must be used to construct an actual current, verify its divergence, and identify the associated conserved charge.

The same standard will be applied to electrodynamics. The electromagnetic field tensor will not be introduced as an isolated matrix to memorize. It will be built from the four-potential, its components will be related explicitly to the electric and magnetic fields, its transformation properties will be examined, and its invariants will be calculated. Maxwell's equations will then be derived from an action, rather than merely restated in covariant notation.

These notes are meant to serve two related purposes. During the lecture, they provide a structured path through the material and a reliable source of notation, derivations, and examples. After the lecture, they should function as a readable reconstruction of the argument. A student who did not follow one step on the board should be able to return to the notes and see how that step fits into the complete calculation.

The exposition will therefore contain several recurring elements.

Motivation before formalism

A new object will be introduced because it solves a problem or expresses a principle. The field concept will be motivated by locality and continuous degrees of freedom. The action will be motivated as a compact way to encode dynamics. Four-dimensional notation will be motivated by Lorentz symmetry. Gauge freedom will be motivated by the non-uniqueness of the potentials that describe the same physical electromagnetic field.

Definitions connected to examples

Definitions will be followed by concrete cases. Scalar fields will be represented by temperature, density, or a real dynamical field. Vector fields will be connected with velocity fields and electric fields. Tensor fields will be introduced through objects that are actually used later, rather than through formal classification alone.

Complete derivations

Important formulas will be derived in sufficient detail that the logical structure is visible. This does not mean that every elementary algebraic step must be expanded indefinitely. It means that no essential conceptual step should be hidden behind phrases such as "it is easy to show" or "after some manipulation." When a sign, boundary condition, index contraction, or change of variables matters, it will be displayed.

Simple proofs and checks

Not every statement in physics requires a theorem-proof format, but important claims should be justified. We will prove elementary identities when they clarify the structure of the theory, verify conservation laws directly, check gauge invariance explicitly, and compare covariant formulas with familiar three-dimensional expressions. Dimensional analysis, symmetry checks, and limiting cases will be used as routine diagnostic tools.

Worked calculations

The notes will include explicit calculations rather than only general rules. A free scalar field will be solved by a plane-wave ansatz. A Noether current will be computed for a specified transformation. The electromagnetic field tensor will be written component by component. The Green function for the Poisson equation will be applied to a point source. The radiation field of an oscillating dipole will be developed far enough to calculate the angular distribution and total emitted power.

Physical interpretation

After a derivation, we will ask what the result means. What quantity is conserved? What propagates? Which variables are measurable? Which variables contain redundancy? What changes under a transformation, and what remains invariant? A correct calculation without interpretation is incomplete, because field theory is not only a formal language but also a way of organizing physical information.

Connections between chapters

The course is designed as a chain of constructions. The action principle for particles becomes the action principle for fields. Translation symmetry becomes energy-momentum conservation. The scalar field provides a simple model before the electromagnetic field is introduced. The electromagnetic potential becomes the first example of a gauge connection. The field strength becomes the first example of curvature. These connections will be stated explicitly so that the course does not appear as twelve unrelated topics.

The level of the notes is therefore deliberately intermediate between a concise reference text and a fully expanded textbook. They will remain mathematically precise, but they will also explain why the mathematics is being used. They will contain derivations and examples, but they will avoid turning every section into a long catalogue of exercises. The objective is a coherent set of lecture notes that can be read, checked, and used.

How Each Topic Will Be Developed

To keep the exposition consistent, most substantial topics will be developed according to a common pattern. The details will vary, but the reader should repeatedly encounter the same sequence of questions: what problem are we solving, what structures are needed, how is the result derived, and how can it be used?

1. The motivating problem

A section should begin with a recognizable task or conceptual difficulty. We may want to describe a continuous system, derive a differential equation from an action, express a conservation law locally, solve a field equation with a source, or understand why two different potentials represent the same electromagnetic field. This opening question determines the purpose of the formalism that follows.

The motivation should be specific enough to guide the calculation. Instead of saying only that fields are important, we will ask how a system with one degree of freedom at every point in space can be described. Instead of saying only that symmetries imply conservation laws, we will ask how an infinitesimal transformation changes the action and how that change produces a divergence equation.

2. The objects, assumptions, and conventions

Before calculating, the relevant objects must be defined. Their domains, codomains, indices, units, transformation rules, and regularity assumptions should be clear. In relativistic calculations, the metric signature and index conventions must be fixed. In variational calculations, the allowed variations and boundary conditions must be stated. In Green-function methods, the differential operator and boundary conditions must be identified.

This stage is essential because many apparent contradictions in field theory are actually differences of convention. A sign in the metric changes signs in scalar products. A definition of the field-strength tensor changes the placement of signs in its components. A choice of units changes the constants appearing in Maxwell's equations. The notes will not hide these choices.

3. The derivation

The main relation will then be derived step by step. The derivation should show every operation that carries conceptual information. If a derivative is moved from a variation to another factor, the integration by parts will be written. If dummy indices are renamed, the reason will be visible. If antisymmetry causes a term to vanish, the antisymmetry will be used explicitly. If a boundary term is discarded, the corresponding boundary condition will be stated.

A representative example is the field-theoretic action

$$S[\phi] = \int_\Omega \mathcal{L}(\phi, \partial_\mu \phi, x)\, \mathrm{d}^4 x.$$

The notes will not jump directly from this expression to the Euler–Lagrange field equation. They will define a varied field $\phi + \varepsilon\eta$, differentiate with respect to ε, separate the variation of the field from the variation of its derivative, integrate by parts, analyze the boundary contribution, and only then infer the differential equation.

4. A proof, verification, or structural check

Once a formula has been derived, it will be checked. The nature of the check depends on the result. A proposed current may be shown to have zero divergence on solutions of the field equation. A covariant Maxwell equation may be expanded into its three-dimensional components. A gauge transformation may be applied directly to verify that the field tensor remains unchanged. A stress-energy tensor may be tested by computing its energy-density component.

These checks serve two purposes. They confirm the algebra, and they reveal what the abstract expression contains. A tensor formula becomes much easier to understand after one calculates its individual components and recovers a familiar physical law.

5. A fully worked example

General formulas become useful only when they are applied. Each major construction should therefore be followed by at least one complete example, and usually by several examples of increasing complexity.

For a scalar field, we may begin with the free massless field, derive the wave equation, substitute a plane wave, and obtain the dispersion relation. We may then add a mass term and compare the result. For Noether's theorem, we may first treat spacetime translations and then a global phase transformation of a complex scalar field. For Green functions, we may begin with a point source and then pass to a distributed source by integration.

A worked example should include the setup, the calculation, and the interpretation. It should not end immediately after the final algebraic expression. The reader should be told what the answer says about propagation, conservation, symmetry, or physical measurement.

6. Limiting cases and alternative forms

An important result should be examined in limits where its meaning is already known. A relativistic formula may be reduced to a nonrelativistic one. A massive scalar field may be compared with the massless case. A retarded solution may be compared with a static solution. A tensor equation may be rewritten in vector notation.

Alternative forms will also be compared when this improves understanding. The electromagnetic field may be described using $\mathbf{E}$ and $\mathbf{B}$, the four-potential A_μ, the antisymmetric tensor $F_{\mu\nu}$, or a differential two-form. These descriptions are not competing theories; they emphasize different structural features of the same theory.

7. Common mistakes and diagnostic questions

Where a calculation is especially sensitive to signs, indices, or assumptions, the notes will identify typical errors. Examples include confusing raised and lowered components, varying a derivative incorrectly, discarding a boundary term without justification, treating gauge-dependent quantities as observables, or using a static Green function for a time-dependent problem.

The reader should learn not only a successful calculation, but also how to detect an unsuccessful one. Useful questions include:

- Are the free indices the same on both sides of the equation?
- Do the dimensions agree?
- Is the expression invariant under the symmetry it is supposed to respect?
- Does the result reduce correctly in a simple limit?
- Have all boundary conditions been used consistently?
- Is the conserved quantity really independent of time?

8. A concise conclusion

A section should close by identifying what has been established and what will be used next. This final paragraph is important in a cumulative subject. The student should know which result is a definition, which is a derived equation, which is a conservation law, and which is a method of solution.

The notes will therefore distinguish clearly between several kinds of statements:

- definitions that introduce notation or structure;
- assumptions that restrict the model;
- identities that hold algebraically;
- field equations that hold for physical solutions;
- conservation laws that follow from symmetries;
- examples that illustrate a method;
- approximations that hold only in a specified regime.

This distinction prevents formulas from appearing as an undifferentiated list. It allows the reader to understand not only what an equation says, but why it is valid and when it may be used.

Course Structure and Chapter Roadmap

The course is organized into twelve principal chapters, corresponding approximately to twelve teaching weeks. The chapters are not independent units. Each one introduces tools that are used later, and the order is chosen so that the formalism grows from familiar mechanics toward relativistic fields, electrodynamics, propagation, radiation, and geometric field theories.

The first three chapters establish the field concept and the variational method. Chapters 4–6 introduce the relativistic language, the scalar-field prototype, and the relation between symmetry and conservation. Chapters 7–9 develop electromagnetism as a complete classical field theory, including gauge freedom and energy-momentum. Chapters 10–11 explain how fields generated by sources are solved and how radiation emerges. Chapter 12 gives a geometric outlook toward gauge theory and gravity.

Chapter 1: Fields, Locality, and Continuous Degrees of Freedom

The opening chapter establishes what a field is and why fields are required. We begin with familiar examples: temperature as a scalar field, the velocity of a fluid as a vector field, the displacement of a string as a time-dependent field, and the electric field as a physical quantity defined throughout space. These examples make clear that a field assigns data to every point of a domain and therefore carries infinitely many degrees of freedom.

The idea of locality is then introduced. A local field theory describes changes at a point in terms of the field and its derivatives near that point. This contrasts with a direct action-at-a-distance description. The distinction will be illustrated through sources, responses, and propagation. We will ask what it means for a source to affect a field locally and how information can move through a continuous medium.

A central worked construction will be the passage from a discrete chain of coupled oscillators to a continuous string. The discrete labels will be replaced by a spatial coordinate, finite differences will become derivatives, and the sum over masses will become an integral. This example will show explicitly how a field theory can emerge as the continuum limit of a many-particle system.

By the end of the chapter, the student should be able to identify scalar, vector, and tensor fields, state their domains and values, distinguish discrete from continuous degrees of freedom, and explain why local differential equations are natural equations of motion for fields.

Chapter 2: Action Principles: From Particles to Fields

The second chapter develops the variational principle in the setting of ordinary mechanics. This is necessary because the field-theoretic action is a direct extension of the mechanical action. We introduce a functional as an object whose input is an entire trajectory rather than a single number. The action is then defined as an integral of the Lagrangian along a path.

The variation of a trajectory will be constructed explicitly. We will compare a physical trajectory with nearby trajectories that share fixed endpoints, differentiate the action with respect to the variation parameter, and use integration by parts. The boundary term will not simply be discarded; the endpoint conditions that make it vanish will be stated and interpreted.

The Euler–Lagrange equations will be derived in full and applied to several concrete systems: a free particle, a particle in a potential, the harmonic oscillator, and a simple constrained coordinate. These examples will show how to move from a Lagrangian to an equation of motion and then verify that a proposed solution satisfies the resulting equation.

This chapter also establishes a working habit that will recur throughout the course: whenever an equation is derived from an action, the allowed variations, the boundary data, and the variables held fixed must be specified.

Chapter 3: Lagrangian Field Theory and Field Equations

The third chapter performs the transition from trajectories to fields. The Lagrangian becomes a Lagrangian density, and the action becomes an integral over spacetime. A variation now changes the field value at every point, and derivatives of the field produce derivatives of the variation.

The Euler–Lagrange field equations will be derived carefully for one field and then generalized to several coupled fields. The calculation will display the spacetime integration by parts and the boundary term on the boundary of the integration region. This provides the basic method used in nearly every later derivation.

Several field equations will be obtained from actions. The vibrating string will produce the wave equation. A static energy functional will lead to Laplace or Poisson equations. A field with a potential $V(\phi)$ will provide a first example of a nonlinear field equation. We will compare the roles of kinetic, gradient, source, and potential terms in a Lagrangian density.

The chapter will close by identifying canonical field momenta and the Hamiltonian density at an introductory level. The main goal, however, is practical: given a Lagrangian density, the student should be able to vary it, manage the boundary term, and derive the corresponding differential equation.

Chapter 4: Spacetime, Lorentz Symmetry, and Covariant Notation

Classical relativistic field theory requires a language in which space and time are combined consistently. This chapter introduces Minkowski spacetime, events, the spacetime interval, proper time, and Lorentz transformations. The metric convention will be fixed and used throughout the remaining chapters.

Four-vectors, covectors, and tensors will be introduced through calculations rather than definitions alone. Students will practice raising and lowering indices, contracting tensors, distinguishing free and dummy indices, and checking whether an expression is a Lorentz scalar. Four-position, four-velocity, four-momentum, and the four-gradient will provide the main examples.

The d'Alembert operator will be constructed from the metric and the four-gradient. Familiar equations such as the wave equation will then be rewritten in covariant form. We will compare the compact tensor notation with the expanded time-and-space components so that the notation does not conceal the underlying differential equation.

Particular attention will be given to sign conventions. The notes will show how the chosen metric signature affects norms, derivatives, and Lagrangian densities. By the end of the chapter, the student should be able to read and manipulate the index notation used in the remainder of the course.

Chapter 5: The Real Scalar Field

The real scalar field is the simplest complete relativistic field theory and will serve as the principal training model. We construct the free scalar-field Lagrangian from Lorentz scalars and derive the massless and massive field equations. The massive equation is the classical Klein–Gordon equation, treated here as a classical partial differential equation rather than as a quantum equation.

Plane-wave solutions will be substituted explicitly into the field equation. This calculation will produce the dispersion relation and clarify the meaning of frequency, wave vector, and the mass parameter. The massless and massive cases will be compared, including phase and group behavior where appropriate.

The chapter will then examine energy and momentum for scalar fields. We will calculate the energy density of simple configurations and ask under what conditions the energy is bounded below. Interacting theories with a potential $V(\phi)$ will be introduced, and equilibrium configurations will be identified as extrema of the potential.

Linearization around an equilibrium will show how a nonlinear theory produces an approximate linear wave equation for small perturbations. This provides a useful method that reappears in many areas of physics: find a background solution, perturb it, keep the leading terms, and analyze stability.

Chapter 6: Symmetries, Currents, and Noether's Theorem

This chapter develops one of the central organizing principles of field theory: continuous symmetries produce local conservation laws. We begin with infinitesimal transformations of coordinates and fields, making clear what changes actively and what changes because the coordinates have been relabeled.

Noether's theorem will be derived for fields with enough detail to identify every term in the current. The derivation will distinguish variations that vanish at the boundary from transformations that represent physical symmetries. The resulting divergence equation will be related to a conserved charge by integration over space.

Several explicit applications will follow. Time translation will be connected with energy, spatial translation with momentum, and rotations with angular momentum. A complex scalar field will provide an example of an internal global phase symmetry and its associated current. For each case, the current will be calculated and its conservation checked using the field equation.

The chapter will also prepare the transition to gauge theory by distinguishing a global symmetry from a local redundancy. The student should leave with a practical algorithm: specify the infinitesimal transformation, compute the variation of the Lagrangian density, identify the Noether current, and verify its divergence.

Chapter 7: Electromagnetic Potentials and Gauge Freedom

Electromagnetism is introduced through potentials because the potentials are the natural variables of the action principle. We begin with the scalar and vector potentials and derive the electric and magnetic fields from them. The relations will be checked against the homogeneous

Maxwell equations.

Gauge transformations will then be performed explicitly. The potentials change, but the electric and magnetic fields remain unchanged. This calculation provides the first concrete example of a description containing more variables than there are physical degrees of freedom. The distinction between a change of description and a change of physical state will be emphasized.

The scalar and vector potentials will be combined into the four-potential A_μ. From it we construct the antisymmetric electromagnetic field tensor $F_{\mu\nu}$. The matrix of components will be written out, and the electric and magnetic components will be identified with careful attention to signs and index positions.

The two standard Lorentz invariants of the electromagnetic field will be calculated and interpreted. Finally, the homogeneous Maxwell equations will be written in covariant form and related to the fact that the field tensor is constructed from a potential.

Chapter 8: Maxwell Theory from an Action

This chapter builds Maxwell theory as a variational field theory. The electromagnetic Lagrangian density is formed from the field tensor, and a coupling term to an external four-current is added. We then vary the action with respect to the four-potential.

The calculation will be shown in full: the variation of the field tensor, the use of antisymmetry, the integration by parts, the treatment of the boundary term, and the extraction of the inhomogeneous Maxwell equations. The covariant result will be expanded into Gauss's law and the Ampere–Maxwell law.

Gauge invariance of the source coupling will be analyzed. We will show that the action changes by a boundary term only when the current is conserved. This makes the relation between gauge invariance and charge conservation explicit rather than merely asserted.

The Lorenz gauge condition will be introduced and used to reduce the equations for the potentials to wave equations. The action for a relativistic charged particle interacting with the electromagnetic potential will then be varied to obtain the Lorentz-force law. This completes a unified action-based description of both the field and its interaction with matter.

Chapter 9: Energy, Momentum, and Stress in Field Theory

Field equations describe evolution, but they do not by themselves show how energy and momentum are distributed and transported. This chapter develops the local energy-momentum description of scalar and electromagnetic fields.

We begin with canonical field momentum and Hamiltonian density. Translation symmetry is then used to construct the canonical energy-momentum tensor. The distinction between canonical and symmetric tensors will be discussed, including why the symmetric form is especially natural in relativistic theories and in coupling to gravity.

For the scalar field, the energy density, momentum density, and flux will be computed. For the electromagnetic field, the components of the energy-momentum tensor will be identified with electromagnetic energy density, the Poynting vector, momentum density, and the Maxwell stress tensor.

Several explicit calculations will show how these quantities are used. We will calculate energy flow in a plane wave, force from the stress tensor in a simple geometry, and radiation pressure. Local conservation equations will be integrated over a finite region to obtain balance laws with flux through the boundary.

Chapter 10: Green Functions, Sources, and Field Propagation

The tenth chapter turns from deriving field equations to solving them. Green functions provide a systematic method for linear differential equations with sources. We introduce the idea through an operator equation and define a Green function as the response to a point source.

The static case will be developed first. The Green function of the Poisson equation in three-dimensional space will be used to recover the Coulomb potential. The solution for a distributed source will then be obtained by superposition. This calculation will connect the abstract Green-function definition with a familiar physical result.

Time-dependent equations require additional choices. We will construct retarded and advanced Green functions for wave equations and explain the boundary conditions encoded by each choice. The retarded solution will be associated with causal propagation from past sources to later fields.

The method will then be applied to electromagnetic potentials. Retarded potentials will be derived as integrals over the source evaluated at retarded time. The chapter will emphasize that a differential equation alone does not determine a unique physical solution; boundary and causal conditions are part of the problem.

Chapter 11: Radiation from Time-Dependent Sources

Radiation is the regime in which a time-dependent source creates a field that carries energy to arbitrarily large distances. We begin by separating near-field terms from radiation-field terms according to their dependence on distance. This makes clear why a $1/r$ field can transport a finite amount of energy through a large sphere.

A multipole expansion will organize the radiation produced by localized sources. The electric dipole will be studied in detail. Starting from an oscillating dipole moment, we will calculate the far-zone fields, the Poynting vector, the angular distribution of emitted power, and the total radiated power.

The Lienard–Wiechert potentials will then be introduced for a moving point charge. Their dependence on retarded time will be analyzed, and the velocity and acceleration contributions to the field will be distinguished. The acceleration field will be identified as the radiative part.

The Larmor formula will be derived or motivated from the radiation field, with its assumptions stated clearly. Energy balance and radiation reaction will be discussed at an introductory level, including the conceptual difficulties that arise when a point charge interacts with its own field.

Chapter 12: Gauge Fields and Gravity: The Geometric Outlook

The final chapter places the preceding material in a broader geometric framework. Electromagnetism will be reinterpreted as a $U(1)$ gauge theory. The four-potential will be viewed as a connection, the field tensor as curvature, and the gauge-covariant derivative as the derivative appropriate for charged fields.

This language will be introduced through the electromagnetic case before any non-Abelian generalization is attempted. The non-Abelian field strength will then be presented as a natural extension in which the gauge potentials themselves interact because the symmetry group is noncommutative.

Gravity will be introduced as a theory in which the metric is a dynamical field. Connections, curvature, geodesics, and the Einstein–Hilbert action will be discussed at a structural level. The variation of the gravitational action will be outlined sufficiently to show why the Einstein tensor appears and why stress-energy acts as a source.

The weak-field form $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ will connect general relativity with linear field theory and the Newtonian potential. The chapter will conclude by comparing electromagnetism and gravity: their dynamical variables, symmetries, sources, field strengths, conservation laws, and geometric interpretations. The purpose is not to complete a course in Yang–Mills theory or general relativity, but to show where the methods developed in the previous eleven chapters lead.

How to Read and Extend These Notes

These notes are intended to complement a lecture presented at the board. The lecture can emphasize the main physical argument, develop a representative derivation, and respond to questions. The written notes can preserve the complete structure: definitions, conventions, intermediate steps, calculations, checks, diagrams, and conclusions. Neither form should merely duplicate the other. Together they should allow the student to see both the main line of reasoning and the details needed for independent study.

A productive way to read a chapter is to separate three levels of understanding.

1. **Conceptual level:** What physical or mathematical problem is being addressed? What new object or principle is introduced?

2. **Derivational level:** How does the main equation follow from the assumptions? Which steps use algebra, calculus, symmetry, or boundary conditions?

3. **Operational level:** Given a specific field, source, or transformation, can the method be applied to obtain a concrete result?

The student should not try to memorize every displayed equation at the first reading. It is more useful to identify the role of each equation. Is it a definition, an identity, an equation of motion, a conservation law, a gauge choice, or a solution under specified boundary conditions? Once that role is understood, the formula is easier to reconstruct and use.

Calculations should be read actively. Before following a derivation, the reader should note the variables, free indices, dimensions, and expected form of the answer. After the derivation, the result should be checked in a simple case. For instance, a covariant equation can be expanded into components, a conservation law can be integrated over space, and a relativistic expression can be examined in a low-velocity limit.

The notes will be developed incrementally. Each chapter is divided into section files, and the substantive content will be written one section at a time. This is not only a repository-management choice. It supports mathematical and pedagogical quality. A single section can be checked for logical continuity, notation, sign conventions, examples, and links to earlier material before the next section is added.

The intended development cycle for future work is:

1. select one section;

2. identify its precise learning objective;

3. write the motivation and definitions;

4. develop the derivation or central argument;

5. add one or more explicit examples;

6. verify notation, equations, and limiting cases;

7. compile and inspect the PDF;

8. continue to the next section only after the current one is coherent.

This section-by-section method prevents a chapter from becoming a long sequence of loosely connected formulas. It also makes it possible to revise the course structure without rewriting unrelated material. If a topic later requires more detail, its section can be expanded or divided. If a topic proves secondary, it can be shortened without disturbing the rest of the chapter.

The twelve chapters describe the intended logical route, but the final notes should remain responsive to the actual needs of students. Some ideas may require additional preliminary explanations. Some calculations may need a second example. A diagram may communicate a geometric relation more effectively than another paragraph. The structure should therefore be stable enough to guide the course, but flexible enough to improve as the material is taught and reviewed.

A successful use of these notes should lead to more than recognition. After studying a topic, the student should be able to explain the problem in words, reproduce the main derivation with stated assumptions, carry out a representative calculation, interpret the result, and recognize the most common mistakes. That standard will guide the construction of every subsequent section.

Chapter 1

Fields, Locality, and Continuous Degrees of Freedom

1.1 What Is a Field?

A field is a way of assigning physical or mathematical information to every point of a region. The word "field" is familiar from phrases such as temperature field, velocity field, electric field, or gravitational field, but the common structure behind these examples is easy to miss. In each case, the question is not only "what is the value of a quantity?" but rather

> What value does the quantity have *here*, and how does that value compare with the value at nearby points?

This spatial dependence is the first essential feature of a field. If the field can also change with time, then we ask the same question at every point and at every instant.

From a single quantity to a distributed quantity

Consider the temperature of a small object that is kept almost uniform. In a simple model, one number may be sufficient:

$$T = 293\,\mathrm{K}.$$

This number describes the whole object only because we have assumed that the temperature is approximately the same everywhere. If one end of the object is heated, that assumption is no longer reasonable. We now need a temperature value for each position. For a thin rod, we may write

$$T = T(x,t),$$

where x labels position along the rod and t denotes time. The symbol T still represents temperature, but it no longer denotes one number. It denotes a rule that returns a number when a position and a time are supplied.

For example, suppose that at one instant the temperature along a one-metre rod is approximated by

$$T(x) = 293\,\mathrm{K} + 10\,\mathrm{K\,m^{-1}}\,x, \qquad 0 \le x \le 1\,\mathrm{m}.$$

At the left end,

$$T(0) = 293\,\mathrm{K},$$

while at $x = 0.40\,\mathrm{m}$,

$$T(0.40\,\mathrm{m}) = 293\,\mathrm{K} + 10\,\mathrm{K\,m^{-1}} \cdot 0.40\,\mathrm{m} = 297\,\mathrm{K}.$$

The calculation is elementary, but it shows the operational meaning of the field: the formula assigns a temperature to every allowed position. A single expression contains an entire spatial distribution.

A mathematical description

Abstractly, a field may be written as a map

$$\Phi : D \longrightarrow V.$$

Here:

- D is the domain on which the field is defined;
- V is the set or space of possible field values;
- $\Phi(p)$ is the field value at the point $p \in D$.

The domain may be a line, a surface, a region of three-dimensional space, or spacetime itself. The value space V may consist of real numbers, vectors, matrices, tensors, or more complicated objects. The distinction between these possibilities will be developed in the next section.

The notation should always reveal what information the field requires. For a static field in ordinary space we may write

$$\Phi = \Phi(\mathbf{x}),$$

where $\mathbf{x} = (x, y, z)$. For a time-dependent field we write

$$\Phi = \Phi(\mathbf{x}, t).$$

Later, in relativistic notation, space and time will be combined into a spacetime point x^μ, and the same field may be written as $\Phi(x^\mu)$.

A field configuration

At a fixed time $t = t_0$, the function

$$\mathbf{x} \longmapsto \Phi(\mathbf{x}, t_0)$$

is called a field configuration, or a snapshot of the field. It tells us the value of the field everywhere in space at that instant.

This is the field-theoretic analogue of specifying the position of a particle at one instant. The important difference is that a particle position is described by finitely many numbers, whereas a field configuration requires one value at every point of the spatial domain.

For a vibrating string, for example, the displacement field may be denoted by

$$u = u(x, t).$$

At a fixed time $t = t_0$, the graph of $u(x, t_0)$ is the actual shape of the string at that instant. Knowing only $u(0.5\,\mathrm{m}, t_0)$ tells us the displacement of one point, not the configuration of the whole string. To specify the complete state of the string, we need the displacement for every x, and generally also its time derivative $\partial u/\partial t$.

This observation is the origin of the statement that a field has continuously many degrees of freedom. The idea will be made precise later in this chapter by comparing a field with a chain of many coupled oscillators.

A field is more than a collection of values

It is useful to distinguish a field from an arbitrary list of unrelated numbers. Physical fields are usually assumed to vary in a sufficiently regular way. Nearby points tend to have related field values, and derivatives such as

$$\frac{\partial \Phi}{\partial x}, \qquad \frac{\partial \Phi}{\partial t}, \qquad \nabla \Phi$$

carry physical information.

For the temperature field, a spatial derivative measures how rapidly temperature changes from one place to another. For the displacement of a string, a spatial derivative measures the local slope, while a time derivative measures the local velocity. These derivatives allow a theory to connect the behaviour at one point with the behaviour in its neighbourhood.

A field theory therefore contains more than the statement that a field exists. It also contains equations that determine how the field may vary. A typical field equation relates the value of the field and its derivatives. For example, the one-dimensional wave equation has the form

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}.$$

We will derive this equation later rather than accept it as a formula to memorize. For the moment, its structure is what matters: the acceleration of one small part of the string is related to the curvature of the string near that point.

Background fields and dynamical fields

Not every field appearing in a calculation is treated in the same way. A useful distinction is the following.

A *background field* is prescribed as part of the environment. We may study the motion of a charged particle in a given electric field $\mathbf{E}(\mathbf{x}, t)$ without asking how that electric field was produced.

A *dynamical field* is an unknown quantity whose evolution is determined by a field equation. In electrodynamics, the electromagnetic field itself becomes dynamical when Maxwell's equations are solved together with specified sources and boundary data.

The same mathematical object can play either role depending on the problem. A gravitational field may be treated as a fixed background when studying a falling particle, or as a dynamical field when solving the equations of general relativity. It is therefore important to ask:

> Is this field given, or is it one of the unknowns of the theory?

Fields and measurement

A field is not identified only by a symbol. Its physical meaning depends on how its values are connected with observations. A temperature field may be sampled by thermometers placed at different positions. A fluid velocity field may be inferred from the motion of small tracer particles. An electric field may be defined operationally through the force on a sufficiently small test charge.

This does not mean that every useful field variable is directly observable. Potentials, for example, may contain a redundancy: different potentials can describe the same electric and magnetic fields. Such cases require care, because a mathematical variable may be indispensable for formulating the theory even when not every part of it represents an independent observable quantity.

For each field introduced in this course, we will therefore ask four questions:

1. On what domain is the field defined?

2. What kind of object is assigned to each point?
3. Is the field prescribed or dynamical?
4. How are its values connected with measurable effects?

What has been established

A field is a rule that assigns an appropriate value to every point of a domain. A time-dependent field assigns such values throughout space and throughout time. A configuration is the complete spatial distribution at one instant, while the full field history describes how configurations change.

The essential conceptual step is the replacement of a finite list of variables by a continuously distributed quantity. The next section examines what kind of value may be attached to each point and why scalar, vector, and tensor fields must be distinguished.

1.2 Scalar, Vector, and Tensor Fields

The previous section described a field as a rule that assigns a value to every point of a domain. We now ask a second question:

> What kind of value is attached to each point?

The answer matters because different kinds of values respond differently when coordinates are changed. A temperature does not acquire a direction when the coordinate axes are rotated. A velocity does have a direction, although its numerical components depend on the chosen axes. A stress field contains still more information: it describes how forces act on surfaces with different orientations.

The terms *scalar*, *vector*, and *tensor* identify these different transformation behaviours.

Scalar fields

A scalar field assigns one number to every point. In three-dimensional space, a scalar field may be written as

$$\phi : D \subseteq \mathbb{R}^3 \longrightarrow \mathbb{R}, \qquad \mathbf{x} \longmapsto \phi(\mathbf{x}).$$

Typical examples include temperature, mass density, pressure, electric potential, and the real scalar field studied later in this course.

The word *scalar* does not merely mean that we have written one numerical component. It means that the physical value is unchanged when we describe the same point using another coordinate system. If a room has temperature $295\,\mathrm{K}$ at a particular event, rotating the coordinate axes does not change that temperature.

As a simple example, consider the field on a flat plate

$$T(x, y) = 300\,\mathrm{K} - 2\,\mathrm{K\,m^{-1}}\,x + 3\,\mathrm{K\,m^{-1}}\,y.$$

At the point $(x, y) = (1\,\mathrm{m}, 2\,\mathrm{m})$,

$$T(1\,\mathrm{m}, 2\,\mathrm{m}) = 300\,\mathrm{K} - 2\,\mathrm{K} + 6\,\mathrm{K} = 304\,\mathrm{K}.$$

Another observer may use rotated coordinates (x', y'), but the temperature assigned to the same physical point is still $304\,\mathrm{K}$. The formula written in terms of x' and y' may look different, yet the scalar value at the point is the same.

A useful distinction is therefore:

- the *coordinate expression* of a field may change;
- the physical scalar assigned to a point does not depend on the coordinate labels.

Vector fields

A vector field assigns a vector to every point:

$$\mathbf{v} : D \subseteq \mathbb{R}^3 \longrightarrow \mathbb{R}^3.$$

Examples include the velocity field of a fluid, the electric field, the magnetic field, and the displacement field of an elastic medium.

In a chosen Cartesian basis, a vector field is written in components as

$$\mathbf{v}(\mathbf{x}) = v^1(\mathbf{x})\,\mathbf{e}_1 + v^2(\mathbf{x})\,\mathbf{e}_2 + v^3(\mathbf{x})\,\mathbf{e}_3.$$

The three component functions v^1, v^2, v^3 are scalar-valued functions, but together they represent one geometric vector. If the coordinate axes are rotated, the components change in a coordinated way so that the physical vector remains the same.

This is why a vector should not be identified with a column of numbers alone. The column

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

represents a vector only after a basis has been specified. Relative to a different basis, the same vector generally has different components.

Consider the two-dimensional velocity field

$$\mathbf{v}(x, y) = -\Omega y\,\mathbf{e}_x + \Omega x\,\mathbf{e}_y,$$

where Ω is a constant with units of inverse time. At the point $(x, y) = (2\,\mathrm{m}, 1\,\mathrm{m})$,

$$\mathbf{v}(2\,\mathrm{m}, 1\,\mathrm{m}) = -\Omega(1\,\mathrm{m})\,\mathbf{e}_x + \Omega(2\,\mathrm{m})\,\mathbf{e}_y.$$

If $\Omega = 0.5\,\mathrm{s}^{-1}$, then

$$\mathbf{v} = -0.5\,\mathrm{m\,s^{-1}}\,\mathbf{e}_x + 1.0\,\mathrm{m\,s^{-1}}\,\mathbf{e}_y.$$

The local speed is the magnitude

$$\|\mathbf{v}\| = \sqrt{(-0.5)^2 + (1.0)^2}\,\mathrm{m\,s^{-1}} = \sqrt{1.25}\,\mathrm{m\,s^{-1}}.$$

This example shows how a vector field is used operationally: choose a point, evaluate each component there, and combine the components into a vector with magnitude and direction. The complete field may be visualized as an arrow attached to every point of the domain.

Tensor fields

A tensor field assigns a tensor to every point. Tensors generalize scalars and vectors. A scalar may be regarded as a tensor of rank zero, and a vector as a tensor of rank one. A rank-two tensor contains information about how one direction is related to another.

In coordinates, a rank-two tensor field may be written as an array of components

$$T^{ij}(\mathbf{x}) \quad \text{or} \quad T_{ij}(\mathbf{x}).$$

However, not every array of numbers is automatically a tensor. The defining property is the transformation law: when coordinates are changed, the components must transform so that the underlying geometric or physical object remains well defined.

A concrete example is the stress tensor $\boldsymbol{\sigma}$ in a continuous material. At each point, it determines the force per unit area acting on a small surface through that point. The force depends not only on position but also on the orientation of the surface. If $\mathbf{n}$ is a unit normal to the surface, the traction vector is

$$\mathbf{t} = \boldsymbol{\sigma}\,\mathbf{n}.$$

In two dimensions, suppose that at one point

$$\boldsymbol{\sigma} = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix} \text{MPa}.$$

For a surface whose unit normal is $\mathbf{n} = \mathbf{e}_x$, represented by

$$\mathbf{n} = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

we obtain

$$\mathbf{t} = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{MPa} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \text{MPa}.$$

For a surface with normal $\mathbf{e}_y$,

$$\mathbf{t} = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{MPa} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \text{MPa}.$$

The tensor therefore contains more information than a single vector: it tells us which traction vector is associated with each possible surface direction. This is a useful model for understanding why tensor fields naturally appear when physical effects depend on orientation.

Other important tensor fields include the electromagnetic field tensor, the energy-momentum tensor, and the spacetime metric. Each will be introduced only when there is a physical problem that requires it.

Transformation behaviour is the essential distinction

It is tempting to distinguish the three cases simply by counting components:

- one component for a scalar;
- several components for a vector;
- a matrix of components for a rank-two tensor.

This is useful as a first picture, but it is not the complete definition. The essential question is how the components change when the basis or coordinates change.

A scalar field obeys

$$\phi'(x') = \phi(x),$$

meaning that both expressions assign the same scalar to the same physical point. A vector transforms by one copy of the coordinate transformation matrix, while a rank-two tensor transforms by two copies. The precise formulas will be developed when tensor notation is introduced in Chapter 4.

For now, the practical lesson is:

> The components depend on the description; the scalar, vector, or tensor represented by those components is the physical or geometric object.

Several fields may describe one system

A physical system is rarely described by only one field. A fluid may require a mass-density field $\rho(\mathbf{x}, t)$, a pressure field $p(\mathbf{x}, t)$, a temperature field $T(\mathbf{x}, t)$, and a velocity field $\mathbf{v}(\mathbf{x}, t)$. These fields have different value types but share the same spatial and temporal domain.

Similarly, electrodynamics uses a scalar charge-density field, a vector current-density field, electric and magnetic vector fields, and later a single antisymmetric tensor field that combines the electric and magnetic information in a relativistic form.

The choice of field variables is part of constructing a theory. We must choose enough fields to describe the relevant physical information, but we should avoid introducing quantities that are redundant unless that redundancy serves a useful structural purpose.

What has been established

A scalar field assigns one coordinate-independent number to each point. A vector field assigns a geometric vector whose components depend on the chosen basis. A tensor field assigns a multilinear object that can relate several directions and whose components obey a definite transformation law.

The number of written components is not the deepest distinction. What matters is how the assigned object behaves under a change of coordinates. In the next section, we turn from the type of field value to the domain on which the field is defined and to the roles played by space and time.

1.3 Space-Time Dependence and Field Domains

A field is not specified completely until its domain is known. The formula

$$\phi(x, t) = A\cos(kx - \omega t)$$

may describe a disturbance on an infinite line, a wave on a finite string, or a local approximation inside a larger system. The same expression can represent different physical problems depending on the allowed values of x, the time interval under consideration, and the conditions imposed at the boundary.

The domain is therefore not a technical detail added after the field has been defined. It is part of the model.

Spatial domains

A field may be defined on spaces of different dimensions.

For a thin rod or string, one spatial coordinate may be sufficient:

$$D = [0, L] \subset \mathbb{R}.$$

A field on the rod is then written as $u(x, t)$, with $0 \leq x \leq L$. The reduction to one dimension is a modelling assumption. It means that variations across the thickness of the rod are neglected or averaged.

For a thin plate or membrane, the domain may be a region $D \subset \mathbb{R}^2$, and a field is written as

$$\phi = \phi(x, y, t).$$

For a field in ordinary three-dimensional space, the domain is a region $D \subset \mathbb{R}^3$, and we write

$$\phi = \phi(x, y, z, t) \quad \text{or} \quad \phi = \phi(\mathbf{x}, t).$$

The notation $\mathbf{x}$ is compact, but it should not hide the fact that three independent spatial coordinates are present.

Some fields live on curved surfaces or curved spaces. For example, the temperature on the surface of a sphere is naturally a field on that sphere, not on the entire surrounding three-dimensional space. Coordinates such as latitude and longitude may be used to label points, but the sphere itself is the geometric domain. Later, when gravity is discussed, spacetime will play this role as a curved domain.

Static and time-dependent fields

A *static field* does not change with time. It is written as

$$\phi = \phi(\mathbf{x}).$$

This does not mean that the field is spatially uniform. A static electric potential may vary strongly from point to point while remaining unchanged in time.

A *time-dependent field* is written as

$$\phi = \phi(\mathbf{x}, t).$$

At each fixed time $t = t_0$, the function $\phi(\mathbf{x}, t_0)$ is one field configuration. As time changes, the system passes through a family of configurations.

It is helpful to separate two related objects:

- the *configuration at time* t_0, which is a function of space only;
- the *field history*, which is a function of both space and time.

For a vibrating string, a photograph at one instant shows the configuration $u(x, t_0)$. A video of the motion represents the history $u(x, t)$.

Spacetime as a single domain

In nonrelativistic notation, space and time are usually written separately. In relativistic field theory, they are combined into a spacetime point. We write

$$x^\mu = (x^0, x^1, x^2, x^3),$$

where the time coordinate is commonly chosen as $x^0 = ct$ or, in units with $c = 1$, simply $x^0 = t$.

A relativistic field may then be written as

$$\phi = \phi(x^\mu).$$

This notation does not erase the physical distinction between time and space. It organizes them into one structure so that Lorentz transformations can be handled consistently. The geometric details will be developed in Chapter 4. At this stage, the important idea is that a field assigns data to events in spacetime, not merely to points in space.

Partial derivatives and what they measure

When a field depends on several independent variables, its rate of change is described by partial derivatives. Consider the field

$$u(x, t) = A\cos(kx)\cos(\omega t).$$

The spatial derivative is

$$\frac{\partial u}{\partial x} = -Ak\sin(kx)\cos(\omega t).$$

This derivative measures how the displacement changes when the position is varied while the time is held fixed. For a string, it is related to the local slope of the string.

The time derivative is

$$\frac{\partial u}{\partial t} = -A\omega\cos(kx)\sin(\omega t).$$

This derivative measures how the displacement at one fixed position changes with time. For a string, it is the transverse velocity of the material point labelled by x.

The two derivatives answer different physical questions. Confusing them would mean confusing a comparison between neighbouring points at one instant with a comparison between different instants at one point.

The dimensions provide a useful check. If u has dimensions of length, then

$$\left[\frac{\partial u}{\partial x}\right] = 1, \qquad \left[\frac{\partial u}{\partial t}\right] = \text{length/time}.$$

The spatial derivative is dimensionless in this example because it represents a slope, while the time derivative has the dimensions of velocity.

The field equation is not the whole problem

A differential equation usually admits many solutions. To determine the physical field, one must also specify the domain and appropriate data.

Consider a string occupying

$$0 \le x \le L$$

and satisfying the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}.$$

This equation alone does not tell us which motion occurs. We also need initial data, such as

$$u(x,0) = f(x), \qquad \frac{\partial u}{\partial t}(x,0) = g(x).$$

The function $f(x)$ gives the initial shape of the string, and $g(x)$ gives the initial velocity of each point.

If the endpoints are fixed, we also impose boundary conditions:

$$u(0,t) = 0, \qquad u(L,t) = 0.$$

The complete mathematical problem therefore consists of four ingredients:

1. the field variable $u(x,t)$;
2. the domain $0 \le x \le L$;
3. the field equation;
4. the initial and boundary data.

Changing the boundary conditions changes the physical problem. A string with fixed endpoints, a string with one free endpoint, and a periodic string may obey the same local differential equation but have different allowed motions.

Boundary conditions encode physical information

A boundary condition is not merely a device for selecting one mathematical solution. It represents how the system interacts with its surroundings.

For a temperature field, fixing T at the boundary may represent contact with a heat reservoir. Fixing the normal heat flux may represent insulation or a prescribed flow of heat. For an electromagnetic field, boundary conditions describe conductors, interfaces between materials, incoming radiation, or behaviour at large distances.

It is therefore useful to ask what physical arrangement produces the chosen boundary condition. A mathematically convenient condition may not describe the experiment under consideration.

Local coordinates and physical points

Coordinates are labels assigned to points of the domain. The physical point is not identical with its coordinate tuple. On a flat Cartesian domain, the distinction may seem unnecessary because the coordinates are especially simple. On curved surfaces or in curvilinear coordinates, it becomes essential.

For example, the north pole of a sphere is one physical point even though longitude is not uniquely defined there. A field value at the north pole must be well defined independently of this coordinate ambiguity.

This observation prepares an important theme of field theory: equations should describe geometric or physical relations, not depend on an arbitrary choice of coordinates. Component formulas may change when coordinates are changed, while the underlying statement remains the same.

A checklist for specifying a field problem

Before manipulating a field formula, identify the following information:

1. What is the spatial domain?
2. Is the field static or time dependent?
3. What type of value is assigned to each point?
4. Which coordinates are being used?
5. What differential equation does the field satisfy?
6. What initial data are specified?
7. What boundary conditions are imposed?

If any of these items is missing, the problem may be incomplete or ambiguous.

What has been established

A field is defined on a specific spatial or spacetime domain. Its dependence on space describes variation from point to point, while its dependence on time describes the evolution of configurations. Partial derivatives express these distinct kinds of change.

A field equation must be interpreted together with its domain, initial data, and boundary conditions. The next section examines another structural property of field theories: locality, the idea that the behaviour at a point is governed by field values and derivatives in its immediate neighbourhood.

1.4 Locality, Sources, and Field Response

A field theory becomes especially useful when it replaces a direct interaction between distant objects by a local description. Instead of saying that one object acts immediately on another object far away, we introduce a field that exists throughout the intervening region. The source changes the field near itself, the field propagates or adjusts according to a differential equation, and another object responds to the field at its own position.

This viewpoint separates three questions:

1. What produces the field?
2. How does the field behave between sources?
3. How does matter respond to the field locally?

The answers are encoded in sources, field equations, and coupling laws.

What locality means

A field equation is called local when the behaviour at a point is determined by the field and a finite number of its derivatives evaluated at that point. A schematic local equation has the form

$$\mathcal{F}(\phi(x), \partial_\mu \phi(x), \partial_\mu \partial_\nu \phi(x), \ldots) = J(x),$$

where $J(x)$ represents a source.

The equation may involve neighbouring values indirectly through derivatives, but it does not require an independent instruction involving every distant point. A derivative is local because it is determined by how the field changes in an arbitrarily small neighbourhood.

For example, the one-dimensional wave equation

$$\frac{\partial^2 u}{\partial t^2}(x,t) = c^2 \frac{\partial^2 u}{\partial x^2}(x,t)$$

relates the acceleration of the string at (x,t) to the curvature of the string at the same point and time. The curvature is calculated from nearby points, not from the entire string at once.

Locality does not mean that distant regions never influence one another. It means that the influence is transmitted through successive local interactions. A disturbance created near one end of a string changes neighbouring parts of the string, and the disturbance then travels along it.

Sources

A source is a quantity that drives or generates a field. The source may be concentrated at isolated points, distributed throughout a region, or imposed at a boundary.

Examples include:

- electric charge density as a source of the electromagnetic field;
- mass-energy as a source of the gravitational field;
- an external force density as a source of displacement in an elastic medium;
- heat production as a source in a temperature equation.

A simple linear field equation may be written as

$$\mathcal{L}\phi = J,$$

where $\mathcal{L}$ is a differential operator, ϕ is the field, and J is the source.

This notation is compact, but each symbol has a different role. The operator tells us how the field responds, the source tells us what drives it, and the boundary or initial data select the physical solution.

A worked static example

Consider a scalar field $\phi(x)$ on the interval

$$0 \leq x \leq L.$$

Suppose that it satisfies

$$\frac{\mathrm{d}^2\phi}{\mathrm{d}x^2} = -\rho_0,$$

where ρ_0 is a constant source density. We impose the boundary conditions

$$\phi(0) = 0, \qquad \phi(L) = 0.$$

The equation can be solved by integrating twice. First,

$$\frac{\mathrm{d}\phi}{\mathrm{d}x} = -\rho_0 x + C_1.$$

Integrating again gives

$$\phi(x) = -\frac{\rho_0}{2}x^2 + C_1 x + C_2.$$

The first boundary condition gives

$$\phi(0) = C_2 = 0.$$

The second boundary condition gives

$$0 = -\frac{\rho_0}{2}L^2 + C_1 L,$$

so

$$C_1 = \frac{\rho_0 L}{2}.$$

The field is therefore

$$\boxed{\phi(x) = \frac{\rho_0}{2}x(L - x)}.$$

We can verify the result directly:

$$\frac{\mathrm{d}\phi}{\mathrm{d}x} = \frac{\rho_0}{2}(L - 2x),$$

and

$$\frac{\mathrm{d}^2\phi}{\mathrm{d}x^2} = -\rho_0.$$

The solution also satisfies $\phi(0) = \phi(L) = 0$.

This example shows several important features of a field problem. The source fixes the local curvature of the field, while the boundary conditions determine the integration constants and therefore the global shape. Neither the source nor the boundary conditions alone are sufficient.

If $\rho_0 > 0$, the field is largest at the centre. Setting the derivative equal to zero,

$$\frac{\mathrm{d}\phi}{\mathrm{d}x} = 0,$$

gives

$$x = \frac{L}{2}.$$

At that point,

$$\phi\left(\frac{L}{2}\right) = \frac{\rho_0 L^2}{8}.$$

Thus the differential equation, the source, and the boundary conditions together determine not only the formula but also qualitative features such as where the maximum occurs.

Source-free does not mean field-free

In a region where the source vanishes, the field may satisfy a homogeneous equation,

$$\mathcal{L}\phi = 0.$$

This does not imply $\phi = 0$. A source located outside the region may produce a nonzero field inside it. Boundary data may also sustain a nonzero solution. Electromagnetic waves can travel through a region containing no charge or current.

The distinction is important:

- *source-free* means that the local source term vanishes;
- *field-free* means that the field itself vanishes.

These are different statements.

Static response and dynamical propagation

A static equation describes a field configuration after time dependence has been neglected. A dynamical equation describes how disturbances evolve.

A static source-field relation may have the form

$$\nabla^2\phi = -\rho.$$

A corresponding dynamical equation may have the form

$$\frac{\partial^2\phi}{\partial t^2} - c^2\nabla^2\phi = J.$$

The dynamical equation contains a propagation speed c. If the source changes, the response at a distant point does not appear immediately; it arrives after the disturbance has propagated.

Static solutions are still useful, but they must be interpreted correctly. They describe time-independent configurations or approximations in which propagation effects are negligible. A static formula should not automatically be read as evidence of instantaneous action at a distance.

Fields replace direct distant action

Newton's law of gravitation and Coulomb's law can be written as forces depending directly on the separation between two objects. Such formulas are effective and often sufficient for static problems. Field theory reorganizes the description.

For electromagnetism, the source produces an electromagnetic field. A test charge at position $\mathbf{x}$ responds to the field evaluated at that same position. The force law is local in this sense:

$$\mathbf{F}(t) = q\big[\mathbf{E}(\mathbf{x}(t), t) + \mathbf{v}(t) \times \mathbf{B}(\mathbf{x}(t), t)\big].$$

The charge does not need direct access to the current state of a distant source. It responds to the field where it is located. Maxwell's equations determine how changes in the sources affect the field and how those changes propagate.

This separation becomes essential when sources move rapidly or radiation is emitted. A direct static force law is then insufficient.

Linear response and superposition

If the field equation is linear,

$$\mathcal{L}\phi = J,$$

and two sources J_1 and J_2 produce fields ϕ_1 and ϕ_2, then

$$\mathcal{L}\phi_1 = J_1, \qquad \mathcal{L}\phi_2 = J_2.$$

By linearity,

$$\mathcal{L}(\phi_1 + \phi_2) = J_1 + J_2.$$

Thus the combined source produces the combined field

$$\phi = \phi_1 + \phi_2.$$

This is the principle of superposition. It allows complicated sources to be decomposed into simpler pieces. Later, Green functions will turn this principle into a systematic method of solution.

Nonlinear field equations do not generally satisfy superposition. In that case, two individual solutions cannot simply be added. Recognizing whether an equation is linear is therefore one of the first practical checks in a field problem.

Local laws and global solutions

A local differential equation constrains the field point by point, but its solution is a global object defined over the entire domain. The global solution depends on local laws together with initial and boundary data.

This distinction explains why local theories can produce rich large-scale behaviour. Waves, resonances, standing patterns, and long-range fields emerge from local differential equations applied across a domain.

The local equation tells us how neighbouring values are related. The domain and boundary conditions determine how those local relations fit together into one complete configuration.

What has been established

A local field theory describes the behaviour at a point using the field, its derivatives, and sources at that point. Distant influence is transmitted through the field rather than imposed as an independent instantaneous rule.

Sources drive fields, but source-free regions may still contain nonzero fields. Static equations describe configurations, while dynamical equations describe propagation. Linear equations admit superposition, which will later support powerful solution methods.

The next section examines why a field represents a system with continuously many degrees of freedom and how this statement should be understood without treating neighbouring field values as completely unrelated.

1.5 Degrees of Freedom in Continuous Systems

A degree of freedom is an independent quantity needed to specify the configuration of a system. For a particle moving on a line, one coordinate $q(t)$ is sufficient. For a particle moving in three-dimensional space, three coordinates are required. For N particles moving on a line, a configuration may be written as

$$\mathbf{q}(t) = \big(q_1(t), q_2(t), \ldots, q_N(t)\big).$$

The configuration is one point in an N-dimensional configuration space.

A continuous system is different. A string, fluid, elastic body, or field occupies a region containing continuously many points. To describe its configuration, we need a function rather than a finite list of numbers. The displacement of a string, for example, is written as

$$u = u(x,t), \qquad 0 \le x \le L.$$

At a fixed time, the complete configuration is the function $u(x)$. Informally, we say that there is one degree of freedom for each value of x. This statement is useful, but it must be interpreted carefully.

From finitely many coordinates to a field

Suppose that a string is approximated by N marked points. Let

$$x_n = n\,a, \qquad n = 0, 1, \ldots, N-1,$$

where

$$a = \frac{L}{N-1}$$

is the spacing between neighbouring points.

We record the displacement at each point:

$$u_n(t) = u(x_n, t).$$

The approximate configuration is then the finite vector

$$\mathbf{u}(t) = \big(u_0(t), u_1(t), \ldots, u_{N-1}(t)\big).$$

For $N = 5$, only five displacement values are stored. The shape between the marked points must be inferred by interpolation. If N is increased, more detail is retained. In the formal continuum limit,

$$N \longrightarrow \infty, \qquad a \longrightarrow 0, \qquad L \text{ fixed},$$

the finite list is replaced by the continuous function $u(x,t)$.

This is one practical meaning of a field: it is the limiting description obtained when a system is resolved at arbitrarily small spatial intervals.

A numerical example of increasing resolution

Take a string of length $L = 1\,\mathrm{m}$ with the configuration

$$u(x) = 0.02\,\mathrm{m}\,\sin\left(\frac{\pi x}{L}\right).$$

If we sample only the endpoints and the midpoint, then

$$x_0 = 0, \qquad x_1 = \frac{L}{2}, \qquad x_2 = L,$$

and

$$(u_0, u_1, u_2) = (0, 0.02\,\mathrm{m}, 0).$$

This three-number description captures the maximum displacement but not the detailed curvature.

With five equally spaced points,

$$x_n = \frac{nL}{4}, \qquad n = 0, 1, 2, 3, 4,$$

we obtain

$$\begin{aligned}
u_0 &= 0, \\
u_1 &= 0.02\sin\left(\frac{\pi}{4}\right)\mathrm{m}, \\
u_2 &= 0.02\,\mathrm{m}, \\
u_3 &= 0.02\sin\left(\frac{3\pi}{4}\right)\mathrm{m}, \\
u_4 &= 0.
\end{aligned}$$

Since

$$\sin\left(\frac{\pi}{4}\right) = \sin\left(\frac{3\pi}{4}\right) = \frac{1}{\sqrt{2}},$$

the sampled configuration is

$$\mathbf{u} = \left(0, \frac{0.02}{\sqrt{2}}, 0.02, \frac{0.02}{\sqrt{2}}, 0\right)\mathrm{m}.$$

The field formula contains all these finite approximations simultaneously. Any desired sampling resolution can be extracted from it.

The values are not arbitrary independent numbers

The phrase "one degree of freedom at each point" does not mean that neighbouring field values are always unrelated. Physical fields are usually required to be continuous or differentiable, and their equations relate values in nearby regions.

If the displacement of a string changed from 1 cm to 100 m over an arbitrarily small distance, the continuum model would usually cease to be physically meaningful. Smoothness assumptions restrict the allowed configurations.

A finite-difference approximation makes this relation visible. The spatial derivative at x_n may be approximated by

$$\frac{\partial u}{\partial x}(x_n, t) \approx \frac{u_{n+1}(t) - u_n(t)}{a}.$$

If the derivative must remain finite as $a \to 0$, then the difference $u_{n+1} - u_n$ must decrease with a. Neighbouring values cannot vary arbitrarily.

Boundary conditions impose further restrictions. For a string fixed at both ends,

$$u(0, t) = 0, \qquad u(L, t) = 0.$$

The endpoint values are not free degrees of freedom. Constraints and symmetries may similarly reduce the number of independent variables.

Configuration space becomes a function space

For a system with N coordinates, the configuration space may be $\mathbb{R}^N$. For a field, the configuration is a function, so the configuration space is a space of functions.

For a fixed string, one possible configuration space is schematically

$$\mathcal{C} = \{u : [0, L] \to \mathbb{R} \mid u(0) = u(L) = 0\},$$

with additional smoothness conditions when derivatives are needed.

The symbol $\mathcal{C}$ does not represent ordinary three-dimensional space. Each point of $\mathcal{C}$ represents one entire shape of the string. Moving through configuration space means changing from one complete field configuration to another.

This is why the action of a field theory is a functional. Its input is not a finite coordinate vector but a complete field history.

A field configuration versus a complete dynamical state

For many second-order dynamical equations, the field value at one instant is not enough to predict the future. We must also specify its time derivative.

For a string, the initial data are

$$u(x, 0) = f(x), \qquad \frac{\partial u}{\partial t}(x, 0) = g(x).$$

The function f gives the initial shape, while g gives the initial velocity distribution. Two strings may have the same shape at one instant but move differently because their velocity fields differ.

Thus we should distinguish:

- a *configuration*, specified by $u(x)$;
- a *dynamical state*, often specified by $u(x)$ together with $\partial u/\partial t$.

Later, in Hamiltonian field theory, the time derivative will be replaced by a conjugate momentum field.

Normal modes provide another set of coordinates

A field may also be described by expanding it in a set of spatial patterns. For a string fixed at $x = 0$ and $x = L$, a useful expansion is

$$u(x,t) = \sum_{n=1}^{\infty} q_n(t) \sin\left(\frac{n\pi x}{L}\right).$$

Each sine function already satisfies the endpoint conditions. The time-dependent coefficient $q_n(t)$ tells us how strongly the n-th normal mode is excited.

In this description, the field is represented not by one value at every point but by the infinite list

$$q_1(t), q_2(t), q_3(t), \ldots$$

The two descriptions contain the same information when the expansion is valid:

- position-space description: $u(x,t)$;
- mode description: the coefficients $q_n(t)$.

For example, the configuration

$$u(x,0) = A\sin\left(\frac{\pi x}{L}\right) + B\sin\left(\frac{2\pi x}{L}\right)$$

has only two nonzero mode amplitudes:

$$q_1(0) = A, \qquad q_2(0) = B, \qquad q_n(0) = 0 \quad \text{for } n \geq 3.$$

The full system permits infinitely many modes, but a particular configuration may involve only a few of them.

This mode viewpoint is important in wave physics and later becomes central in quantum field theory. It also clarifies that "infinitely many degrees of freedom" can be organized into a sequence of ordinary time-dependent coordinates.

Continuum models are approximations with a scale of validity

Many classical fields describe matter that is microscopically discrete. A string is made of atoms, a fluid is made of molecules, and an elastic solid has a crystal or molecular structure. The continuum description is effective when the length scales of interest are much larger than the microscopic spacing.

If a wave has wavelength λ and the microscopic spacing is a, the continuum approximation is usually most reliable when

$$\lambda \gg a.$$

At wavelengths comparable with the microscopic scale, the discrete structure may become observable, and the continuum field equation may need correction.

A field theory can therefore be exact within a mathematical model while still being an approximation to a more microscopic physical system. The domain of validity should always be kept in mind.

Discretization as a computational method

The same idea used to motivate the continuum limit is also used in the opposite direction for numerical calculations. A continuous field is replaced by values on a finite grid, derivatives are approximated by finite differences, and the field equation becomes a large system of algebraic or ordinary differential equations.

Increasing the number of grid points should improve the approximation when the numerical method is stable and convergent. Thus the relation

$$\text{finite system} \longleftrightarrow \text{continuous field}$$

works in both directions:

- a dense discrete system motivates a continuum field;
- a continuum field is discretized for numerical solution.

What has been established

A field represents a system whose configuration requires a function rather than a finite list of coordinates. This can be understood by sampling the field at N points and taking the limit of increasing resolution.

The field values are not generally arbitrary because smoothness, field equations, constraints, and boundary conditions relate them. The infinite-dimensional configuration space can be described either through values in position space or through infinitely many mode amplitudes.

The next section makes the discrete-to-continuum passage explicit. Starting from a chain of masses connected by springs, we will derive the wave equation and the corresponding continuum energy.

1.6 From Discrete Oscillators to Continuous Media

The continuum description of a string can be derived from a discrete mechanical model. This derivation is important because it shows that a field equation is not an arbitrary formula. It may emerge from ordinary Newtonian mechanics when a system contains many closely spaced degrees of freedom.

We consider a chain of identical masses connected by identical springs. The masses are separated by a fixed equilibrium distance a. The transverse displacement of the n-th mass is denoted by

$$u_n(t).$$

The equilibrium position of that mass is

$$x_n = na.$$

The goal is to replace the discrete variables

$$u_0(t), u_1(t), u_2(t), \ldots$$

by a continuous displacement field

$$u = u(x,t).$$

Assumptions of the model

The derivation uses several assumptions that should be stated explicitly.

1. Every mass has the same mass m.
2. Neighbouring masses are connected by identical springs with spring constant κ.
3. Only nearest neighbours interact.
4. The transverse displacements are small enough that the restoring force is linear.
5. The displacement changes smoothly over distances comparable with the spacing a.

The final continuum equation is valid only within the regime where these assumptions are reasonable.

The discrete elastic energy

For small relative displacements, the spring between masses n and $n+1$ stores the potential energy

$$V_{n,n+1} = \frac{\kappa}{2}\big(u_{n+1} - u_n\big)^2.$$

The two springs connected to the n-th mass contribute

$$V = \frac{\kappa}{2}\big(u_n - u_{n-1}\big)^2 + \frac{\kappa}{2}\big(u_{n+1} - u_n\big)^2 + \text{terms independent of } u_n.$$

The force on the n-th mass is

$$F_n = -\frac{\partial V}{\partial u_n}.$$

Differentiating the first spring term gives

$$\frac{\partial}{\partial u_n}\left[\frac{\kappa}{2}(u_n - u_{n-1})^2\right] = \kappa(u_n - u_{n-1}),$$

while the second gives

$$\frac{\partial}{\partial u_n}\left[\frac{\kappa}{2}(u_{n+1} - u_n)^2\right] = -\kappa(u_{n+1} - u_n).$$

Therefore

$$\begin{aligned} F_n &= -\kappa(u_n - u_{n-1}) + \kappa(u_{n+1} - u_n) \\ &= \kappa\big(u_{n+1} - 2u_n + u_{n-1}\big). \end{aligned}$$

Newton's second law gives the discrete equation of motion

$$\boxed{m\frac{\mathrm{d}^2 u_n}{\mathrm{d}t^2} = \kappa\big(u_{n+1} - 2u_n + u_{n-1}\big)}.$$

The combination

$$u_{n+1} - 2u_n + u_{n-1}$$

is the discrete second difference. It measures whether the displacement of the n-th mass lies above or below the average of its neighbours.

If

$$u_n = \frac{u_{n-1} + u_{n+1}}{2},$$

then the second difference vanishes, and the two neighbouring springs exert no net transverse restoring force. If u_n lies above the average, the net force points downward. The force therefore tends to reduce local curvature.

Introducing a continuous field

We now assume that the discrete displacements are samples of a smooth function:

$$u_n(t) = u(x_n, t), \qquad x_n = na.$$

The neighbouring displacements are

$$u_{n+1}(t) = u(x+a, t), \qquad u_{n-1}(t) = u(x-a, t),$$

where $x = x_n$.

For a sufficiently smooth field, Taylor expansion about x gives

$$\begin{aligned} u(x+a,t) &= u(x,t) + a\frac{\partial u}{\partial x} + \frac{a^2}{2}\frac{\partial^2 u}{\partial x^2} + \frac{a^3}{6}\frac{\partial^3 u}{\partial x^3} + O(a^4), \\ u(x-a,t) &= u(x,t) - a\frac{\partial u}{\partial x} + \frac{a^2}{2}\frac{\partial^2 u}{\partial x^2} - \frac{a^3}{6}\frac{\partial^3 u}{\partial x^3} + O(a^4). \end{aligned}$$

Adding these expressions and subtracting $2u(x,t)$, the terms linear and cubic in a cancel:

$$u(x+a,t) - 2u(x,t) + u(x-a,t) = a^2\frac{\partial^2 u}{\partial x^2} + O(a^4).$$

To leading order,

$$u_{n+1} - 2u_n + u_{n-1} \approx a^2\frac{\partial^2 u}{\partial x^2}.$$

The discrete second difference becomes the continuous second derivative.

Substituting this approximation into the discrete equation gives

$$m\frac{\partial^2 u}{\partial t^2} = \kappa a^2\frac{\partial^2 u}{\partial x^2}.$$

Identifying continuum parameters

The mass per unit length is

$$\mu = \frac{m}{a}.$$

We also define

$$T = \kappa a.$$

The quantity T has the dimensions of force:

$$[T] = [\kappa][a] = \frac{\mathrm{N}}{\mathrm{m}}\,\mathrm{m} = \mathrm{N}.$$

In the continuum string model, T plays the role of the tension.

Using

$$m = \mu a, \qquad \kappa = \frac{T}{a},$$

the discrete approximation becomes

$$\mu a \frac{\partial^2 u}{\partial t^2} = \frac{T}{a} a^2 \frac{\partial^2 u}{\partial x^2}.$$

Cancelling the common factor a, we obtain

$$\boxed{\mu \frac{\partial^2 u}{\partial t^2} = T \frac{\partial^2 u}{\partial x^2}}.$$

Dividing by μ,

$$\frac{\partial^2 u}{\partial t^2} = \frac{T}{\mu} \frac{\partial^2 u}{\partial x^2}.$$

Defining

$$c^2 = \frac{T}{\mu},$$

we arrive at the one-dimensional wave equation

$$\boxed{\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}}.$$

Dimensional check

The ratio T/μ has dimensions

$$\left[\frac{T}{\mu}\right] = \frac{\mathrm{kg\,m\,s^{-2}}}{\mathrm{kg\,m^{-1}}} = \mathrm{m^2\,s^{-2}}.$$

Therefore

$$[c] = \mathrm{m\,s^{-1}},$$

so c is a speed. This is consistent with its interpretation as the propagation speed of small disturbances along the string.

The dimensions of both sides of the wave equation also agree:

$$\left[\frac{\partial^2 u}{\partial t^2}\right] = \mathrm{m\,s^{-2}},$$

and

$$\left[c^2 \frac{\partial^2 u}{\partial x^2}\right] = \mathrm{m^2\,s^{-2}} \cdot \mathrm{m^{-1}} = \mathrm{m\,s^{-2}}.$$

Physical interpretation of the wave equation

The left-hand side,

$$\frac{\partial^2 u}{\partial t^2},$$

is the local transverse acceleration of the string.

The right-hand side contains

$$\frac{\partial^2 u}{\partial x^2},$$

which measures local curvature. A straight segment has zero second derivative and therefore no net transverse restoring force. A curved segment is accelerated toward a straighter configuration.

The coefficient T/μ compares the restoring effect of tension with the inertia of the string. Increasing the tension increases the propagation speed, while increasing the mass density decreases it:

$$c = \sqrt{\frac{T}{\mu}}.$$

The equation therefore has a direct mechanical interpretation. It is not merely a formal relation between derivatives.

Discrete waves and the long-wavelength limit

The continuum approximation can be checked by solving the discrete model with a wave-like ansatz. Let

$$u_n(t) = Ae^{i(kna-\omega t)}.$$

Then

$$\frac{\mathrm{d}^2 u_n}{\mathrm{d}t^2} = -\omega^2 u_n,$$

and

$$u_{n+1} = e^{ika}u_n, \qquad u_{n-1} = e^{-ika}u_n.$$

Substitution into the discrete equation gives

$$-m\omega^2 u_n = \kappa\left(e^{ika} - 2 + e^{-ika}\right)u_n.$$

Using

$$e^{ika} + e^{-ika} = 2\cos(ka),$$

we find

$$-m\omega^2 = 2\kappa\left(\cos(ka) - 1\right).$$

Since

$$1 - \cos(ka) = 2\sin^2\left(\frac{ka}{2}\right),$$

the exact discrete dispersion relation is

$$\boxed{\omega^2 = \frac{4\kappa}{m}\sin^2\left(\frac{ka}{2}\right)}.$$

For long wavelengths,

$$ka \ll 1,$$

and

$$\sin\left(\frac{ka}{2}\right) \approx \frac{ka}{2}.$$

Therefore

$$\omega^2 \approx \frac{4\kappa}{m}\frac{k^2a^2}{4} = \frac{\kappa a^2}{m}k^2.$$

Because

$$\frac{\kappa a^2}{m} = \frac{T}{\mu} = c^2,$$

we obtain

$$\boxed{\omega^2 \approx c^2k^2}.$$

This is the dispersion relation of the continuum wave equation. The derivation shows exactly when the continuum model is valid: the wavelength must be much larger than the lattice spacing,

$$\lambda \gg a.$$

At shorter wavelengths, the sine function in the discrete dispersion relation cannot be replaced by its linear approximation, and the discrete structure becomes physically relevant.

From discrete energy to field energy

The total energy of the discrete chain is

$$E = \sum_n \left[\frac{m}{2}\dot{u}_n^2 + \frac{\kappa}{2}(u_{n+1} - u_n)^2\right].$$

In the continuum approximation,

$$\dot{u}_n \approx \frac{\partial u}{\partial t},$$

and

$$u_{n+1} - u_n \approx a\frac{\partial u}{\partial x}.$$

Using $m = \mu a$ and $\kappa = T/a$, the energy associated with one lattice interval becomes

$$\frac{m}{2}\dot{u}_n^2 + \frac{\kappa}{2}(u_{n+1} - u_n)^2 \approx a\left[\frac{\mu}{2}\left(\frac{\partial u}{\partial t}\right)^2 + \frac{T}{2}\left(\frac{\partial u}{\partial x}\right)^2\right].$$

The sum therefore becomes a Riemann sum:

$$E \approx \sum_n a\left[\frac{\mu}{2}\left(\frac{\partial u}{\partial t}\right)^2 + \frac{T}{2}\left(\frac{\partial u}{\partial x}\right)^2\right].$$

In the continuum limit,

$$\boxed{E = \int_0^L \left[\frac{\mu}{2}\left(\frac{\partial u}{\partial t}\right)^2 + \frac{T}{2}\left(\frac{\partial u}{\partial x}\right)^2\right] \mathrm{d}x}.$$

The first term is kinetic energy density, and the second is elastic energy density. The energy is no longer assigned to individual masses and springs; it is distributed continuously along the string.

This expression will later help motivate a Lagrangian density. The transition from a mechanical sum to a spatial integral is one of the basic structural steps from particle mechanics to field theory.

What the continuum limit keeps and what it forgets

The continuum field retains the long-wavelength collective motion of the discrete chain. It forgets details that depend on the individual masses and the precise lattice spacing.

The continuum approximation is therefore not obtained by simply replacing an index by a coordinate. Three coordinated replacements occur:

1. the discrete displacement $u_n(t)$ becomes the field $u(x,t)$;
2. finite differences become spatial derivatives;
3. sums of local contributions become spatial integrals.

The parameters must also be rescaled so that finite continuum quantities such as $\mu = m/a$ and $T = \kappa a$ remain meaningful.

What has been established

Starting from Newton's law for a chain of coupled oscillators, we derived the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \qquad c = \sqrt{\frac{T}{\mu}}.$$

The derivation displayed the physical meaning of each term, the dimensional consistency of the result, and the assumptions behind the continuum approximation. The discrete dispersion relation showed that the field description is the long-wavelength limit of the microscopic model.

We also derived the continuum energy as an integral of an energy density. This prepares the later transition from mechanical Lagrangians to Lagrangian densities. The final section of the chapter now compares several physical fields and applies a common set of questions to each of them.

1.7 Basic Physical Examples of Fields

The concept of a field is useful because the same mathematical structure appears in many different physical systems. The symbols and units change, but the underlying questions remain similar:

1. What is the domain of the field?
2. What type of value is attached to each point?
3. Is the field static or time dependent?
4. Is it prescribed, or must it be solved for?
5. What produces it, and how is it measured?

This section applies these questions to several important examples. The purpose is not to develop the complete theory of each field. It is to learn how to recognize the mathematical and physical role played by a field.

Temperature as a scalar field

A temperature field assigns one real number to every point of a material body:

$$T = T(\mathbf{x}, t).$$

Its domain is the body together with the time interval under consideration. Its value is a scalar measured in kelvin.

Consider a thin rod of length L with the model temperature

$$T(x,t) = T_{\text{eq}} + \Delta T\, e^{-t/\tau} \cos\left(\frac{\pi x}{L}\right).$$

Here T_{eq} is an equilibrium temperature, ΔT is the initial temperature variation, and τ is a characteristic decay time.

At the left end,

$$T(0,t) = T_{\text{eq}} + \Delta T e^{-t/\tau}.$$

At the midpoint,

$$T\left(\frac{L}{2}, t\right) = T_{\text{eq}} + \Delta T e^{-t/\tau} \cos\left(\frac{\pi}{2}\right) = T_{\text{eq}}.$$

At the right end,

$$T(L,t) = T_{\text{eq}} - \Delta T e^{-t/\tau}.$$

The formula therefore represents a rod whose left and right ends initially lie above and below equilibrium by equal amounts. The exponential factor causes the variation to decrease with time.

The field can be sampled with thermometers, but a finite set of measurements gives only an approximation to the complete field. A theoretical temperature field interpolates between measurements and predicts values at unmeasured points.

The spatial gradient

$$\nabla T$$

points in the direction of the most rapid temperature increase. In heat conduction, this gradient helps determine the local heat flux. Thus derivatives of the field have direct physical meaning.

Density and velocity fields in a fluid

A fluid requires more than one field. Its mass density is a scalar field,

$$\rho = \rho(\mathbf{x}, t),$$

while its velocity is a vector field,

$$\mathbf{v} = \mathbf{v}(\mathbf{x}, t).$$

The density tells us how much mass is stored locally. If V is a region of space, the total mass inside it is

$$M_V(t) = \int_V ho(\mathbf{x}, t)\, \mathrm{d}^3 x.$$

For a uniform density ρ_0 inside a rectangular box of side lengths a, b, and c,

$$M = \int_0^a \int_0^b \int_0^c \rho_0 \,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x = \rho_0 abc.$$

This calculation shows how a local field quantity produces a global observable by integration.

The velocity field gives the velocity of the fluid element passing through each point. As an example, consider

$$\mathbf{v}(x, y) = \alpha x\,\mathbf{e}_x - \alpha y\,\mathbf{e}_y,$$

where α has units of inverse time.

At $(x, y) = (2\,\mathrm{m}, 1\,\mathrm{m})$,

$$\mathbf{v} = 2\alpha\,\mathrm{m}\,\mathbf{e}_x - \alpha\,\mathrm{m}\,\mathbf{e}_y.$$

The flow stretches material in the x-direction and compresses it in the y-direction. The divergence is

$$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = \alpha - \alpha = 0.$$

The vanishing divergence indicates that local expansion in one direction is balanced by compression in the other. This is an example of how a derivative of a vector field describes a local property of a flow.

A complete fluid model may also contain pressure, temperature, and concentration fields. These fields interact through local differential equations.

Displacement of a string

The transverse displacement of a string is a scalar field on a one-dimensional spatial domain:

$$u = u(x, t), \qquad 0 \le x \le L.$$

Although the physical string lies in space, the coordinate x labels position along its equilibrium length, and u records displacement in a chosen transverse direction.

A travelling wave may be written as

$$u(x, t) = A\cos(kx - \omega t).$$

Its local transverse velocity is

$$\frac{\partial u}{\partial t} = A\omega \sin(kx - \omega t),$$

and its local slope is

$$\frac{\partial u}{\partial x} = -Ak\sin(kx - \omega t).$$

The two quantities are proportional for this right-moving wave:

$$\frac{\partial u}{\partial t} = -\frac{\omega}{k}\frac{\partial u}{\partial x}.$$

If the dispersion relation is $\omega = ck$, then

$$\frac{\partial u}{\partial t} = -c\frac{\partial u}{\partial x}.$$

This relation expresses the rigid translation of the wave profile to the right with speed c. A local time change is linked to the local spatial slope because the same shape is moving through the coordinate system.

The string example also shows that a scalar field need not represent a scalar quantity in all of three-dimensional space. The field value is scalar relative to the reduced one-dimensional model because only one transverse component has been retained.

Electric potential and electric field

Electrostatics uses both a scalar field and a vector field. The electric potential is a scalar field

$$\Phi = \Phi(\mathbf{x}),$$

while the electric field is a vector field

$$\mathbf{E} = \mathbf{E}(\mathbf{x}).$$

They are related by

$$\mathbf{E} = -\nabla\Phi.$$

For a point charge Q at the origin, the electrostatic potential in vacuum is

$$\Phi(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r},$$

where $r = \|\mathbf{x}\|$.
Because the potential depends only on the radial coordinate, its gradient is

$$\nabla\Phi = \frac{\mathrm{d}\Phi}{\mathrm{d}r}\,\widehat{\mathbf{r}}.$$

Differentiating,

$$\frac{\mathrm{d}\Phi}{\mathrm{d}r} = -\frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}.$$

Therefore

$$\boxed{\mathbf{E}(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\widehat{\mathbf{r}}}.$$

For $Q > 0$, the field points radially outward. For $Q < 0$, it points inward.

This calculation illustrates how a vector field can be generated from the spatial variation of a scalar field. The potential describes an energy-like quantity per unit charge, while its gradient determines the local force direction.

A test charge q placed at $\mathbf{x}$ experiences

$$\mathbf{F} = q\mathbf{E}(\mathbf{x}).$$

The response is local: the charge responds to the field evaluated at its own position.

The electromagnetic field

When electric and magnetic effects vary with time, the theory requires the electric vector field $\mathbf{E}(\mathbf{x}, t)$ and magnetic vector field $\mathbf{B}(\mathbf{x}, t)$. These fields are coupled: a changing magnetic field is related to the circulation of the electric field, and a changing electric field contributes to the magnetic field.

In relativistic notation, $\mathbf{E}$ and $\mathbf{B}$ are not independent geometric objects. They are components of one antisymmetric rank-two tensor field,

$$F_{\mu\nu}(x).$$

Different inertial observers may divide the same tensor into electric and magnetic components differently. This is one reason tensor notation is not merely a compressed way of writing equations; it reveals which quantities belong to one relativistic field.

The detailed construction of $F_{\mu\nu}$ from the electromagnetic potential will be carried out in Chapter 7.

Gravitational potential and gravitational field

In Newtonian gravity, a mass distribution produces a scalar potential Φ_g. For a point mass M,

$$\Phi_g(r) = -\frac{GM}{r}.$$

The gravitational acceleration field is

$$\mathbf{g} = -\nabla \Phi_g.$$

Since

$$\frac{\mathrm{d}\Phi_g}{\mathrm{d}r} = \frac{GM}{r^2},$$

we obtain

$$\boxed{\mathbf{g} = -\frac{GM}{r^2}\widehat{\mathbf{r}}}.$$

The minus sign indicates that the field points toward the mass.

The mathematical resemblance to electrostatics is clear. Both theories use a scalar potential whose gradient produces a force field. The physical interpretation differs: electric charges can have either sign, while ordinary mass is positive and Newtonian gravity is attractive.

General relativity replaces the Newtonian gravitational field by a dynamical spacetime metric tensor. The metric determines intervals, motion, and curvature. The final chapter will outline this geometric field description.

Stress as a tensor field

In a continuous material, the force acting across an internal surface depends on the orientation of that surface. A single force vector at each point is therefore insufficient. The stress tensor field

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}(\mathbf{x}, t)$$

assigns a linear map to each point. Given a unit normal $\mathbf{n}$, it produces the traction vector

$$\mathbf{t}(\mathbf{n}) = \boldsymbol{\sigma}\mathbf{n}.$$

The field value is a tensor because it relates one direction, the surface normal, to another vector, the force per unit area. The same tensor describes the traction on every possible surface orientation through the point.

This example demonstrates why tensors arise naturally in continuum physics. They encode directional relationships that cannot be represented by one scalar or one vector.

One system may require several coupled fields

Consider a heated fluid moving through a pipe. A useful model may require

$$\rho(\mathbf{x}, t), \qquad p(\mathbf{x}, t), \qquad T(\mathbf{x}, t), \qquad \mathbf{v}(\mathbf{x}, t).$$

The density ρ, pressure p, and temperature T are scalar fields. The velocity $\mathbf{v}$ is a vector field.

These fields are not separate descriptions of unrelated phenomena. The density and velocity are connected by mass conservation. Pressure gradients influence acceleration. Temperature may change density and pressure. Motion transports heat.

A field theory often consists of a coupled system of equations rather than one equation for one field. The first task is to identify the required fields and the physical principles that connect them.

A practical field-identification procedure

When a new field appears, analyze it in the following order.

1. **Name the field variable.** For example, T, ρ, $\mathbf{v}$, or $F_{\mu\nu}$.
2. **State the domain.** Is it a line, surface, spatial region, or spacetime?
3. **Identify the value type.** Is it scalar, vector, or tensor?
4. **Give the physical units.**
5. **Decide whether it is prescribed or dynamical.**
6. **Identify sources and boundary data.**
7. **State how the field is measured or how it affects matter.**
8. **Identify the relevant derivatives or integrals.**

This procedure prevents a field from becoming an unexplained symbol. It connects the mathematical object with the physical problem it is meant to describe.

Chapter summary

This chapter established the basic language of classical field theory.

A field assigns physical or mathematical information to every point of a domain. Depending on the value assigned, the field may be scalar, vector, or tensor. The classification is determined by transformation behaviour, not merely by the number of written components.

A field may depend on space and time. Its physical problem is not specified by a differential equation alone; the domain, initial data, and boundary conditions are also required. Local field equations relate values and derivatives at the same point, while sources drive the field and local coupling laws determine how matter responds.

Continuous systems contain infinitely many degrees of freedom in the sense that a configuration is a function rather than a finite coordinate vector. This structure can be understood by discretizing space or by expanding the field in infinitely many normal modes.

Most importantly, the wave equation was derived from a chain of coupled oscillators. Finite differences became derivatives, sums became integrals, and microscopic parameters combined into the continuum quantities μ and T. The derivation showed how a local field equation and a field energy emerge from ordinary mechanics.

The next chapter develops the action principle for particles. That may appear to be a temporary return from fields to mechanics, but it provides the variational language needed to derive field equations systematically. Once the mechanical argument is understood in detail, it can be extended from finitely many coordinates to an entire field.

Chapter 2

Action Principles: From Particles to Fields

2.1 Action as a Functional

The equations of mechanics are often introduced through forces. One specifies the forces acting on a particle and then uses Newton's second law to determine its acceleration. The action principle organizes the same dynamics in a different way. Instead of asking for the force at each instant, it assigns one number to an entire possible history of the system and then identifies the physical history through a stationarity condition.

This change of viewpoint is essential for field theory. A field configuration is already a function of space and time, so the natural dynamical object is not an ordinary function of a few variables but a *functional*: an object whose input is itself a function.

From instantaneous states to complete trajectories

Consider a particle moving on a line. Its position is described by a function

$$q = q(t), \qquad t_1 \leq t \leq t_2.$$

At one instant, the value $q(t)$ tells us where the particle is. The complete function q tells us the whole trajectory between the initial time t_1 and the final time t_2.

It is important to distinguish these two kinds of input:

- an ordinary function may take a number as input, for example $V(q)$;
- a functional takes an entire function as input, for example $S[q]$.

The square brackets in $S[q]$ emphasize that the argument is the complete trajectory, not merely the value of the coordinate at one time.

For example, the expression

$$F[q] = \int_{t_1}^{t_2} q(t)\,\mathrm{d}t$$

is a functional. Once a complete function $q(t)$ is supplied, the integral produces one number. If

$$q_1(t) = t, \qquad 0 \leq t \leq 1,$$

then

$$F[q_1] = \int_0^1 t\,\mathrm{d}t = \frac{1}{2}.$$

For a different trajectory,

$$q_2(t) = t^2, \qquad 0 \leq t \leq 1,$$

we obtain

$$F[q_2] = \int_0^1 t^2\,\mathrm{d}t = \frac{1}{3}.$$

The output is one number in each case, but that number depends on the entire shape of the input function over the interval.

The Lagrangian

In ordinary mechanics, the dynamical information is encoded in a function called the Lagrangian. For one generalized coordinate, it is usually written as

$$L = L(q, \dot{q}, t),$$

where

$$\dot{q} = \frac{\mathrm{d}q}{\mathrm{d}t}.$$

For a large class of mechanical systems,

$$L = T - V,$$

where T is the kinetic energy and V is the potential energy.

For a particle of mass m moving in a potential $V(q)$, the Lagrangian is

$$L(q, \dot{q}) = \frac{m}{2}\dot{q}^2 - V(q).$$

The Lagrangian is an ordinary function. At a specified time, it can be evaluated once the instantaneous position and velocity are known. Along a complete trajectory $q(t)$, however, it becomes a function of time:

$$t \longmapsto L(q(t), \dot{q}(t), t).$$

The action is obtained by integrating this quantity over the time interval.

Definition of the action

The action functional is defined by

$$\boxed{S[q] = \int_{t_1}^{t_2} L(q(t), \dot{q}(t), t)\,\mathrm{d}t}.$$

The input of S is a trajectory $q(t)$. The output is a single number. Different candidate trajectories generally produce different values of the action.

The action depends not only on the geometric curve traced by the particle but also on how the trajectory is parametrized in time, because the velocity $\dot{q}(t)$ enters the Lagrangian.

The endpoints and the time interval are part of the variational problem. When we later compare nearby trajectories, we must state whether the initial and final times are fixed and whether the endpoint positions are fixed. These choices determine which variations are allowed.

Dimensions of the action

Because the Lagrangian has the dimensions of energy,

$$[L] = \mathrm{J},$$

the action has dimensions

$$[S] = [L][t] = \mathrm{J\,s}.$$

In base SI units,

$$[S] = \mathrm{kg\,m^2\,s^{-1}}.$$

This dimensional check is useful. Every term in a Lagrangian must have the dimensions of energy, and every contribution to the action must have the dimensions of energy multiplied by time.

The same dimensions are carried by Planck's constant $\hbar$. Classical mechanics does not require $\hbar$, but the comparison becomes important in quantum mechanics, where the dimensionless ratio $S/\hbar$ appears naturally.

Example: a free particle along a prescribed path

For a free particle moving on a line,

$$V(q) = 0,$$

so the Lagrangian is

$$L = \frac{m}{2}\dot{q}^2.$$

Consider the prescribed linear trajectory

$$q(t) = q_1 + \frac{q_2 - q_1}{t_2 - t_1}(t - t_1).$$

Its velocity is constant:

$$\dot{q}(t) = \frac{q_2 - q_1}{t_2 - t_1}.$$

Substitution into the action gives

$$\begin{aligned} S[q] &= \int_{t_1}^{t_2} \frac{m}{2}\left(\frac{q_2 - q_1}{t_2 - t_1}\right)^2 \mathrm{d}t \\ &= \frac{m}{2}\left(\frac{q_2 - q_1}{t_2 - t_1}\right)^2 (t_2 - t_1) \\ &= \boxed{\frac{m(q_2 - q_1)^2}{2(t_2 - t_1)}}. \end{aligned}$$

This calculation does not yet prove that the linear trajectory is physical. It only evaluates the action for one candidate path. To identify the physical trajectory, we must compare it with nearby paths and determine whether the first-order change of the action vanishes.

Example: action of a trial trajectory

To see explicitly that the action depends on the whole path, consider a free particle on the interval

$$0 \le t \le T$$

with fixed endpoints

$$q(0) = 0, \qquad q(T) = a.$$

One admissible path is the linear trajectory

$$q_0(t) = \frac{a}{T}t.$$

A family of alternative paths with the same endpoints is

$$q_\varepsilon(t) = \frac{a}{T}t + \varepsilon \sin\left(\frac{\pi t}{T}\right),$$

because the sine term vanishes at both endpoints. The velocity is

$$\dot{q}_\varepsilon(t) = \frac{a}{T} + \varepsilon\frac{\pi}{T}\cos\left(\frac{\pi t}{T}\right).$$

For the free-particle Lagrangian,

$$S[q_\varepsilon] = \frac{m}{2}\int_0^T \dot{q}_\varepsilon^2(t)\,\mathrm{d}t.$$

Expanding the square,

$$\dot{q}_\varepsilon^2 = \frac{a^2}{T^2} + 2\varepsilon\frac{a\pi}{T^2}\cos\left(\frac{\pi t}{T}\right) + \varepsilon^2\frac{\pi^2}{T^2}\cos^2\left(\frac{\pi t}{T}\right).$$

The term linear in ε integrates to zero because

$$\int_0^T \cos\left(\frac{\pi t}{T}\right)\,\mathrm{d}t = 0.$$

Also,

$$\int_0^T \cos^2\left(\frac{\pi t}{T}\right)\,\mathrm{d}t = \frac{T}{2}.$$

Therefore

$$\boxed{S[q_\varepsilon] = \frac{ma^2}{2T} + \frac{m\pi^2}{4T}\varepsilon^2}.$$

The action changes only at second order in ε near $\varepsilon = 0$. This is the simplest preview of stationarity: the first derivative of the action with respect to the deformation parameter vanishes at the linear path.

Stationary does not always mean minimal

The physical trajectory is characterized by a stationary action, not universally by the smallest possible action. In symbols, the condition is written

$$\delta S = 0.$$

This means that the first-order change of the action vanishes for all allowed infinitesimal variations of the trajectory. Depending on the system and the time interval, the stationary trajectory may correspond to a local minimum, a local maximum, or a saddle point of the action.

The phrase "principle of least action" is historically common, but "principle of stationary action" is more precise. The essential statement is not

$$S[q_{\text{physical}}] < S[q]$$

for every other path. The essential statement is that sufficiently small admissible changes of the physical path produce no first-order change in S.

Local dynamics from a global quantity

The action is an integral over an entire time interval, so it may appear to define dynamics globally. Nevertheless, when the action is stationary under arbitrary local deformations of the trajectory, the result is a differential equation that holds at every time.

This is one of the central structural facts of variational mechanics:

> A condition imposed on the action of a complete trajectory produces a local equation of motion.

The mechanism behind this statement will be developed in the next sections. We will introduce a varied trajectory, calculate the first variation of the action, use integration by parts, control the boundary term, and derive the Euler–Lagrange equation.

Why this formulation is useful for field theory

For a particle, the action is a functional of a trajectory $q(t)$. For a field, the action will be a functional of a field configuration $\phi(x)$ over spacetime:

$$S[\phi] = \int \mathcal{L}(\phi, \partial_\mu \phi, x)\, \mathrm{d}^4 x.$$

The mechanical Lagrangian L is replaced by a Lagrangian density $\mathcal{L}$, and the time integral is replaced by a spacetime integral. The conceptual structure remains the same:

1. choose a candidate history;
2. evaluate its action;
3. compare it with nearby admissible histories;
4. require the first-order change of the action to vanish.

This is why the variational principle is first developed for particles. The particle calculation contains, in a simpler setting, the same ideas that will later be applied to fields.

What has been established

A functional maps an entire function to a number. In mechanics, the action functional assigns a number to every admissible trajectory:

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, \mathrm{d}t.$$

The Lagrangian is evaluated locally along the path, while the action summarizes the complete history over the chosen time interval. The physical path is not selected by evaluating one instantaneous state; it is selected through the stationarity of the action under allowed variations.

The next section constructs those variations explicitly and explains how a one-parameter family of nearby trajectories is used to calculate the first variation of the action.

2.2 Variations of Particle Trajectories

The action principle compares one candidate trajectory with nearby trajectories. To perform this comparison systematically, we introduce a parameter that labels a family of paths. The derivative with respect to that parameter defines the *variation* of the trajectory.

This construction is the central technical step of variational mechanics. Ordinary differentiation measures how a quantity changes when time changes. A variation measures how a complete trajectory changes when we move from one candidate path to another.

A one-parameter family of paths

Let

$$q = q(t)$$

be a reference trajectory on the interval

$$t_1 \le t \le t_2.$$

We construct nearby trajectories by writing

$$\boxed{q_\varepsilon(t) = q(t) + \varepsilon\eta(t)}.$$

Here:

- ε is a real parameter;
- $\eta(t)$ is a chosen deformation function;
- $q_0(t) = q(t)$ is the reference trajectory.

For small ε, the path q_ε lies close to q. Positive and negative values of ε deform the path in opposite directions.

At a fixed time t, the number $\eta(t)$ tells us how strongly the trajectory is displaced at that instant. The entire function η specifies the shape of the deformation over the complete interval.

Definition of the variation

The variation of the trajectory is defined by differentiating the family q_ε with respect to ε and then setting $\varepsilon = 0$:

$$\delta q(t) := \left.\frac{\partial q_\varepsilon(t)}{\partial \varepsilon}\right|_{\varepsilon=0}.$$

For the family

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t),$$

we obtain

$$\boxed{\delta q(t) = \eta(t)}.$$

The symbol δ does not represent a new time derivative. It denotes differentiation in the space of trajectories.

The distinction is therefore

$$\frac{\mathrm{d}q}{\mathrm{d}t} \quad \text{versus} \quad \delta q.$$

The first quantity compares the values of one trajectory at nearby times. The second compares nearby trajectories at the same time.

Variation of the velocity

The velocity along the varied trajectory is

$$\dot{q}_\varepsilon(t) = \frac{\mathrm{d}}{\mathrm{d}t} q_\varepsilon(t).$$

Using

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t),$$

we find

$$\dot{q}_\varepsilon(t) = \dot{q}(t) + \varepsilon\dot{\eta}(t).$$

The variation of the velocity is therefore

$$\begin{aligned} \delta\dot{q}(t) &:= \left.\frac{\partial \dot{q}_\varepsilon(t)}{\partial \varepsilon}\right|_{\varepsilon=0} \\ &= \dot{\eta}(t). \end{aligned}$$

Since $\delta q = \eta$, this may be written as

$$\boxed{\delta\dot{q} = \frac{\mathrm{d}}{\mathrm{d}t}(\delta q)}.$$

Thus the variation and the time derivative commute for the class of smooth variations considered here:

$$\delta\left(\frac{\mathrm{d}q}{\mathrm{d}t}\right) = \frac{\mathrm{d}}{\mathrm{d}t}(\delta q).$$

This identity will be used repeatedly. It is the reason that varying a Lagrangian containing $\dot{q}$ produces a term involving the time derivative of δq.

A geometric picture

A trajectory can be represented as a curve in the (t, q)-plane. The family

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t)$$

is then a family of nearby curves. For each fixed value of t, changing ε moves vertically from one curve to another.

The reference path is the curve corresponding to $\varepsilon = 0$. The function $\eta(t)$ gives the tangent direction of the family in the space of curves.

This geometric picture explains why a variation is not one additional trajectory by itself. It is an infinitesimal direction in which the reference trajectory may be deformed.

Variations with fixed endpoints

In the standard derivation of the Euler–Lagrange equation, the initial and final positions are held fixed:

$$q_\varepsilon(t_1) = q(t_1), \qquad q_\varepsilon(t_2) = q(t_2).$$

Substituting

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t)$$

gives

$$q(t_1) + \varepsilon\eta(t_1) = q(t_1),$$

and

$$q(t_2) + \varepsilon\eta(t_2) = q(t_2).$$

For these relations to hold for all sufficiently small ε, we require

$$\boxed{\eta(t_1) = 0, \qquad \eta(t_2) = 0}.$$

Equivalently,

$$\delta q(t_1) = 0, \qquad \delta q(t_2) = 0.$$

The deformation is otherwise arbitrary in the interior of the interval. It may be concentrated in a small region, oscillatory, or spread over the entire path, provided it is sufficiently smooth and satisfies the endpoint conditions.

The role of these conditions will be examined in detail in the next section. They are not decorative assumptions: they determine the boundary term produced by integration by parts.

Example of an admissible deformation

Suppose that

$$0 \le t \le T$$

and the endpoints are fixed. A simple admissible deformation is

$$\eta(t) = A \sin\left(\frac{\pi t}{T}\right),$$

because

$$\eta(0) = 0, \qquad \eta(T) = 0.$$

The corresponding family of trajectories is

$$q_\varepsilon(t) = q(t) + \varepsilon A \sin\left(\frac{\pi t}{T}\right).$$

The deformation is largest near the middle of the interval and vanishes at the endpoints. Another admissible choice is

$$\eta(t) = At(T - t).$$

Again,

$$\eta(0) = 0, \qquad \eta(T) = 0.$$

These examples show that the allowed deformation is not unique. The stationarity condition must hold for every admissible $\eta(t)$, not merely for one convenient choice.

A localized variation

To understand why a variational principle produces a local differential equation, it is useful to consider a deformation that is nonzero only in a small part of the interval.

Let t_a and t_b satisfy

$$t_1 < t_a < t_b < t_2.$$

We may choose a smooth function $\eta(t)$ such that

$$\eta(t) = 0$$

outside the interval (t_a, t_b). The varied and reference trajectories then agree everywhere except in that small time region.

If the action is stationary under all such localized deformations, the coefficient multiplying $\eta(t)$ in the first variation must vanish at every interior time. This idea will later be formalized by the fundamental lemma of the calculus of variations.

The varied action

The action of the varied path is

$$S[q_\varepsilon] = \int_{t_1}^{t_2} L\big(q_\varepsilon(t), \dot{q}_\varepsilon(t), t\big)\, \mathrm{d}t.$$

Once the deformation function $\eta(t)$ is chosen, this expression is an ordinary function of the parameter ε:

$$\varepsilon \longmapsto S[q_\varepsilon].$$

The first variation of the action is defined by

$$\boxed{\delta S[q;\eta] := \frac{\mathrm{d}}{\mathrm{d}\varepsilon} S[q_\varepsilon]\bigg|_{\varepsilon=0}}.$$

The notation $\delta S[q;\eta]$ makes two dependences visible:

- the variation is evaluated at the reference trajectory q;
- its value depends on the chosen deformation direction η.

When the context is clear, this is abbreviated to δS.

First-order expansion of the Lagrangian

For small ε, the Lagrangian along the varied path may be expanded to first order:

$$\begin{aligned} L(q_\varepsilon, \dot{q}_\varepsilon, t) &= L(q + \varepsilon\eta, \dot{q} + \varepsilon\dot{\eta}, t) \\ &= L(q, \dot{q}, t) + \varepsilon \left(\frac{\partial L}{\partial q}\eta + \frac{\partial L}{\partial \dot{q}}\dot{\eta} \right) + O(\varepsilon^2). \end{aligned}$$

Integrating over time gives

$$S[q_\varepsilon] = S[q] + \varepsilon \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q}\eta + \frac{\partial L}{\partial \dot{q}}\dot{\eta} \right) \mathrm{d}t + O(\varepsilon^2).$$

Therefore

$$\boxed{\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q}\, \delta q + \frac{\partial L}{\partial \dot{q}}\, \delta \dot{q} \right) \mathrm{d}t}.$$

Using $\delta q = \eta$ and $\delta \dot{q} = \dot{\eta}$, we may also write

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \eta + \frac{\partial L}{\partial \dot{q}} \dot{\eta} \right) \mathrm{d}t.$$

At this stage, the derivative $\dot{\eta}$ still appears. To obtain a condition involving the arbitrary function η itself, we will transfer the time derivative from η to $\partial L / \partial \dot{q}$ by integration by parts.

Linearity of the first variation

The first variation is linear in the deformation direction. If

$$\eta(t) = a\eta_1(t) + b\eta_2(t),$$

then

$$\delta S[q; a\eta_1 + b\eta_2] = a\, \delta S[q; \eta_1] + b\, \delta S[q; \eta_2].$$

This follows directly from the first-order expression for δS. Linearity is important because the first variation is the functional analogue of a directional derivative.

The second-order and higher-order changes of the action are generally not linear in η. They become relevant when one asks whether a stationary path is a minimum, maximum, or saddle point.

Several generalized coordinates

For a system described by coordinates

$$q^1(t), q^2(t), \ldots, q^N(t),$$

we introduce one variation for each coordinate:

$$q^i_\varepsilon(t) = q^i(t) + \varepsilon \eta^i(t).$$

Then

$$\delta q^i = \eta^i, \qquad \delta \dot{q}^i = \dot{\eta}^i.$$

The first variation becomes

$$\delta S = \int_{t_1}^{t_2} \sum_{i=1}^{N} \left(\frac{\partial L}{\partial q^i} \eta^i + \frac{\partial L}{\partial \dot{q}^i} \dot{\eta}^i \right) \mathrm{d}t.$$

Each deformation function $\eta^i(t)$ may be chosen independently when the coordinates are independent. This independence will later produce one Euler–Lagrange equation for each generalized coordinate.

Common mistakes

Several distinctions must be kept clear.

1. The variation δq is not the same as a small time increment $\mathrm{d}q$. It compares different paths at the same time.

2. The parameter ε is not physical time. It labels trajectories in a family.

3. The function $\eta(t)$ is not chosen after the calculation to force a desired result. Stationarity must hold for every admissible η.
4. If the endpoints are fixed, the variation must vanish there. This does not imply that $\dot{\eta}$ also vanishes at the endpoints.
5. The equality $\delta\dot{q} = \mathrm{d}(\delta q)/\mathrm{d}t$ must be used explicitly when varying a velocity-dependent Lagrangian.

These points prevent the variational calculation from becoming a formal manipulation without a clear meaning.

Preparation for field variations

The same construction extends directly to a field. A reference field $\phi(x)$ is replaced by a family

$$\phi_\varepsilon(x) = \phi(x) + \varepsilon\eta(x),$$

and the field variation is

$$\delta\phi(x) = \frac{\partial\phi_\varepsilon(x)}{\partial\varepsilon}\bigg|_{\varepsilon=0} = \eta(x).$$

Derivatives of the field vary according to

$$\delta(\partial_\mu\phi) = \partial_\mu(\delta\phi).$$

Thus the particle calculation is not an unrelated preliminary exercise. It is the one-dimensional version of the field-theoretic variation that will be used throughout the course.

What has been established

A variation is defined by embedding a reference trajectory in a one-parameter family

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t).$$

The deformation function is

$$\delta q(t) = \eta(t),$$

and its velocity variation is

$$\delta\dot{q}(t) = \dot{\eta}(t).$$

The first variation of the action is the derivative of $S[q_\varepsilon]$ with respect to the path parameter at $\varepsilon = 0$. It is linear in the deformation direction and, before integration by parts, contains both η and $\dot{\eta}$.

The next section examines the endpoint and boundary conditions that determine which variations are admissible and what happens to the boundary term in the variational calculation.

2.3 Boundary Conditions in Variational Problems

A variational problem is not defined by the action functional alone. One must also specify which trajectories are allowed to compete. The boundary data determine the admissible class of paths and therefore determine which variations may be used.

This point is essential because the derivation of the equations of motion produces a boundary term. Whether that term vanishes, survives, or imposes an additional condition depends entirely on the boundary assumptions.

What is held fixed?

Consider the action

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, \mathrm{d}t.$$

Before varying it, we must decide which quantities are fixed. The most common mechanical problem fixes

$$t_1, \qquad t_2, \qquad q(t_1), \qquad q(t_2).$$

Thus every admissible trajectory begins at the same initial event and ends at the same final event.

If

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t),$$

then fixed endpoint positions require

$$q_\varepsilon(t_1) = q(t_1), \qquad q_\varepsilon(t_2) = q(t_2).$$

Therefore

$$\boxed{\eta(t_1) = 0, \qquad \eta(t_2) = 0}.$$

Equivalently,

$$\delta q(t_1) = 0, \qquad \delta q(t_2) = 0.$$

These conditions apply only at the endpoints. Inside the interval, the deformation remains arbitrary.

The admissible class of trajectories

The action is evaluated on a collection of functions rather than on all imaginable curves. A typical admissible class consists of trajectories that

- are continuous on $[t_1, t_2]$;
- possess the derivatives required by the Lagrangian;
- satisfy the prescribed endpoint data;
- obey any additional constraints of the mechanical system.

For a Lagrangian depending on q and $\dot{q}$, one usually assumes at least that the trajectory is piecewise continuously differentiable. To derive the classical Euler–Lagrange equation pointwise, greater smoothness is normally assumed.

The exact function space matters in rigorous treatments. In an introductory derivation, the main requirement is that all differentiations and integrations by parts are legitimate.

Admissible and inadmissible deformations

Suppose the interval is

$$0 \leq t \leq T$$

and the endpoints are fixed. The function

$$\eta_1(t) = A \sin\left(\frac{\pi t}{T}\right)$$

is admissible because

$$\eta_1(0) = 0, \qquad \eta_1(T) = 0.$$

Similarly,

$$\eta_2(t) = At(T - t)$$

is admissible.

By contrast,

$$\eta_3(t) = A$$

is not admissible when the endpoint positions are fixed, because it shifts both endpoints.

The function

$$\eta_4(t) = At$$

also fails, since

$$\eta_4(T) = AT \neq 0$$

unless $A = 0$.

The stationarity condition must hold for every admissible deformation, not for deformations that violate the stated boundary data.

Fixed endpoint values do not fix endpoint slopes

From

$$\eta(t_1) = 0, \qquad \eta(t_2) = 0,$$

it does not follow that

$$\dot{\eta}(t_1) = 0, \qquad \dot{\eta}(t_2) = 0.$$

For example,

$$\eta(t) = A \sin\left(\frac{\pi(t - t_1)}{t_2 - t_1}\right)$$

vanishes at both endpoints, but its derivative is generally nonzero there.

This distinction is important. A first-order Lagrangian $L(q, \dot{q}, t)$ produces a boundary term proportional to δq, so fixing q at the endpoints is sufficient. If the Lagrangian depended on higher derivatives, additional endpoint data could be required.

How the boundary term appears

The first variation of the action is

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \eta + \frac{\partial L}{\partial \dot{q}} \dot{\eta} \right) \mathrm{d}t.$$

The second term contains $\dot{\eta}$. Integration by parts gives

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}} \dot{\eta} \, \mathrm{d}t = \left[\frac{\partial L}{\partial \dot{q}} \eta \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) \eta \, \mathrm{d}t.$$

Therefore

$$\boxed{\delta S = \left[\frac{\partial L}{\partial \dot{q}} \eta \right]_{t_1}^{t_2} + \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \eta \, \mathrm{d}t}.$$

The first term is the boundary contribution. It is evaluated only at t_1 and t_2.

Why the boundary term vanishes for fixed endpoints

When the endpoint positions are fixed,

$$\eta(t_1) = 0, \qquad \eta(t_2) = 0.$$

Hence

$$\left[\frac{\partial L}{\partial \dot{q}} \eta \right]_{t_1}^{t_2} = 0.$$

The first variation then reduces to an integral over the interior:

$$\delta S = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \eta \, \mathrm{d}t.$$

The boundary term has not disappeared by algebraic coincidence. It vanishes because the class of admissible paths was chosen to have fixed endpoint positions.

A correct derivation should always state this reason explicitly.

What happens if an endpoint is free?

Suppose the final position is not fixed. Then $\eta(t_2)$ need not vanish. If the initial position remains fixed, we still have

$$\eta(t_1) = 0,$$

but the boundary contribution becomes

$$\left[\frac{\partial L}{\partial \dot{q}} \eta \right]_{t_1}^{t_2} = \left. \frac{\partial L}{\partial \dot{q}} \right|_{t_2} \eta(t_2).$$

For the action to be stationary for arbitrary endpoint displacement $\eta(t_2)$, one must require

$$\boxed{\left. \frac{\partial L}{\partial \dot{q}} \right|_{t_2} = 0}.$$

This is an example of a *natural boundary condition*. It is not imposed before the variation. It follows from stationarity when the corresponding endpoint value is free.

If both endpoint positions are free, analogous conditions arise at both ends.

Canonical momentum at the boundary

The quantity

$$p := \frac{\partial L}{\partial \dot{q}}$$

is the canonical momentum conjugate to q. The boundary contribution may therefore be written as

$$[p\,\delta q]_{t_1}^{t_2}.$$

This form reveals the pairing between endpoint position variations and canonical momentum. For the standard Lagrangian

$$L = \frac{m}{2}\dot{q}^2 - V(q),$$

we have

$$p = m\dot{q}.$$

Thus a free endpoint can produce a condition on the endpoint momentum.

This structure becomes even more important in field theory, where boundary terms pair field variations with normal derivatives or canonical field momenta on the boundary of a spacetime region.

Fixed times and variable times

The standard derivation assumes that t_1 and t_2 are fixed. If an endpoint time is also allowed to vary, the variation of the action contains additional terms because both the path value and the integration limit change.

Such problems lead to transversality conditions and require a more careful treatment. They are not needed for the basic derivation developed here, but they illustrate a general rule:

> Every quantity allowed to vary can contribute its own boundary term and its own boundary condition.

In the remainder of this chapter, unless stated otherwise, the endpoint times and endpoint positions are fixed.

Boundary conditions and uniqueness

The variational principle produces differential equations in the interior, but differential equations alone do not usually determine a unique trajectory. Boundary conditions or initial conditions are also required.

For a second-order equation of motion, two independent pieces of data are normally needed. One may specify

$$q(t_1) \quad \text{and} \quad q(t_2),$$

as in a fixed-endpoint variational problem, or one may specify

$$q(t_1) \quad \text{and} \quad \dot{q}(t_1),$$

as in an initial-value problem.

These are different ways of selecting a particular solution of the same differential equation. The variational derivation commonly uses endpoint data because they make the boundary term transparent.

Boundary data in field theory

For a field $\phi(x)$ defined on a spacetime region Ω, the particle endpoints are replaced by the boundary $\partial\Omega$. A standard field variation satisfies

$$\delta\phi = 0 \qquad \text{on } \partial\Omega.$$

Spacetime integration by parts then produces a surface term over $\partial\Omega$. The condition $\delta\phi = 0$ makes that term vanish, just as $\delta q(t_1) = \delta q(t_2) = 0$ removes the endpoint term in mechanics.

Other field boundary conditions are possible. If the field value is not fixed, stationarity may produce a natural boundary condition involving derivatives of the field. The particle problem is therefore the simplest model of the boundary analysis required later.

Diagnostic questions

Before discarding a boundary term, one should ask:

1. What are the boundaries of the integration region?
2. Which endpoint or boundary values are fixed?
3. Which variations are therefore required to vanish?
4. Are the endpoint times fixed or variable?
5. Does the action contain higher derivatives that require additional boundary data?
6. If a boundary value is free, what natural boundary condition follows?

These questions prevent one of the most common errors in variational calculations: setting a boundary term to zero without stating why.

What has been established

Boundary conditions are part of the definition of a variational problem. For fixed endpoint positions, admissible variations satisfy

$$\delta q(t_1) = 0, \qquad \delta q(t_2) = 0.$$

These conditions eliminate the endpoint contribution

$$\left[\frac{\partial L}{\partial \dot{q}}\delta q\right]_{t_1}^{t_2}.$$

If an endpoint is free, the boundary term does not vanish automatically and stationarity produces a natural boundary condition. The next section uses the fixed-endpoint problem to derive the Euler–Lagrange equation in full.

2.4 Euler–Lagrange Equations in Mechanics

We now derive the differential equation satisfied by a stationary trajectory. The derivation begins with the action, varies the complete path, isolates the boundary contribution, and then uses the arbitrariness of the interior deformation.

The result is the Euler–Lagrange equation. It is the basic equation of motion in Lagrangian mechanics and the direct prototype of the field equations derived later.

The variational problem

Consider one generalized coordinate $q(t)$ with action

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, \mathrm{d}t.$$

We compare the reference trajectory with the family

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t),$$

where the endpoint positions are fixed:

$$\eta(t_1) = 0, \qquad \eta(t_2) = 0.$$

The stationary-action condition is

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} S[q_\varepsilon]\bigg|_{\varepsilon=0} = 0$$

for every admissible deformation $\eta(t)$.

Differentiating the action

The varied action is

$$S[q_\varepsilon] = \int_{t_1}^{t_2} L(q_\varepsilon, \dot{q}_\varepsilon, t)\, \mathrm{d}t.$$

Differentiating with respect to ε gives

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} S[q_\varepsilon] = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \frac{\partial q_\varepsilon}{\partial \varepsilon} + \frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}_\varepsilon}{\partial \varepsilon} \right) \mathrm{d}t.$$

At $\varepsilon = 0$,

$$\frac{\partial q_\varepsilon}{\partial \varepsilon}\bigg|_{\varepsilon=0} = \eta,$$

and

$$\frac{\partial \dot{q}_\varepsilon}{\partial \varepsilon}\bigg|_{\varepsilon=0} = \dot{\eta}.$$

Therefore the first variation is

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \eta + \frac{\partial L}{\partial \dot{q}} \dot{\eta} \right) \mathrm{d}t.$$

All derivatives of L are evaluated along the reference trajectory $q(t)$.

Integration by parts

The term containing $\dot{\eta}$ must be rewritten so that the arbitrary function appears without a derivative. Integration by parts gives

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}} \dot{\eta}\, \mathrm{d}t = \left[\frac{\partial L}{\partial \dot{q}} \eta \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) \eta\, \mathrm{d}t.$$

Substituting this result into δS,

$$\delta S = \left[\frac{\partial L}{\partial \dot{q}} \eta \right]_{t_1}^{t_2} + \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \eta\, \mathrm{d}t.$$

Because the endpoint positions are fixed,

$$\eta(t_1) = \eta(t_2) = 0,$$

so the boundary term vanishes. Hence

$$\delta S = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) \right] \eta(t)\,\mathrm{d}t.$$

The fundamental lemma

We now use a basic result from the calculus of variations.

If a continuous function $f(t)$ satisfies

$$\int_{t_1}^{t_2} f(t)\eta(t)\,\mathrm{d}t = 0$$

for every sufficiently smooth function $\eta(t)$ that vanishes at the endpoints, then

$$f(t) = 0$$

throughout the interior of the interval.

The reason is that η can be chosen to be localized in any small region. If f were positive or negative in some neighbourhood, one could choose η with the same sign there and obtain a nonzero integral.

Applying the lemma to the first variation yields

$$\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) = 0.$$

Equivalently,

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0}.$$

This is the Euler–Lagrange equation for one generalized coordinate.

Canonical momentum and generalized force

Define the canonical momentum conjugate to q by

$$p := \frac{\partial L}{\partial \dot{q}}.$$

Then the Euler–Lagrange equation becomes

$$\dot{p} = \frac{\partial L}{\partial q}.$$

The quantity $\partial L/\partial q$ plays the role of a generalized force. For standard Cartesian coordinates with

$$L = \frac{m}{2}\dot{q}^2 - V(q),$$

we have

$$p = m\dot{q}$$

and

$$\frac{\partial L}{\partial q} = -\frac{\mathrm{d}V}{\mathrm{d}q}.$$

Thus

$$m\ddot{q} = -\frac{\mathrm{d}V}{\mathrm{d}q},$$

which is Newton's second law with force

$$F(q) = -\frac{\mathrm{d}V}{\mathrm{d}q}.$$

The variational principle therefore reproduces the familiar mechanical equation.

Several generalized coordinates

For a system with coordinates

$$q^1(t), \dots, q^N(t),$$

the action is

$$S[\mathbf{q}] = \int_{t_1}^{t_2} L(q^1, \dots, q^N, \dot{q}^1, \dots, \dot{q}^N, t)\,\mathrm{d}t.$$

We vary each coordinate independently:

$$q^i_\varepsilon(t) = q^i(t) + \varepsilon\eta^i(t).$$

The first variation is

$$\delta S = \int_{t_1}^{t_2} \sum_{i=1}^{N} \left(\frac{\partial L}{\partial q^i}\eta^i + \frac{\partial L}{\partial \dot{q}^i}\dot{\eta}^i \right) \mathrm{d}t.$$

After integration by parts and use of the endpoint conditions,

$$\delta S = \int_{t_1}^{t_2} \sum_{i=1}^{N} \left[\frac{\partial L}{\partial q^i} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^i} \right) \right] \eta^i\,\mathrm{d}t.$$

Since the functions η^i are independent, each coefficient must vanish:

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^i} \right) - \frac{\partial L}{\partial q^i} = 0, \qquad i = 1, \dots, N}.$$

Thus one generalized coordinate produces one Euler–Lagrange equation.

Example: particle in one dimension

Let

$$L(q, \dot{q}) = \frac{m}{2}\dot{q}^2 - V(q).$$

We calculate

$$\frac{\partial L}{\partial \dot{q}} = m\dot{q},$$

so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) = m\ddot{q}.$$

Also,

$$\frac{\partial L}{\partial q} = -\frac{\mathrm{d}V}{\mathrm{d}q}.$$

The Euler–Lagrange equation is therefore

$$m\ddot{q} + \frac{\mathrm{d}V}{\mathrm{d}q} = 0.$$

For the harmonic potential

$$V(q) = \frac{k}{2}q^2,$$

we obtain

$$m\ddot{q} + kq = 0.$$

For the linear potential

$$V(q) = mgq,$$

we obtain

$$m\ddot{q} + mg = 0,$$

or

$$\ddot{q} = -g.$$

Cyclic coordinates

If the Lagrangian does not depend explicitly on a coordinate q^j, then

$$\frac{\partial L}{\partial q^j} = 0.$$

The corresponding Euler–Lagrange equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^j}\right) = 0.$$

Therefore the conjugate momentum is conserved:

$$\boxed{p_j = \frac{\partial L}{\partial \dot{q}^j} = \text{constant}}.$$

Such a coordinate is called cyclic or ignorable. This is an early example of the relation between absence of coordinate dependence and a conservation law. The general connection between continuous symmetries and conserved quantities will later be developed through Noether's theorem.

Explicit time dependence

The Euler–Lagrange equation remains valid when the Lagrangian depends explicitly on time:

$$L = L(q, \dot{q}, t).$$

The partial derivatives $\partial L/\partial q$ and $\partial L/\partial \dot{q}$ are taken while holding the other arguments fixed. The total time derivative

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right)$$

then includes all time dependence along the trajectory.

If L has no explicit time dependence, an energy-like quantity is conserved. That result will be established later using symmetry arguments.

A practical derivation algorithm

Given a mechanical Lagrangian, the equations of motion may be obtained as follows:

1. identify the generalized coordinates q^i;
2. write the Lagrangian $L(q^i, \dot{q}^i, t)$;
3. compute $\partial L/\partial q^i$;
4. compute $\partial L/\partial \dot{q}^i$;
5. take the total time derivative of $\partial L/\partial \dot{q}^i$;
6. substitute into the Euler–Lagrange equation;
7. simplify and check dimensions, signs, and limiting cases.

One should not replace the total derivative in step 5 by a partial derivative. The quantity $\partial L/\partial \dot{q}^i$ is evaluated along the trajectory and may depend on coordinates, velocities, and time.

Verification of a proposed solution

After deriving an equation of motion, a proposed trajectory must be checked by substitution.

For the free particle,

$$L = \frac{m}{2}\dot{q}^2,$$

and the Euler–Lagrange equation gives

$$\ddot{q} = 0.$$

The general solution is

$$q(t) = A + Bt.$$

Indeed,

$$\dot{q} = B, \qquad \ddot{q} = 0.$$

If endpoint values are specified, the constants A and B are determined by those data.

This final check is part of the calculation. Deriving the differential equation and solving it are distinct steps.

Common errors

The most frequent mistakes in the derivation are:

1. omitting the variation of $\dot{q}$;
2. writing $\delta\dot{q} = \delta q$ instead of $\delta\dot{q} = \mathrm{d}(\delta q)/\mathrm{d}t$;
3. discarding the boundary term without stating the endpoint conditions;
4. treating $\eta(t)$ as one special function rather than an arbitrary admissible deformation;
5. confusing the total derivative $\mathrm{d}/\mathrm{d}t$ with a partial derivative;
6. reversing the sign of $\partial L/\partial q$.

A reliable derivation should make each of these operations visible.

Preparation for field equations

The particle equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0$$

becomes, in field theory,

$$\partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0.$$

The time derivative is replaced by a spacetime divergence, the trajectory is replaced by a field, and the Lagrangian is replaced by a Lagrangian density. The logical structure of the derivation remains unchanged.

What has been established

Stationarity of the action under all smooth fixed-endpoint variations implies the Euler–Lagrange equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0.$$

The derivation depends on four explicit ingredients: the varied path, the commutation of variation with the time derivative, integration by parts, and the vanishing of the endpoint variation. For several independent coordinates, one equation is obtained for each coordinate.

The next section explains why generalized coordinates are useful and how constraints can be incorporated into the choice of variables.

2.5 Generalized Coordinates and Constraints

The coordinates used to describe a mechanical system need not be Cartesian positions. They should be chosen to represent the independent degrees of freedom as directly as possible. Coordinates adapted to the geometry and constraints of a system often make the Lagrangian formulation much simpler than a force-based Cartesian description.

A generalized coordinate may be a distance, an angle, or any parameter that uniquely labels the allowed configurations within the region under consideration.

Degrees of freedom

The number of degrees of freedom is the number of independent parameters required to specify a configuration.

A free particle in three-dimensional space has three degrees of freedom, which may be represented by

$$x, \qquad y, \qquad z.$$

Two free particles have six Cartesian coordinates. If constraints relate some of these coordinates, the number of independent degrees of freedom is reduced.

For example, a particle constrained to move on a sphere of radius R satisfies

$$x^2 + y^2 + z^2 = R^2.$$

Although three Cartesian coordinates appear, they are connected by one independent constraint. The particle therefore has two degrees of freedom.

A convenient choice is

$$q^1 = \theta, \qquad q^2 = \varphi,$$

with

$$\begin{aligned} x &= R \sin\theta \cos\varphi, \\ y &= R \sin\theta \sin\varphi, \\ z &= R \cos\theta. \end{aligned}$$

The constraint is then built directly into the coordinate representation.

Generalized coordinates

A set of generalized coordinates is denoted by

$$q^1, q^2, \ldots, q^N.$$

The Cartesian positions of all particles are expressed as functions of these coordinates and possibly time:

$$\mathbf{r}_a = \mathbf{r}_a(q^1, \ldots, q^N, t).$$

Here the index a labels particles. The explicit time dependence allows for moving constraints, such as a bead constrained to a wire whose shape changes with time.

The generalized coordinates need not have the dimensions of length. An angular coordinate is dimensionless in SI units, while other coordinates may have dimensions adapted to the problem.

The Euler–Lagrange equations retain the same form for every generalized coordinate:

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^i}\right) - \frac{\partial L}{\partial q^i} = 0}.$$

This coordinate flexibility is one of the principal advantages of Lagrangian mechanics.

Velocities in generalized coordinates

If

$$\mathbf{r}_a = \mathbf{r}_a(q^1, \ldots, q^N, t),$$

then the particle velocity is obtained by the chain rule:

$$\boxed{\dot{\mathbf{r}}_a = \sum_{i=1}^{N} \frac{\partial \mathbf{r}_a}{\partial q^i} \dot{q}^i + \frac{\partial \mathbf{r}_a}{\partial t}}.$$

For a time-independent coordinate transformation,

$$\frac{\partial \mathbf{r}_a}{\partial t} = 0,$$

so

$$\dot{\mathbf{r}}_a = \sum_{i=1}^{N} \frac{\partial \mathbf{r}_a}{\partial q^i} \dot{q}^i.$$

The kinetic energy can then be expressed in terms of the generalized coordinates and velocities.

For particles with masses m_a,

$$T = \frac{1}{2} \sum_a m_a \dot{\mathbf{r}}_a^2.$$

After substitution, the kinetic energy commonly takes the form

$$T = \frac{1}{2} \sum_{i,j} g_{ij}(q) \dot{q}^i \dot{q}^j,$$

where the coefficients $g_{ij}(q)$ depend on the chosen coordinates. In geometric language, these coefficients form the metric on configuration space.

Example: a particle constrained to a circle

Consider a particle of mass m constrained to a circle of radius R in the plane. Cartesian coordinates satisfy

$$x^2 + y^2 = R^2.$$

Using Cartesian coordinates would require carrying this constraint throughout the calculation. Instead, introduce the angular coordinate θ:

$$x = R\cos\theta, \qquad y = R\sin\theta.$$

The velocity components are

$$\dot{x} = -R\sin\theta\,\dot{\theta}, \qquad \dot{y} = R\cos\theta\,\dot{\theta}.$$

Therefore

$$\begin{aligned}\dot{x}^2 + \dot{y}^2 &= R^2\left(\sin^2\theta + \cos^2\theta\right)\dot{\theta}^2 \\ &= R^2\dot{\theta}^2.\end{aligned}$$

The kinetic energy is

$$T = \frac{mR^2}{2}\dot{\theta}^2.$$

If there is no potential, the Lagrangian is

$$L = \frac{mR^2}{2}\dot{\theta}^2.$$

The Euler–Lagrange equation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(mR^2\dot{\theta}\right) = 0,$$

so

$$\ddot{\theta} = 0.$$

Uniform angular motion follows without introducing the radial constraint force. The normal force that keeps the particle on the circle does not appear explicitly because the coordinate θ already describes only allowed motion.

Holonomic constraints

A holonomic constraint can be written as a relation among coordinates and time:

$$f_\alpha(q^1, \dots, q^M, t) = 0, \qquad \alpha = 1, \dots, r.$$

If the r constraints are independent, they reduce the number of degrees of freedom from M to

$$N = M - r.$$

Examples include:

- a particle on a circle, $x^2 + y^2 - R^2 = 0$;
- a particle on a sphere, $x^2 + y^2 + z^2 - R^2 = 0$;
- two particles connected by a rigid rod, $\|\mathbf{r}_1 - \mathbf{r}_2\| - \ell = 0$.

When possible, the most direct method is to solve the constraints by introducing independent generalized coordinates.

Example: the simple pendulum

A point mass m attached to a massless rigid rod of length ℓ moves in a vertical plane. Its Cartesian coordinates may be written as

$$x = \ell \sin\theta, \qquad y = -\ell\cos\theta.$$

The rigid-length constraint

$$x^2 + y^2 = \ell^2$$

is automatically satisfied.
The velocity components are

$$\dot{x} = \ell\cos\theta\,\dot{\theta}, \qquad \dot{y} = \ell\sin\theta\,\dot{\theta}.$$

Thus

$$T = \frac{m}{2}\left(\dot{x}^2 + \dot{y}^2\right) = \frac{m\ell^2}{2}\dot{\theta}^2.$$

Taking the zero of potential energy at the lowest point,

$$V = mg\ell(1 - \cos\theta).$$

The Lagrangian is

$$L = \frac{m\ell^2}{2}\dot{\theta}^2 - mg\ell(1 - \cos\theta).$$

We calculate

$$\frac{\partial L}{\partial \dot{\theta}} = m\ell^2\dot{\theta},$$

and

$$\frac{\partial L}{\partial \theta} = -mg\ell\sin\theta.$$

The Euler–Lagrange equation gives

$$m\ell^2\ddot{\theta} + mg\ell\sin\theta = 0.$$

Dividing by $m\ell^2$,

$$\boxed{\ddot{\theta} + \frac{g}{\ell}\sin\theta = 0}.$$

The tension in the rod does not appear in this equation. It enforces the constraint but performs no virtual work along an allowed angular displacement.

Constraint forces and virtual displacements

A virtual displacement is an infinitesimal change of configuration consistent with the constraints at a fixed time. In generalized coordinates,

$$\delta\mathbf{r}_a = \sum_i \frac{\partial \mathbf{r}_a}{\partial q^i}\delta q^i.$$

For ideal holonomic constraints, the constraint forces do no virtual work:

$$\sum_a \mathbf{F}_a^{\text{constraint}} \cdot \delta\mathbf{r}_a = 0.$$

This is why such forces can disappear from the equations when independent generalized coordinates are used. The coordinates describe motion tangent to the constraint surface, while the constraint forces act in directions that maintain the restriction.

This statement must be applied with care. Friction and other non-ideal forces may perform work along allowed displacements and cannot be removed merely by changing coordinates.

Constraints imposed with Lagrange multipliers

Sometimes solving the constraints explicitly is inconvenient. One may then retain redundant coordinates and introduce Lagrange multipliers.

Suppose the coordinates q^a satisfy

$$f_\alpha(q,t) = 0.$$

Define the augmented Lagrangian

$$L_{\mathrm{aug}} = L(q,\dot{q},t) + \sum_\alpha \lambda_\alpha(t) f_\alpha(q,t).$$

Variation with respect to λ_α gives

$$f_\alpha(q,t) = 0,$$

while variation with respect to q^a gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^a}\right) - \frac{\partial L}{\partial q^a} = \sum_\alpha \lambda_\alpha \frac{\partial f_\alpha}{\partial q^a}.$$

The multiplier terms represent generalized constraint forces.

Example: circle constraint with a multiplier

For a free particle in the plane constrained by

$$f(x,y) = x^2 + y^2 - R^2 = 0,$$

take

$$L_{\mathrm{aug}} = \frac{m}{2}\left(\dot{x}^2 + \dot{y}^2\right) + \lambda(t)\left(x^2 + y^2 - R^2\right).$$

The equations for x and y are

$$m\ddot{x} - 2\lambda x = 0,$$

and

$$m\ddot{y} - 2\lambda y = 0.$$

Variation with respect to λ restores the circle constraint.

The vector form is

$$m\ddot{\mathbf{r}} = 2\lambda \mathbf{r}.$$

The right-hand side is radial, as expected for the force that keeps the particle on the circle. The multiplier determines the magnitude of that constraint force.

Time-dependent constraints

A holonomic constraint may depend explicitly on time:

$$f(q,t) = 0.$$

An example is a bead constrained to a moving wire. Coordinates adapted to such a constraint can produce a Lagrangian with explicit time dependence.

In that case, energy need not be conserved because the moving constraint can transfer energy to or from the system. The Euler–Lagrange equations remain valid, but the physical interpretation of conserved quantities must account for the external time dependence.

Non-holonomic constraints

Not every constraint can be reduced to an equation involving coordinates alone. A constraint may involve velocities:

$$g_\alpha(q, \dot{q}, t) = 0.$$

Such constraints are called non-holonomic when they cannot be integrated into relations of the form $f_\alpha(q, t) = 0$.

Rolling without slipping is a standard example. The treatment of non-holonomic systems requires care because simply substituting the velocity constraint into the Lagrangian may produce incorrect equations.

The present course focuses mainly on holonomic constraints, which provide the cleanest preparation for field-theoretic variables and gauge redundancies.

Coordinate singularities and coordinate ranges

Generalized coordinates may fail to be valid globally. Spherical coordinates, for example, are singular at the poles, and an angular coordinate is periodic:

$$\varphi \sim \varphi + 2\pi.$$

A coordinate description must therefore include its range and any identifications. A singularity of the coordinate system need not represent a physical singularity of the mechanical system.

This distinction reappears in field theory, where different coordinate systems or gauge choices may describe the same physical configuration.

A practical coordinate-selection procedure

For a constrained mechanical system:

1. list the Cartesian variables;
2. identify the independent constraints;
3. count the remaining degrees of freedom;
4. choose generalized coordinates adapted to the allowed configurations;
5. express positions and velocities in those coordinates;
6. compute T and V;
7. form $L = T - V$;
8. apply one Euler–Lagrange equation to each independent coordinate.

If the constraints cannot be solved conveniently, introduce Lagrange multipliers and vary the augmented action.

Connection with field variables

A field may be regarded as a system with one or more generalized coordinates at every point of space. The field components are the dynamical variables, and relations among them may play the role of constraints or redundancies.

The lesson from mechanics is that the choice of variables matters. Variables adapted to the structure of the theory can remove unnecessary equations and expose the true independent degrees of freedom.

In gauge theories, however, the variables often contain deliberate redundancy rather than an ordinary mechanical constraint. Distinguishing physical degrees of freedom from descriptive redundancy will become a central issue later.

What has been established

Generalized coordinates parameterize the independent configurations of a mechanical system. Holonomic constraints reduce the number of degrees of freedom and can often be incorporated directly into the coordinates. When this is done, ideal constraint forces need not appear explicitly in the Euler–Lagrange equations.

Alternatively, constraints may be enforced with Lagrange multipliers, whose equations restore the constraints and whose contributions represent constraint forces.

The next section examines integration by parts and boundary terms as a general mathematical mechanism, preparing the transition from one-dimensional mechanical actions to spacetime field actions.

2.6 Integration by Parts and Boundary Terms

Integration by parts is the operation that converts the variation of a derivative into the derivative of a coefficient. In variational mechanics, this step separates the first variation into two distinct pieces:

- an interior term that produces the differential equation of motion;
- a boundary term that records what happens at the endpoints.

The same structure reappears in every later field-theoretic derivation. Understanding it in one time dimension is therefore more important than memorizing the final Euler–Lagrange formula.

The ordinary integration-by-parts identity

For sufficiently differentiable functions $f(t)$ and $g(t)$,

$$\int_{t_1}^{t_2} f(t)\dot{g}(t)\,\mathrm{d}t = [f(t)g(t)]_{t_1}^{t_2} - \int_{t_1}^{t_2} \dot{f}(t)g(t)\,\mathrm{d}t.$$

This follows from the product rule

$$\frac{\mathrm{d}}{\mathrm{d}t}(fg) = \dot{f}g + f\dot{g}.$$

Rearranging,

$$f\dot{g} = \frac{\mathrm{d}}{\mathrm{d}t}(fg) - \dot{f}g,$$

and integrating over the interval gives the identity.

The notation

$$[fg]_{t_1}^{t_2} = f(t_2)g(t_2) - f(t_1)g(t_1)$$

makes the endpoint contribution explicit.

Application to the first variation

For the action

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, \mathrm{d}t,$$

the first variation is

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right) \mathrm{d}t.$$

Since

$$\delta \dot{q} = \frac{\mathrm{d}}{\mathrm{d}t}(\delta q),$$

the second term has the form required for integration by parts. Set

$$f(t) = \frac{\partial L}{\partial \dot{q}}, \qquad g(t) = \delta q(t).$$

Then

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}} \delta \dot{q}\, \mathrm{d}t = \left[\frac{\partial L}{\partial \dot{q}} \delta q \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) \delta q\, \mathrm{d}t.$$

Therefore

$$\boxed{\delta S = \left[\frac{\partial L}{\partial \dot{q}} \delta q \right]_{t_1}^{t_2} + \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \delta q\, \mathrm{d}t}.$$

The derivative has been transferred from δq to $\partial L / \partial \dot{q}$. This is the decisive step in the derivation.

Why the derivative must be moved

Before integration by parts, the first variation contains both δq and $\delta \dot{q}$. These are not independent functions because

$$\delta \dot{q} = \frac{\mathrm{d}}{\mathrm{d}t}(\delta q).$$

The fundamental lemma of the calculus of variations applies to an integral of the form

$$\int f(t) \delta q(t)\, \mathrm{d}t.$$

Integration by parts rewrites the complete interior contribution in precisely this form. The arbitrariness of δq can then be used to conclude that its coefficient vanishes.

Without this step, one cannot correctly infer separate conditions from the coefficients of δq and $\delta \dot{q}$.

The boundary term as a separate object

The boundary contribution is

$$B = \left[\frac{\partial L}{\partial \dot{q}} \delta q \right]_{t_1}^{t_2}.$$

Using the canonical momentum

$$p = \frac{\partial L}{\partial \dot{q}},$$

we may write

$$B = [p\,\delta q]_{t_1}^{t_2}.$$

This term contains no time integral. It depends only on endpoint data.
For fixed endpoint positions,

$$\delta q(t_1) = \delta q(t_2) = 0,$$

so

$$B = 0.$$

For free endpoints, the term survives and produces natural boundary conditions. Thus one should never write "the boundary term is zero" without first stating the conditions that make it zero.

A direct example

Consider

$$L = \frac{m}{2}\dot{q}^2 - V(q).$$

The first variation is

$$\delta S = \int_{t_1}^{t_2} \left(-V'(q)\delta q + m\dot{q}\,\delta\dot{q}\right)\,\mathrm{d}t.$$

Integrating the second term by parts,

$$\int_{t_1}^{t_2} m\dot{q}\,\delta\dot{q}\,\mathrm{d}t = [m\dot{q}\,\delta q]_{t_1}^{t_2} - \int_{t_1}^{t_2} m\ddot{q}\,\delta q\,\mathrm{d}t.$$

Hence

$$\delta S = [m\dot{q}\,\delta q]_{t_1}^{t_2} - \int_{t_1}^{t_2} \left(m\ddot{q} + V'(q)\right)\delta q\,\mathrm{d}t.$$

For fixed endpoints, the first term vanishes. Stationarity then implies

$$m\ddot{q} + V'(q) = 0.$$

Every sign in this equation can be traced back to the product rule and the minus sign introduced by integration by parts.

Sign checking

A reliable way to check the sign is to start from

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\delta q\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right)\delta q + \frac{\partial L}{\partial \dot{q}}\delta\dot{q}.$$

Solving for the last term,

$$\frac{\partial L}{\partial \dot{q}}\delta\dot{q} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\delta q\right) - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right)\delta q.$$

The minus sign is therefore unavoidable.

Adding a total time derivative to the Lagrangian

Suppose two Lagrangians differ by a total derivative:

$$L'(q,\dot q,t) = L(q,\dot q,t) + \frac{\mathrm{d}}{\mathrm{d}t}F(q,t).$$

Their actions satisfy

$$\begin{aligned} S'[q] &= \int_{t_1}^{t_2} L'\,\mathrm{d}t \\ &= S[q] + \int_{t_1}^{t_2} \frac{\mathrm{d}F}{\mathrm{d}t}\,\mathrm{d}t \\ &= S[q] + F(q(t_2),t_2) - F(q(t_1),t_1). \end{aligned}$$

Thus the two actions differ only by an endpoint term.

For fixed endpoint values and fixed endpoint times, the variation of this additional term vanishes. Therefore L and L' produce the same Euler–Lagrange equations.

This establishes an important equivalence:

$$\boxed{L \sim L + \frac{\mathrm{d}F}{\mathrm{d}t}}.$$

The Lagrangian is not unique. Different Lagrangians can encode the same equations of motion when they differ by a total derivative.

Example of equivalent Lagrangians

Consider

$$L = \frac{m}{2}\dot q^2 - V(q)$$

and choose

$$F(q,t) = \alpha q t,$$

where α is constant. Then

$$\frac{\mathrm{d}F}{\mathrm{d}t} = \alpha q + \alpha t \dot q.$$

The modified Lagrangian is

$$L' = \frac{m}{2}\dot q^2 - V(q) + \alpha q + \alpha t \dot q.$$

Although the individual partial derivatives of L' differ from those of L, the extra contributions cancel in the Euler–Lagrange equation.

Indeed,

$$\frac{\partial L'}{\partial q} = -V'(q) + \alpha,$$

and

$$\frac{\partial L'}{\partial \dot q} = m\dot q + \alpha t.$$

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L'}{\partial \dot{q}}\right) = m\ddot{q} + \alpha.$$

The Euler–Lagrange equation becomes

$$m\ddot{q} + \alpha - (-V'(q) + \alpha) = 0,$$

so

$$m\ddot{q} + V'(q) = 0.$$

The dynamics is unchanged.

Boundary terms are not always negligible

A boundary term may be physically or mathematically important when

- endpoint values are not fixed;
- the integration region has a physical boundary;
- one studies conserved charges or canonical momenta;
- the action contains higher derivatives;
- the theory is defined on a noncompact region and fall-off conditions matter;
- surface contributions encode measurable quantities.

Thus "boundary term" does not mean "unimportant term." It means a term supported on the boundary rather than in the interior.

Repeated integration by parts

If an action depends on higher derivatives, for example

$$S[q] = \int L(q, \dot{q}, \ddot{q}, t)\,\mathrm{d}t,$$

then varying $\ddot{q}$ produces

$$\delta\ddot{q} = \frac{\mathrm{d}^2}{\mathrm{d}t^2}(\delta q).$$

Two integrations by parts are required to remove derivatives from δq. The resulting boundary terms contain both δq and $\delta\dot{q}$.

This explains why higher-derivative actions generally require more boundary data and produce higher-order equations of motion.

The present course will mainly use first-derivative Lagrangians, but the pattern is worth recognizing.

Several coordinates

For generalized coordinates q^i,

$$\delta S = \int_{t_1}^{t_2} \sum_i \left(\frac{\partial L}{\partial q^i} \delta q^i + \frac{\partial L}{\partial \dot{q}^i} \delta \dot{q}^i \right) \mathrm{d}t.$$

Integrating each velocity term by parts gives

$$\delta S = \left[\sum_i \frac{\partial L}{\partial \dot{q}^i} \delta q^i \right]_{t_1}^{t_2} + \int_{t_1}^{t_2} \sum_i \left[\frac{\partial L}{\partial q^i} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}^i} \right) \right] \delta q^i \, \mathrm{d}t.$$

The boundary term is a sum of momentum-coordinate pairings:

$$\left[\sum_i p_i \delta q^i \right]_{t_1}^{t_2}.$$

The field-theory analogue

For a scalar field with action

$$S[\phi] = \int_\Omega \mathcal{L}(\phi, \partial_\mu \phi, x) \, \mathrm{d}^4 x,$$

the first variation contains

$$\int_\Omega \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu (\delta \phi) \, \mathrm{d}^4 x.$$

Spacetime integration by parts gives

$$\begin{aligned} &\int_\Omega \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu (\delta \phi) \, \mathrm{d}^4 x \\ &\qquad = \int_{\partial\Omega} \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta \phi \, n_\mu \, \mathrm{d}\Sigma - \int_\Omega \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \delta \phi \, \mathrm{d}^4 x. \end{aligned}$$

The endpoint term of mechanics becomes a surface integral over the boundary $\partial\Omega$. The one-dimensional formula is therefore not merely analogous to the field calculation; it is its simplest special case.

A diagnostic procedure

Whenever integration by parts appears in a variational derivation:

1. identify the derivative acting on the variation;
2. write the product-rule identity explicitly;
3. separate the boundary and interior terms;
4. state the boundary conditions;
5. justify whether the boundary term vanishes;
6. check the sign of the interior derivative;
7. only then apply the fundamental lemma.

Following this sequence prevents hidden assumptions and sign errors.

What has been established

Integration by parts converts the derivative of a variation into an interior derivative acting on the canonical momentum and an explicit boundary contribution. The interior term yields the Euler–Lagrange equation, while the boundary term records the endpoint assumptions.

Lagrangians that differ by a total time derivative produce actions that differ only by endpoint terms and therefore give the same fixed-endpoint equations of motion.

The next section applies the full method to several concrete particle models, from free motion to oscillation and constrained dynamics.

2.7 Worked Particle Models

The purpose of this section is to apply the variational method as a complete calculation. For each model, we identify the coordinate, write the Lagrangian, derive the Euler–Lagrange equation, solve or simplify the equation, and interpret the result.

The examples are deliberately standard. Their value lies not in novelty but in making the procedure operational.

Model 1: the free particle

Consider a particle of mass m moving on a line with no potential energy. The generalized coordinate is $q(t)$, and the Lagrangian is

$$L(q, \dot{q}) = \frac{m}{2}\dot{q}^2.$$

The required derivatives are

$$\frac{\partial L}{\partial q} = 0,$$

and

$$\frac{\partial L}{\partial \dot{q}} = m\dot{q}.$$

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) = m\ddot{q}.$$

The Euler–Lagrange equation gives

$$m\ddot{q} = 0,$$

or

$$\boxed{\ddot{q} = 0}.$$

Integrating once,

$$\dot{q} = v_0,$$

where v_0 is constant. Integrating again,

$$\boxed{q(t) = q_0 + v_0 t}.$$

The canonical momentum is

$$p = \frac{\partial L}{\partial \dot{q}} = m\dot{q},$$

and the equation of motion states

$$\dot{p} = 0.$$

Thus free motion corresponds to constant momentum and constant velocity.

Fixed-endpoint solution

Suppose

$$q(t_1) = q_1, \qquad q(t_2) = q_2.$$

The solution satisfying these conditions is

$$\boxed{q(t) = q_1 + \frac{q_2 - q_1}{t_2 - t_1}(t - t_1)}.$$

The physical path is the straight line in the (t, q)-plane connecting the two endpoint events.

Check by substitution

Differentiating twice,

$$\dot{q} = \frac{q_2 - q_1}{t_2 - t_1}, \qquad \ddot{q} = 0.$$

The trajectory therefore satisfies the Euler–Lagrange equation exactly.

Model 2: a particle in a general potential

Let

$$L(q, \dot{q}) = \frac{m}{2}\dot{q}^2 - V(q).$$

Then

$$\frac{\partial L}{\partial q} = -V'(q),$$

and

$$\frac{\partial L}{\partial \dot{q}} = m\dot{q}.$$

The Euler–Lagrange equation is

$$m\ddot{q} + V'(q) = 0.$$

Defining the force by

$$F(q) = -V'(q),$$

we obtain

$$\boxed{m\ddot{q} = F(q)}.$$

This is Newton's second law. The Lagrangian and Newtonian formulations therefore describe the same dynamics for conservative forces.

Energy check

Multiply the equation of motion by $\dot{q}$:

$$m\ddot{q}\dot{q} + V'(q)\dot{q} = 0.$$

Using

$$m\ddot{q}\dot{q} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{m}{2}\dot{q}^2\right),$$

and

$$V'(q)\dot{q} = \frac{\mathrm{d}V}{\mathrm{d}t},$$

we find

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{m}{2}\dot{q}^2 + V(q)\right) = 0.$$

Thus

$$\boxed{E = \frac{m}{2}\dot{q}^2 + V(q) = \text{constant}}.$$

This conservation law provides a useful independent check of solutions.

Model 3: motion in a uniform gravitational field

Choose the vertical coordinate y, positive upward. The potential energy is

$$V(y) = mgy.$$

The Lagrangian is

$$L(y, \dot{y}) = \frac{m}{2}\dot{y}^2 - mgy.$$

We calculate

$$\frac{\partial L}{\partial y} = -mg,$$

and

$$\frac{\partial L}{\partial \dot{y}} = m\dot{y}.$$

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{y}}\right) = m\ddot{y}.$$

The Euler–Lagrange equation gives

$$m\ddot{y} + mg = 0.$$

Dividing by m,

$$\boxed{\ddot{y} = -g}.$$

Integrating,

$$\dot{y}(t) = v_0 - gt,$$

and

$$\boxed{y(t) = y_0 + v_0 t - \frac{g}{2}t^2}.$$

Physical interpretation

The mass cancels from the equation of motion. In this idealized model, all particles have the same gravitational acceleration regardless of mass.

The conserved energy is

$$E = \frac{m}{2}\dot{y}^2 + mgy.$$

Substituting the solution verifies that the decrease in kinetic energy during upward motion equals the increase in gravitational potential energy.

Model 4: the harmonic oscillator

Consider a particle attached to a spring with spring constant k. The potential energy is

$$V(q) = \frac{k}{2}q^2.$$

The Lagrangian is

$$L(q, \dot{q}) = \frac{m}{2}\dot{q}^2 - \frac{k}{2}q^2.$$

We have

$$\frac{\partial L}{\partial q} = -kq,$$

and

$$\frac{\partial L}{\partial \dot{q}} = m\dot{q}.$$

The Euler–Lagrange equation gives

$$m\ddot{q} + kq = 0.$$

Define

$$\omega^2 = \frac{k}{m}.$$

Then

$$\boxed{\ddot{q} + \omega^2 q = 0}.$$

The general solution is

$$\boxed{q(t) = A\cos(\omega t) + B\sin(\omega t)}.$$

Equivalently,

$$q(t) = C\cos(\omega t - \varphi).$$

Verification

Differentiating twice,

$$\ddot{q}(t) = -\omega^2 \left[A\cos(\omega t) + B\sin(\omega t)\right] = -\omega^2 q(t).$$

Thus the proposed solution satisfies the equation.

Energy

The conserved energy is

$$E = \frac{m}{2}\dot{q}^2 + \frac{k}{2}q^2.$$

For

$$q(t) = C\cos(\omega t - \varphi),$$

we have

$$\dot{q}(t) = -C\omega\sin(\omega t - \varphi).$$

Therefore

$$\begin{aligned}
E &= \frac{m}{2}C^2\omega^2\sin^2(\omega t - \varphi) + \frac{k}{2}C^2\cos^2(\omega t - \varphi) \\
&= \frac{kC^2}{2}\left[\sin^2(\omega t - \varphi) + \cos^2(\omega t - \varphi)\right] \\
&= \boxed{\frac{kC^2}{2}}.
\end{aligned}$$

The total energy remains constant while kinetic and potential energies exchange periodically.

Model 5: a particle in two-dimensional polar coordinates

Consider a particle moving in a plane. Introduce polar coordinates

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

The velocity components are

$$\dot{x} = \dot{r}\cos\theta - r\dot{\theta}\sin\theta,$$

and

$$\dot{y} = \dot{r}\sin\theta + r\dot{\theta}\cos\theta.$$

Squaring and adding,

$$\dot{x}^2 + \dot{y}^2 = \dot{r}^2 + r^2\dot{\theta}^2.$$

For a central potential $V(r)$, the Lagrangian is

$$\boxed{L(r,\theta,\dot{r},\dot{\theta}) = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - V(r)}.$$

Angular equation

The coordinate θ does not appear explicitly in L:

$$\frac{\partial L}{\partial \theta} = 0.$$

The conjugate momentum is

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta}.$$

The Euler–Lagrange equation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(mr^2\dot{\theta}\right) = 0.$$

Therefore

$$\boxed{mr^2\dot{\theta} = \ell = \text{constant}}.$$

The conserved quantity ℓ is the angular momentum about the origin.

Radial equation

For the radial coordinate,

$$\frac{\partial L}{\partial \dot{r}} = m\dot{r},$$

so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{r}}\right) = m\ddot{r}.$$

Also,

$$\frac{\partial L}{\partial r} = mr\dot{\theta}^2 - V'(r).$$

The radial Euler–Lagrange equation is

$$m\ddot{r} - mr\dot{\theta}^2 + V'(r) = 0.$$

Equivalently,

$$\boxed{m\ddot{r} = mr\dot{\theta}^2 - V'(r)}.$$

The term

$$mr\dot{\theta}^2$$

arises from the coordinate dependence of the kinetic energy. It is not inserted as an additional force by hand.

Effective potential

Using

$$\dot{\theta} = \frac{\ell}{mr^2},$$

the energy becomes

$$E = \frac{m}{2}\dot{r}^2 + \frac{\ell^2}{2mr^2} + V(r).$$

Define the effective potential

$$\boxed{V_{\text{eff}}(r) = V(r) + \frac{\ell^2}{2mr^2}}.$$

Then the radial motion takes the one-dimensional energy form

$$E = \frac{m}{2}\dot{r}^2 + V_{\text{eff}}(r).$$

This reduction shows how a multidimensional problem can be simplified using a cyclic coordinate and its conserved momentum.

Model 6: the simple pendulum and the small-angle limit

For a pendulum of length ℓ and mass m, the Lagrangian is

$$L = \frac{m\ell^2}{2}\dot{\theta}^2 - mg\ell(1 - \cos\theta).$$

The equation of motion is

$$\boxed{\ddot{\theta} + \frac{g}{\ell}\sin\theta = 0}.$$

This equation is nonlinear because of the sine function.
For small angular displacements,

$$|\theta| \ll 1,$$

we use

$$\sin\theta \approx \theta.$$

The equation becomes

$$\ddot{\theta} + \frac{g}{\ell}\theta = 0.$$

This is the harmonic-oscillator equation with

$$\omega_0 = \sqrt{\frac{g}{\ell}}.$$

The approximate solution is

$$\boxed{\theta(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t)}.$$

The corresponding small-amplitude period is

$$\boxed{T = 2\pi\sqrt{\frac{\ell}{g}}}.$$

This example illustrates linearization: a nonlinear equation is approximated near an equilibrium by retaining only the leading term in a series expansion.

Comparing the models

The examples reveal a common pattern.

1. The free particle has a cyclic position coordinate and conserved linear momentum.
2. A potential introduces a generalized force through $-V'(q)$.
3. A quadratic potential produces linear harmonic motion.
4. Curvilinear coordinates make the kinetic energy coordinate dependent.
5. A cyclic angular coordinate produces conserved angular momentum.
6. A nonlinear system may reduce to a linear oscillator near equilibrium.

These are not isolated tricks. The same structural ideas recur in field theory: kinetic terms, potentials, symmetries, conserved quantities, coordinate choices, and linearization around simple backgrounds.

A complete checking procedure

For every model derived from an action, one should check:

1. Are all terms in the Lagrangian dimensionally consistent?
2. Were the generalized coordinates and constraints identified correctly?
3. Was the total time derivative in the Euler–Lagrange equation calculated fully?
4. Does the sign of the force agree with the physical potential?
5. Does a proposed solution satisfy the differential equation?
6. Are conserved quantities constant along the solution?
7. Does the result reduce correctly in a simple limit?

These checks often reveal errors more quickly than repeating the entire derivation.

Chapter summary

This chapter developed the action principle first as a mathematical construction and then as a practical method of deriving dynamics.

The action is a functional of the complete trajectory:

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, \mathrm{d}t.$$

A varied path is written as

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t),$$

and fixed endpoint positions imply

$$\eta(t_1) = \eta(t_2) = 0.$$

Integration by parts separates the variation into a boundary term and an interior term. Stationarity for arbitrary admissible deformations produces the Euler–Lagrange equation

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^i}\right) - \frac{\partial L}{\partial q^i} = 0}.$$

Generalized coordinates incorporate constraints and adapt the description to the geometry of the system. The worked models showed how the same procedure produces free motion, uniform acceleration, harmonic oscillation, angular-momentum conservation, effective radial dynamics, and the small-angle pendulum.

The next chapter extends every part of this construction from particle trajectories to fields. The generalized coordinate $q(t)$ becomes a field $\phi(x)$, the Lagrangian becomes a Lagrangian density, the time integral becomes a spacetime integral, and ordinary integration by parts becomes spacetime integration by parts.

Chapter 3

Lagrangian Field Theory and Field Equations

3.1 Lagrangian Density and Field Action

The action principle developed for particles can be extended to systems with continuously many degrees of freedom. The essential change is that a particle trajectory is replaced by a field configuration, and the mechanical Lagrangian is replaced by a Lagrangian density.

The purpose of this section is to make that transition explicit. We will identify the dynamical variables, define the field action, explain why a density is required, and compare the structure of particle mechanics with the structure of field theory.

From finitely many coordinates to a field

A mechanical system with generalized coordinates

$$q^1(t), \ldots, q^N(t)$$

has finitely many configuration variables at each instant. Its action is

$$S[q] = \int_{t_1}^{t_2} L(q^i, \dot{q}^i, t)\,\mathrm{d}t.$$

A field assigns a value to every point of space. For a real scalar field in three spatial dimensions, we write

$$\phi = \phi(\mathbf{x}, t), \qquad \mathbf{x} = (x^1, x^2, x^3).$$

At a fixed time, the complete function

$$\mathbf{x} \longmapsto \phi(\mathbf{x}, t)$$

is the configuration of the field. Each spatial point carries its own field value, so the field may be viewed informally as a system with one generalized coordinate at every point of space.

This viewpoint explains why the mechanical sum over degrees of freedom becomes a spatial integral. A discrete chain contains a sum of local contributions,

$$L_{\text{discrete}} = \sum_n L_n,$$

whereas a continuum contains a contribution from every spatial point,

$$L_{\text{field}}(t) = \int_\Sigma \mathcal{L}(\mathbf{x}, t)\,\mathrm{d}^3x.$$

Here Σ denotes the spatial region occupied by the system, and $\mathcal{L}$ is the Lagrangian density.

Why a density is needed

The mechanical Lagrangian L has dimensions of energy. A Lagrangian density must therefore have dimensions of energy per unit volume:

$$[\mathcal{L}] = \frac{\mathrm{J}}{\mathrm{m}^3}$$

in three spatial dimensions.

Integrating over space gives the ordinary Lagrangian at time t:

$$L(t) = \int_\Sigma \mathcal{L}(\phi, \partial_t \phi, \nabla\phi, \mathbf{x}, t)\, \mathrm{d}^3 x.$$

The action is then obtained by integrating over time:

$$\begin{aligned} S[\phi] &= \int_{t_1}^{t_2} L(t)\, \mathrm{d}t \\ &= \int_{t_1}^{t_2} \int_\Sigma \mathcal{L}\, \mathrm{d}^3 x\, \mathrm{d}t. \end{aligned}$$

Thus the action is an integral over a spacetime region.

The field action

Let Ω denote the spacetime region

$$\Omega = [t_1, t_2] \times \Sigma.$$

Using spacetime coordinates

$$x^\mu = (x^0, x^1, x^2, x^3),$$

we write

$$\partial_\mu \phi := \frac{\partial \phi}{\partial x^\mu}.$$

At this stage, the index notation is only a compact way to list derivatives with respect to all spacetime coordinates. The metric and relativistic conventions will be introduced in the next chapter.

The action of one real field is written as

$$\boxed{S[\phi] = \int_\Omega \mathcal{L}(\phi, \partial_\mu \phi, x)\, \mathrm{d}^4 x}.$$

The square brackets emphasize that S depends on the complete field history, not on the field value at one point.

If the system contains several fields $\phi^a(x)$, the action becomes

$$S[\phi^1, \ldots, \phi^N] = \int_\Omega \mathcal{L}(\phi^a, \partial_\mu \phi^a, x)\, \mathrm{d}^4 x.$$

The label a distinguishes the fields, while μ labels the spacetime derivative.

Local Lagrangian densities

A local field theory uses a Lagrangian density evaluated from the field and a finite number of its derivatives at the same spacetime point. The simplest class has the form

$$\mathcal{L} = \mathcal{L}(\phi, \partial_\mu \phi, x).$$

The value of $\mathcal{L}$ at x depends on

- the field value $\phi(x)$;
- the derivatives $\partial_\mu \phi(x)$;

- prescribed background quantities or sources at x;
- possibly the point x explicitly.

It does not directly depend on the field value at a distant point y. Spatial coupling is nevertheless present because gradient terms compare neighbouring values through derivatives.

A nonlocal action could contain terms such as

$$\int\int K(x,y)\phi(x)\phi(y)\,\mathrm{d}^4x\,\mathrm{d}^4y,$$

but such theories are not the subject of the basic variational derivation developed here.

Typical terms in a Lagrangian density

A useful nonrelativistic scalar-field model has the form

$$\boxed{\mathcal{L} = \frac{\rho}{2}\left(\frac{\partial\phi}{\partial t}\right)^2 - \frac{\kappa}{2}|\nabla\phi|^2 - V(\phi) + J\phi}.$$

The four terms have distinct roles.

1. The time-derivative term describes local inertia or kinetic behaviour.
2. The gradient term penalizes spatial variation and couples neighbouring points.
3. The potential term determines local restoring forces or self-interactions.
4. The source term describes an externally prescribed influence on the field.

The signs are part of the model. They are chosen so that the associated energy has the expected positive kinetic and gradient contributions. The physical dimensions of ρ, κ, V, and J depend on the dimensions assigned to ϕ.

Example: the vibrating string

For a string with transverse displacement $u(x,t)$, mass density per unit length μ, and tension T, the Lagrangian density per unit length is

$$\boxed{\mathcal{L} = \frac{\mu}{2}\left(\frac{\partial u}{\partial t}\right)^2 - \frac{T}{2}\left(\frac{\partial u}{\partial x}\right)^2}.$$

The first term is the kinetic-energy density of the string. The second is minus the elastic-energy density associated with the local slope.

The Lagrangian at time t is

$$L(t) = \int_0^L \left[\frac{\mu}{2}u_t^2 - \frac{T}{2}u_x^2\right]\mathrm{d}x,$$

where

$$u_t = \frac{\partial u}{\partial t}, \qquad u_x = \frac{\partial u}{\partial x}.$$

The field action is

$$S[u] = \int_{t_1}^{t_2}\int_0^L \left[\frac{\mu}{2}u_t^2 - \frac{T}{2}u_x^2\right]\mathrm{d}x\,\mathrm{d}t.$$

Later in this chapter, varying this action will produce the wave equation.

The action as a functional

A field action maps a complete field history to one number:

$$\phi(x) \longmapsto S[\phi].$$

Two fields may agree at one point and still have different actions because they differ elsewhere. They may also have the same field values but different derivatives, producing different values of the gradient or kinetic terms.

The field action therefore depends on the shape of the field throughout Ω. The physical field is selected by a stationarity condition,

$$\delta S = 0,$$

for every allowed infinitesimal field variation satisfying the specified boundary conditions.

As in mechanics, stationary does not necessarily mean globally minimal. It means that the first-order change of the action vanishes.

Adding a spacetime divergence

Suppose the Lagrangian density is changed by a total divergence:

$$\mathcal{L}' = \mathcal{L} + \partial_\mu K^\mu.$$

The actions differ by

$$S' - S = \int_\Omega \partial_\mu K^\mu \, \mathrm{d}^4 x.$$

By the divergence theorem, this becomes a boundary integral over $\partial\Omega$. Under boundary conditions that keep this surface contribution fixed, $\mathcal{L}$ and $\mathcal{L}'$ produce the same interior field equations.

This is the field-theoretic analogue of adding a total time derivative to a mechanical Lagrangian.

Particle mechanics and field theory compared

The basic correspondence is

$$q^i(t) \quad \longrightarrow \quad \phi^a(\mathbf{x}, t),$$

$$L(q, \dot{q}, t) \quad \longrightarrow \quad \mathcal{L}(\phi, \partial_\mu \phi, x),$$

and

$$\int \mathrm{d}t \quad \longrightarrow \quad \int \mathrm{d}^4 x.$$

The ordinary derivative of a generalized coordinate becomes a collection of spacetime derivatives. The endpoint conditions of mechanics become boundary conditions on the spacetime region.

The conceptual logic remains unchanged:

1. choose a candidate history;
2. evaluate its action;
3. compare it with nearby admissible histories;
4. require the first-order change to vanish.

Common mistakes

Several distinctions should be maintained.

1. The Lagrangian density $\mathcal{L}$ is not the same object as the spatially integrated Lagrangian $L(t)$.
2. The action is a functional of the complete field, not an ordinary function of one field value.
3. A local theory may contain derivatives without being nonlocal; derivatives encode behaviour in an arbitrarily small neighbourhood.
4. The dimensions of $\mathcal{L}$ depend on the integration measure.
5. A total divergence affects the boundary contribution even when it does not change the interior equations.

What has been established

A field theory replaces finitely many generalized coordinates by one or more functions defined throughout space and time. Its action is

$$S[\phi] = \int_\Omega \mathcal{L}(\phi, \partial_\mu \phi, x)\, \mathrm{d}^4 x.$$

The Lagrangian density contains local kinetic, gradient, potential, and source contributions. Spatial integration produces the mechanical Lagrangian at each time, and time integration produces the action.

The next section constructs a family of neighbouring fields and defines the first variation of the field action.

3.2 Variation of Fields

The action principle compares a candidate field history with neighbouring field histories. In mechanics, a trajectory is deformed by adding a small function of time. In field theory, the deformation is a function of all spacetime coordinates.

The purpose of this section is to define field variations precisely, determine how derivatives of a field vary, and construct the first variation of a general local action.

A one-parameter family of fields

Let

$$\phi = \phi(x)$$

be a real field defined on a spacetime region Ω. We embed it in a one-parameter family

$$\boxed{\phi_\varepsilon(x) = \phi(x) + \varepsilon\eta(x)}.$$

Here

- ε is a real parameter;
- $\eta(x)$ is a smooth deformation function;
- $\phi_0(x) = \phi(x)$ is the reference field.

At each spacetime point, changing ε changes the field value in the direction specified by $\eta(x)$. The same parameter labels the deformation throughout the complete region.

The variation of the field is defined by

$$\delta\phi(x) := \left.\frac{\partial\phi_\varepsilon(x)}{\partial\varepsilon}\right|_{\varepsilon=0}.$$

For the linear family above,

$$\boxed{\delta\phi(x) = \eta(x)}.$$

The variation is not a derivative with respect to a spacetime coordinate. It compares different fields at the same spacetime point.

Variation and spacetime differentiation

The derivative of the varied field is

$$\partial_\mu\phi_\varepsilon = \partial_\mu\phi + \varepsilon\partial_\mu\eta.$$

Therefore

$$\begin{aligned}\delta(\partial_\mu\phi) &:= \left.\frac{\partial}{\partial\varepsilon}(\partial_\mu\phi_\varepsilon)\right|_{\varepsilon=0} \\ &= \partial_\mu\eta.\end{aligned}$$

Since $\eta = \delta\phi$,

$$\boxed{\delta(\partial_\mu\phi) = \partial_\mu(\delta\phi)}.$$

For the smooth families considered here, variation and spacetime differentiation commute. In time-and-space notation, this gives

$$\delta\left(\frac{\partial\phi}{\partial t}\right) = \frac{\partial}{\partial t}(\delta\phi),$$

and

$$\delta(\nabla\phi) = \nabla(\delta\phi).$$

These identities are the field-theoretic analogues of

$$\delta\dot{q} = \frac{\mathrm{d}}{\mathrm{d}t}(\delta q)$$

in mechanics.

Admissible field variations

The allowed deformation functions are determined by the boundary data of the variational problem. If the field value is prescribed on the boundary $\partial\Omega$, then every varied field must satisfy the same boundary condition:

$$\phi_\varepsilon|_{\partial\Omega} = \phi|_{\partial\Omega}.$$

Substitution of

$$\phi_\varepsilon = \phi + \varepsilon\eta$$

gives

$$\boxed{\eta|_{\partial\Omega} = 0}.$$

Equivalently,

$$\delta\phi = 0 \qquad \text{on } \partial\Omega.$$

The variation is otherwise arbitrary in the interior of Ω.

Another useful choice is to take η to have compact support inside Ω. Then η vanishes in a neighbourhood of the boundary, so every boundary contribution vanishes automatically. Compactly supported variations are especially useful when deriving local field equations.

A localized field variation

A deformation may be concentrated in a small spacetime region. Let U be a small open subset of Ω. We may choose a smooth function $\eta(x)$ such that

$$\eta(x) = 0 \qquad \text{for } x \notin U.$$

The fields ϕ and ϕ_ε then agree outside U. If the action is stationary under every such local deformation, the coefficient multiplying $\eta(x)$ in the first variation must vanish at every interior point.

This is the mechanism by which a global condition on the action produces a local partial differential equation.

The varied action

Consider the action

$$S[\phi] = \int_\Omega \mathcal{L}(\phi, \partial_\mu\phi, x)\, \mathrm{d}^4x.$$

For the varied field,

$$S[\phi_\varepsilon] = \int_\Omega \mathcal{L}(\phi_\varepsilon, \partial_\mu\phi_\varepsilon, x)\, \mathrm{d}^4x.$$

Once $\eta(x)$ is chosen, this is an ordinary function of ε. The first variation is

$$\boxed{\delta S[\phi;\eta] := \frac{\mathrm{d}}{\mathrm{d}\varepsilon} S[\phi_\varepsilon]\bigg|_{\varepsilon=0}}.$$

The notation records both the reference field ϕ and the deformation direction η. When no ambiguity is possible, we write simply δS.

First-order expansion of the density

Using the multivariable chain rule,

$$\begin{aligned}\mathcal{L}(\phi + \varepsilon\eta, \partial_\mu\phi + \varepsilon\partial_\mu\eta, x) = \mathcal{L}(\phi, \partial_\mu\phi, x) \\ + \varepsilon\frac{\partial\mathcal{L}}{\partial\phi}\eta \\ + \varepsilon\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu\eta \\ + O(\varepsilon^2).\end{aligned}$$

Repeated μ indicates a sum over the spacetime coordinates. Integrating gives

$$
\begin{aligned}
S[\phi_\varepsilon] = S[\phi] \\
&+ \varepsilon \int_\Omega \left[\frac{\partial \mathcal{L}}{\partial \phi} \eta + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu \eta \right] \mathrm{d}^4 x \\
&+ O(\varepsilon^2).
\end{aligned}
$$

Therefore

$$
\boxed{\delta S = \int_\Omega \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu (\delta\phi) \right] \mathrm{d}^4 x}.
$$

This is the first variation before spacetime integration by parts.

What is held fixed in the partial derivatives

The symbols

$$
\frac{\partial \mathcal{L}}{\partial \phi} \qquad \text{and} \qquad \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}
$$

are partial derivatives with respect to different arguments of the Lagrangian density. When calculating

$$
\frac{\partial \mathcal{L}}{\partial \phi},
$$

the derivatives $\partial_\mu \phi$ are treated as independent arguments and held fixed. When calculating

$$
\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)},
$$

the field value and all other derivative components are held fixed.

This is the same formal rule used in mechanics when q and $\dot{q}$ are treated as independent arguments of $L(q, \dot{q}, t)$.

Example: variation of a kinetic term

Consider

$$
\mathcal{L}_{\text{kin}} = \frac{\rho}{2} (\partial_t \phi)^2.
$$

Its variation is

$$
\begin{aligned}
\delta \mathcal{L}_{\text{kin}} &= \frac{\rho}{2} \delta \left[(\partial_t \phi)^2 \right] \\
&= \rho (\partial_t \phi) \delta (\partial_t \phi) \\
&= \rho (\partial_t \phi) \partial_t (\delta \phi).
\end{aligned}
$$

The factor of two from differentiating the square cancels the factor $1/2$.

Example: variation of a gradient term

For

$$
\mathcal{L}_{\text{grad}} = -\frac{\kappa}{2} |\nabla \phi|^2,
$$

write

$$|\nabla\phi|^2 = \sum_{i=1}^{3}(\partial_i\phi)^2.$$

Then

$$\begin{aligned}\delta\mathcal{L}_{\text{grad}} &= -\kappa\sum_{i=1}^{3}(\partial_i\phi)\partial_i(\delta\phi)\\ &= -\kappa\nabla\phi\cdot\nabla(\delta\phi).\end{aligned}$$

A spatial derivative acts on the variation. It will later be transferred to $\nabla\phi$ by integration by parts.

Example: variation of a potential term

For a local potential contribution

$$\mathcal{L}_{\text{pot}} = -V(\phi),$$

we obtain

$$\boxed{\delta\mathcal{L}_{\text{pot}} = -V'(\phi)\delta\phi}.$$

No derivative of $\delta\phi$ appears because the potential depends only on the field value at the same point.

For a source term

$$\mathcal{L}_{\text{source}} = J(x)\phi(x),$$

where J is prescribed and not varied,

$$\boxed{\delta\mathcal{L}_{\text{source}} = J(x)\delta\phi(x)}.$$

Linearity of the first variation

The first variation is linear in the deformation direction. If

$$\eta = a\eta_1 + b\eta_2,$$

then

$$\delta S[\phi; a\eta_1 + b\eta_2] = a\,\delta S[\phi;\eta_1] + b\,\delta S[\phi;\eta_2].$$

This makes δS the field-theoretic analogue of a directional derivative in an infinite-dimensional configuration space.

Terms of order ε^2 determine whether a stationary configuration is a minimum, maximum, or saddle point, but they are not needed to derive the field equation.

Several fields

For fields $\phi^a(x)$, introduce

$$\phi^a_\varepsilon(x) = \phi^a(x) + \varepsilon\eta^a(x).$$

Then

$$\delta\phi^a = \eta^a, \qquad \delta(\partial_\mu\phi^a) = \partial_\mu\eta^a.$$

The first variation is

$$\delta S = \int_\Omega \sum_a \left[\frac{\partial \mathcal{L}}{\partial \phi^a} \eta^a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^a)} \partial_\mu \eta^a \right] \mathrm{d}^4 x.$$

When the fields are independent, the functions η^a may be chosen independently. This will produce one field equation for each field component.

Common mistakes

The following errors are especially common.

1. Confusing $\delta\phi$ with a small coordinate displacement $\mathrm{d}\phi$.
2. Forgetting that derivatives of the field also vary.
3. Writing $\delta(\partial_\mu \phi) = \delta\phi$ instead of $\partial_\mu(\delta\phi)$.
4. Varying a prescribed source or background quantity when it is meant to remain fixed.
5. Treating $\delta\phi$ and $\partial_\mu(\delta\phi)$ as independent functions.
6. Failing to state the boundary conditions imposed on the variation.

What has been established

A field variation is constructed from a family

$$\phi_\varepsilon(x) = \phi(x) + \varepsilon\eta(x),$$

with

$$\delta\phi = \eta$$

and

$$\delta(\partial_\mu \phi) = \partial_\mu(\delta\phi).$$

For a local first-derivative action, the first variation contains both $\delta\phi$ and derivatives of $\delta\phi$. The next section performs spacetime integration by parts and derives the Euler–Lagrange field equation.

3.3 Euler–Lagrange Field Equations

We now derive the differential equation satisfied by a stationary field configuration. The calculation is the spacetime extension of the mechanical derivation, but it contains several derivative directions and a boundary surrounding an entire spacetime region.

Every essential step will be displayed: the varied field, the variation of its derivatives, spacetime integration by parts, the boundary term, and the use of arbitrary interior variations.

Statement of the variational problem

Consider one real field $\phi(x)$ with action

$$S[\phi] = \int_\Omega \mathcal{L}(\phi, \partial_\mu \phi, x)\, \mathrm{d}^4 x.$$

We introduce the family

$$\phi_\varepsilon(x) = \phi(x) + \varepsilon\eta(x),$$

where η is smooth and satisfies the chosen boundary condition. For the standard fixed-boundary problem,

$$\eta = 0 \qquad \text{on } \partial\Omega.$$

The stationary-action condition is

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} S[\phi_\varepsilon]\bigg|_{\varepsilon=0} = 0$$

for every admissible $\eta(x)$.

The first variation

From the chain rule,

$$\delta S = \int_\Omega \left[\frac{\partial \mathcal{L}}{\partial \phi}\eta + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\partial_\mu \eta \right] \mathrm{d}^4x.$$

Define

$$P^\mu := \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}.$$

This notation is temporary and simplifies the integration-by-parts step. The first variation becomes

$$\delta S = \int_\Omega \left[\frac{\partial \mathcal{L}}{\partial \phi}\eta + P^\mu \partial_\mu \eta \right] \mathrm{d}^4x.$$

Spacetime integration by parts

The product rule gives

$$\partial_\mu(P^\mu \eta) = (\partial_\mu P^\mu)\eta + P^\mu \partial_\mu \eta.$$

Therefore

$$P^\mu \partial_\mu \eta = \partial_\mu(P^\mu \eta) - (\partial_\mu P^\mu)\eta.$$

Substitution into the first variation gives

$$\begin{aligned} \delta S = &\int_\Omega \partial_\mu(P^\mu \eta)\, \mathrm{d}^4x \\ &+ \int_\Omega \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu P^\mu \right] \eta\, \mathrm{d}^4x. \end{aligned}$$

The first integral is a spacetime divergence. By the divergence theorem,

$$\int_\Omega \partial_\mu(P^\mu \eta)\, \mathrm{d}^4x = \int_{\partial\Omega} P^\mu n_\mu \eta\, \mathrm{d}\Sigma,$$

where n_μ is normal to the boundary and $\mathrm{d}\Sigma$ is the induced boundary measure.
Thus

$$\boxed{\begin{aligned}\delta S = &\int_{\partial\Omega} \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} n_\mu \eta \, \mathrm{d}\Sigma \\ &+ \int_\Omega \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \right] \eta \, \mathrm{d}^4 x.\end{aligned}}.$$

The first line is the boundary contribution. The second is the interior contribution.

Using the boundary condition

For fixed field values on $\partial\Omega$,

$$\eta|_{\partial\Omega} = 0.$$

Therefore

$$\int_{\partial\Omega} \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} n_\mu \eta \, \mathrm{d}\Sigma = 0.$$

The variation reduces to

$$\delta S = \int_\Omega \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \right] \eta \, \mathrm{d}^4 x.$$

The boundary term has not been ignored. It vanishes because the admissible variations were required to vanish on the boundary.

The fundamental lemma in spacetime

Suppose a continuous function $F(x)$ satisfies

$$\int_\Omega F(x) \eta(x) \, \mathrm{d}^4 x = 0$$

for every smooth compactly supported function η. Then

$$F(x) = 0$$

at every interior point.

The reason is local. If F were positive or negative in a small neighbourhood, one could choose η supported in that neighbourhood with the same sign, producing a nonzero integral.

Applying this result to $\delta S = 0$ gives

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) = 0.$$

Equivalently,

$$\boxed{\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0}.$$

This is the Euler–Lagrange field equation for one real field.

Comparison with the mechanical equation

The mechanical Euler–Lagrange equation is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0.$$

The field equation is

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0.$$

The correspondence is direct:

$$q(t) \longrightarrow \phi(x),$$

$$\dot{q} \longrightarrow \partial_\mu \phi,$$

and

$$\frac{\mathrm{d}}{\mathrm{d}t} \longrightarrow \partial_\mu \quad \text{with summation over } \mu.$$

The single time derivative in mechanics becomes a spacetime divergence.

Time-and-space form

For a density written as

$$\mathcal{L} = \mathcal{L}(\phi, \partial_t\phi, \partial_1\phi, \partial_2\phi, \partial_3\phi, x),$$

the field equation expands to

$$\boxed{\frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial(\partial_t \phi)}\right) + \sum_{i=1}^{3} \frac{\partial}{\partial x^i}\left(\frac{\partial \mathcal{L}}{\partial(\partial_i \phi)}\right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0}.$$

No metric convention is needed for this formula. The signs are determined by the signs already present in $\mathcal{L}$.

Example: kinetic, gradient, potential, and source terms

Consider

$$\mathcal{L} = \frac{\rho}{2}(\partial_t\phi)^2 - \frac{\kappa}{2}|\nabla\phi|^2 - V(\phi) + J\phi.$$

First,

$$\frac{\partial \mathcal{L}}{\partial(\partial_t \phi)} = \rho\partial_t\phi,$$

so

$$\frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial(\partial_t \phi)}\right) = \rho\partial_t^2\phi.$$

For each spatial derivative,

$$\frac{\partial \mathcal{L}}{\partial(\partial_i \phi)} = -\kappa\partial_i\phi.$$

Therefore

$$\sum_{i=1}^{3} \partial_i \left(\frac{\partial \mathcal{L}}{\partial(\partial_i \phi)} \right) = -\kappa \nabla^2 \phi.$$

Finally,

$$\frac{\partial \mathcal{L}}{\partial \phi} = -V'(\phi) + J.$$

The Euler–Lagrange equation is

$$\rho \partial_t^2 \phi - \kappa \nabla^2 \phi + V'(\phi) - J = 0.$$

Thus

$$\boxed{\rho \phi_{tt} - \kappa \nabla^2 \phi + V'(\phi) = J}.$$

This single calculation displays the role of every standard term:

- the time-derivative term produces local acceleration;
- the gradient term produces the Laplacian;
- the potential produces the local force $V'(\phi)$;
- the source appears on the right-hand side.

Functional derivative notation

The coefficient of $\delta\phi$ after integration by parts is called the functional derivative of the action:

$$\boxed{\frac{\delta S}{\delta \phi(x)} = \frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right)}.$$

The first variation may then be written, after the boundary term has been controlled, as

$$\delta S = \int_\Omega \frac{\delta S}{\delta \phi(x)} \delta\phi(x)\, \mathrm{d}^4 x.$$

The field equation is

$$\frac{\delta S}{\delta \phi(x)} = 0.$$

This notation is compact, but it should not hide the integration-by-parts calculation from which it arises.

Structural checks

A derived field equation should be checked systematically.

1. Every term must have the same physical dimensions.
2. The equation should have the same number and type of free indices on both sides.
3. The highest derivative order should agree with the derivative order of the Lagrangian density.
4. Removing a source or potential should give the expected simpler equation.

5. Constant fields should behave consistently when all derivative terms vanish.

For the example above, setting $V = 0$ and $J = 0$ gives

$$\rho\phi_{tt} - \kappa\nabla^2\phi = 0,$$

a wave equation. Setting the time derivatives to zero gives a static elliptic equation.

Common mistakes

The main errors in the derivation are:

1. omitting the variation of $\partial_\mu\phi$;
2. failing to sum over the derivative index μ;
3. treating $\delta\phi$ and $\partial_\mu\delta\phi$ as independent;
4. losing the minus sign introduced by integration by parts;
5. discarding the boundary term without a stated boundary condition;
6. applying a metric sign convention before one has been defined;
7. confusing the partial derivative of $\mathcal{L}$ with the functional derivative of S.

What has been established

For a local action

$$S[\phi] = \int_\Omega \mathcal{L}(\phi, \partial_\mu\phi, x)\,\mathrm{d}^4x,$$

stationarity under arbitrary admissible variations gives

$$\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right) - \frac{\partial\mathcal{L}}{\partial\phi} = 0.$$

The equation is local because the variation can be localized around any interior point. The next section generalizes the derivation to several interacting fields and coupled equations.

3.4 Multiple Fields and Coupled Equations

Many field theories contain several dynamical fields or several components of one field. The variational method extends directly: every independent field is varied independently, and stationarity produces one Euler–Lagrange equation for each field component.

Coupling appears when the Lagrangian density contains terms involving more than one field. The resulting equations must then be solved together.

Several independent fields

Let the dynamical variables be

$$\phi^1(x), \phi^2(x), \ldots, \phi^N(x).$$

We use the collective notation ϕ^a, where

$$a = 1, \ldots, N.$$

The action is

$$\boxed{S[\phi^1,\dots,\phi^N] = \int_\Omega \mathcal{L}(\phi^a,\partial_\mu\phi^a,x)\,\mathrm{d}^4x}.$$

The index a labels fields or components. It is distinct from the spacetime derivative index μ.

For each field, introduce an independent deformation:

$$\phi^a_\varepsilon(x) = \phi^a(x) + \varepsilon\eta^a(x).$$

Then

$$\delta\phi^a = \eta^a, \qquad \delta(\partial_\mu\phi^a) = \partial_\mu\eta^a.$$

First variation for several fields

The chain rule gives

$$\delta S = \int_\Omega \sum_{a=1}^N \left[\frac{\partial\mathcal{L}}{\partial\phi^a}\eta^a + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)}\partial_\mu\eta^a\right]\mathrm{d}^4x.$$

Integrating the derivative term by parts for every value of a,

$$\begin{aligned}\delta S = &\int_{\partial\Omega}\sum_{a=1}^N \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)} n_\mu\eta^a\,\mathrm{d}\Sigma \\ &+ \int_\Omega \sum_{a=1}^N \left[\frac{\partial\mathcal{L}}{\partial\phi^a} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)}\right)\right]\eta^a\,\mathrm{d}^4x.\end{aligned}$$

For fixed boundary values,

$$\eta^a = 0 \qquad \text{on } \partial\Omega$$

for every a, so the boundary term vanishes.

Because the interior functions $\eta^a(x)$ are independent, every coefficient must vanish separately:

$$\boxed{\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)}\right) - \frac{\partial\mathcal{L}}{\partial\phi^a} = 0, \qquad a = 1,\dots,N}.$$

Thus N independent field components produce N field equations.

What coupling means

The fields are uncoupled if the Lagrangian density separates as

$$\mathcal{L} = \mathcal{L}_1(\phi^1,\partial\phi^1) + \cdots + \mathcal{L}_N(\phi^N,\partial\phi^N).$$

Then the equation for ϕ^a contains only ϕ^a.

The fields are coupled when $\mathcal{L}$ contains mixed terms such as

$$g\phi^1\phi^2, \qquad h\partial_\mu\phi^1\partial_\mu\phi^2, \qquad V(\phi^1,\phi^2).$$

A mixed term contributes to more than one Euler–Lagrange equation. A disturbance in one field can therefore influence the evolution of another.

Example: two linearly coupled scalar fields

Consider two real fields $\phi(\mathbf{x},t)$ and $\chi(\mathbf{x},t)$ with density

$$\begin{aligned}\mathcal{L} = \frac{1}{2}\phi_t^2 - \frac{c_\phi^2}{2}|\nabla\phi|^2 - \frac{m_\phi^2}{2}\phi^2 \\ + \frac{1}{2}\chi_t^2 - \frac{c_\chi^2}{2}|\nabla\chi|^2 - \frac{m_\chi^2}{2}\chi^2 \\ - g\phi\chi + J_\phi\phi + J_\chi\chi.\end{aligned}$$

The sources J_ϕ and J_χ are prescribed and are not varied.
For ϕ,

$$\frac{\partial\mathcal{L}}{\partial\phi_t} = \phi_t, \qquad \frac{\partial\mathcal{L}}{\partial(\partial_i\phi)} = -c_\phi^2\partial_i\phi,$$

and

$$\frac{\partial\mathcal{L}}{\partial\phi} = -m_\phi^2\phi - g\chi + J_\phi.$$

Therefore

$$\boxed{\phi_{tt} - c_\phi^2\nabla^2\phi + m_\phi^2\phi + g\chi = J_\phi}.$$

Similarly, variation with respect to χ gives

$$\boxed{\chi_{tt} - c_\chi^2\nabla^2\chi + m_\chi^2\chi + g\phi = J_\chi}.$$

The equations are coupled because each contains the other field.

Consistency check from the interaction term

The interaction density is

$$\mathcal{L}_{\text{int}} = -g\phi\chi.$$

Its variation is

$$\delta\mathcal{L}_{\text{int}} = -g\chi\,\delta\phi - g\phi\,\delta\chi.$$

Thus the same coupling constant appears symmetrically in both equations. This provides a direct check of the signs and coefficients.
If $g = 0$, the equations decouple. Each field then evolves independently.

Normal fields in a symmetric model

Suppose

$$c_\phi = c_\chi = c, \qquad m_\phi = m_\chi = m, \qquad J_\phi = J_\chi = 0.$$

Define

$$\psi_+ = \frac{\phi+\chi}{\sqrt{2}}, \qquad \psi_- = \frac{\phi-\chi}{\sqrt{2}}.$$

Adding the two coupled equations gives

$$(\phi+\chi)_{tt} - c^2\nabla^2(\phi+\chi) + (m^2+g)(\phi+\chi) = 0.$$

Therefore

$$\boxed{(\psi_+)_{tt} - c^2\nabla^2\psi_+ + (m^2 + g)\psi_+ = 0}.$$

Subtracting the equations gives

$$\boxed{(\psi_-)_{tt} - c^2\nabla^2\psi_- + (m^2 - g)\psi_- = 0}.$$

The original fields are coupled, but the combinations ψ_+ and ψ_- evolve independently. They are normal fields, analogous to normal coordinates in coupled-oscillator mechanics.

This example shows that coupling may depend on the chosen variables. A suitable linear transformation can diagonalize a quadratic theory.

Matrix form of a quadratic theory

Let

$$\mathbf{\Phi} = \begin{pmatrix} \phi^1 \\ \vdots \\ \phi^N \end{pmatrix}.$$

A quadratic potential may be written as

$$V(\mathbf{\Phi}) = \frac{1}{2}\mathbf{\Phi}^\mathsf{T} M\mathbf{\Phi},$$

where M is a symmetric matrix. Its gradient in field space is

$$\frac{\partial V}{\partial \mathbf{\Phi}} = M\mathbf{\Phi}.$$

The coupled field equations can then be written schematically as

$$\mathbf{\Phi}_{tt} - c^2\nabla^2\mathbf{\Phi} + M\mathbf{\Phi} = \mathbf{J}.$$

Diagonalizing M identifies combinations of fields that propagate independently in the linear theory.

Field components and genuine independent fields

A vector-valued field may be written as

$$\mathbf{A}(x) = (A^1(x), A^2(x), A^3(x)).$$

For the purpose of variation, each independent component can initially be treated as a separate field. One writes one Euler–Lagrange equation for every component.

However, components need not always represent independent physical degrees of freedom. Constraints or gauge redundancies may relate them. The variational derivative still produces component equations, but additional structure is required to identify the physically independent content.

This distinction will become important in electromagnetism, where the four-potential contains gauge redundancy.

Derivative coupling

Fields may also be coupled through derivatives. For example,

$$\mathcal{L}_{\text{mix}} = -h\nabla\phi \cdot \nabla\chi.$$

Its variation with respect to ϕ contains

$$-h\nabla\chi \cdot \nabla(\delta\phi).$$

After integration by parts, this contributes

$$h\nabla^2\chi$$

to the coefficient of $\delta\phi$. The equation for one field then contains derivatives of the other.

Derivative coupling must therefore be handled through the full Euler–Lagrange formula, not by differentiating only the potential.

External fields and dynamical fields

A symbol appearing in $\mathcal{L}$ is varied only if it is declared dynamical. A prescribed source $J(x)$, background density, or external field remains fixed:

$$\delta J = 0.$$

If the same object is instead promoted to a dynamical field, it must receive its own variation and its own equation of motion.

A correct calculation should therefore begin by listing all dynamical variables and all prescribed backgrounds.

Common mistakes

Typical errors include:

1. varying only one field in an interaction term;
2. assuming coupled equations can be solved independently;
3. confusing the field label a with a spacetime index;
4. varying a prescribed source;
5. forgetting mixed derivative terms;
6. changing variables without transforming the complete Lagrangian density;
7. assuming every component is physically independent when constraints or gauge freedom are present.

What has been established

For several fields, stationarity gives one Euler–Lagrange equation for each independent component:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^a)} \right) - \frac{\partial \mathcal{L}}{\partial \phi^a} = 0.$$

Mixed terms in the Lagrangian density produce coupled equations. In quadratic theories, suitable linear combinations may diagonalize the coupling and reveal independent normal fields.

The next section studies the spacetime boundary term in greater detail and compares fixed-value, free-boundary, and natural boundary conditions.

3.5 Boundary Terms in Spacetime

The derivation of a field equation always separates the variation into an interior term and a boundary term. The interior term produces the differential equation. The boundary term determines which boundary data are compatible with the variational principle.

A boundary term is not automatically zero. It vanishes only after the admissible variations, fall-off conditions, or natural boundary conditions have been specified.

The general boundary contribution

For one field with action

$$S[\phi] = \int_\Omega \mathcal{L}(\phi, \partial_\mu \phi, x)\, \mathrm{d}^4 x,$$

the first variation is

$$\delta S = \int_\Omega \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \right] \delta\phi\, \mathrm{d}^4 x + \int_{\partial\Omega} \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} n_\mu \delta\phi\, \mathrm{d}\Sigma.$$

The second integral is the boundary contribution. It pairs the boundary variation $\delta\phi$ with the normal component of

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}.$$

The variational problem is well posed only when this boundary contribution is controlled.

The geometry of the boundary

For a region

$$\Omega = [t_1, t_2] \times \Sigma,$$

the boundary generally contains three parts:

1. the initial spatial surface at $t = t_1$;
2. the final spatial surface at $t = t_2$;
3. the timelike side boundary generated by $\partial\Sigma$ during the time interval.

Thus boundary conditions may be required both in time and in space.

Fixing the initial and final field configurations means

$$\delta\phi(\mathbf{x}, t_1) = 0, \qquad \delta\phi(\mathbf{x}, t_2) = 0.$$

Fixing the field on the spatial boundary means

$$\delta\phi = 0 \qquad \text{on } \partial\Sigma$$

throughout the time interval.

These conditions play different physical roles and should not be combined without explanation.

Dirichlet boundary conditions

A Dirichlet condition prescribes the field value on the boundary:

$$\phi|_{\partial\Omega} = f(x),$$

where f is given.

Every varied field must have the same boundary value, so

$$\boxed{\delta\phi|_{\partial\Omega} = 0}.$$

The boundary integral then vanishes directly.

This is the field-theoretic analogue of fixing the endpoints of a particle trajectory.

Natural boundary conditions

Suppose the field value is not fixed on part of the boundary. Then $\delta\phi$ is arbitrary there. Stationarity requires the coefficient of that arbitrary boundary variation to vanish:

$$\boxed{n_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = 0 \qquad \text{on the free part of } \partial\Omega}.$$

This is a natural boundary condition. It is derived from the action rather than imposed before the variation.

For a standard gradient energy, the natural condition is often a Neumann condition involving the normal derivative of the field.

Neumann and mixed conditions

A Neumann condition prescribes a normal derivative:

$$\frac{\partial\phi}{\partial n} = g(x) \qquad \text{on } \partial\Omega.$$

Here

$$\frac{\partial\phi}{\partial n} := n^\mu \partial_\mu \phi.$$

A mixed or Robin condition combines the field and its normal derivative:

$$a(x)\phi + b(x)\frac{\partial\phi}{\partial n} = c(x).$$

Such conditions may arise after adding an explicit boundary term to the action. The complete variational principle then includes both bulk and boundary contributions.

Explicit example: the vibrating string

Consider

$$S[u] = \int_{t_1}^{t_2} \int_0^L \left[\frac{\mu}{2}u_t^2 - \frac{T}{2}u_x^2\right] \mathrm{d}x\, \mathrm{d}t.$$

The first variation is

$$\delta S = \int_{t_1}^{t_2} \int_0^L \left[\mu u_t \partial_t(\delta u) - T u_x \partial_x(\delta u)\right] \mathrm{d}x\, \mathrm{d}t.$$

Integrating the time term by parts gives

$$
\begin{aligned}
&\int_{t_1}^{t_2}\int_0^L \mu u_t \partial_t(\delta u)\,\mathrm{d}x\,\mathrm{d}t \\
&\quad = \left[\int_0^L \mu u_t \delta u\,\mathrm{d}x\right]_{t_1}^{t_2} - \int_{t_1}^{t_2}\int_0^L \mu u_{tt}\delta u\,\mathrm{d}x\,\mathrm{d}t.
\end{aligned}
$$

Integrating the spatial term by parts gives

$$
\begin{aligned}
&-\int_{t_1}^{t_2}\int_0^L T u_x \partial_x(\delta u)\,\mathrm{d}x\,\mathrm{d}t \\
&\quad = -\int_{t_1}^{t_2} [T u_x \delta u]_0^L\,\mathrm{d}t + \int_{t_1}^{t_2}\int_0^L T u_{xx}\delta u\,\mathrm{d}x\,\mathrm{d}t.
\end{aligned}
$$

Therefore

$$
\begin{aligned}
\delta S = {}&\left[\int_0^L \mu u_t \delta u\,\mathrm{d}x\right]_{t_1}^{t_2} \\
&- \int_{t_1}^{t_2} [T u_x \delta u]_0^L\,\mathrm{d}t \\
&+ \int_{t_1}^{t_2}\int_0^L (-\mu u_{tt} + T u_{xx})\delta u\,\mathrm{d}x\,\mathrm{d}t.
\end{aligned}
$$

This formula displays the time boundary, the spatial boundary, and the interior equation separately.

Fixed initial and final configurations

If the initial and final string shapes are fixed, then

$$
\delta u(x, t_1) = 0, \qquad \delta u(x, t_2) = 0.
$$

The time-boundary contribution vanishes:

$$
\left[\int_0^L \mu u_t \delta u\,\mathrm{d}x\right]_{t_1}^{t_2} = 0.
$$

This condition does not require u_t to vanish at the initial or final time. It requires only the variation of the field value to vanish there.

Fixed string endpoints

For a string attached at both ends, the endpoint displacement is prescribed:

$$
u(0, t) = 0, \qquad u(L, t) = 0.
$$

Therefore

$$
\delta u(0, t) = 0, \qquad \delta u(L, t) = 0.
$$

The spatial boundary term vanishes independently of the value of u_x.

Free string endpoints

If an endpoint is free to move transversely, δu is arbitrary there. Stationarity then requires

$$Tu_x = 0$$

at that endpoint. For nonzero tension,

$$\boxed{u_x = 0 \qquad \text{at a free endpoint}}.$$

This condition has a direct physical interpretation. The transverse component of the tension at the endpoint must vanish.

The variational principle therefore produces both the interior wave equation and the physically appropriate natural boundary condition.

Boundary forces

An external force applied at the endpoint can be included through a boundary action. For example, a time-dependent endpoint force $F(t)$ at $x = L$ may contribute

$$S_{\text{boundary}} = \int_{t_1}^{t_2} F(t)u(L,t)\,\mathrm{d}t.$$

Its variation is

$$\delta S_{\text{boundary}} = \int_{t_1}^{t_2} F(t)\delta u(L,t)\,\mathrm{d}t.$$

Combining this with the bulk boundary term gives a force-balance condition at $x = L$. The precise sign depends on the orientation convention for the endpoint normal.

This example shows that boundary terms can encode actual interactions rather than merely mathematical corrections.

Fields on unbounded regions

If space is unbounded, the boundary is formally at spatial infinity. One must then impose fall-off conditions strong enough to make the surface integral vanish.

For example, a scalar field may be required to satisfy

$$\phi(\mathbf{x},t) \longrightarrow 0$$

and

$$|\nabla\phi| \longrightarrow 0$$

sufficiently rapidly as $|\mathbf{x}| \to \infty$.

The required rate depends on the spatial dimension and on the surface area growth. Saying only that the field becomes small is not always sufficient; the complete surface integral must tend to zero.

Adding a total divergence

If

$$\mathcal{L}' = \mathcal{L} + \partial_\mu K^\mu,$$

then

$$S' - S = \int_{\partial\Omega} K^\mu n_\mu \, d\Sigma.$$

The interior Euler–Lagrange equations are unchanged when the variation of this boundary contribution vanishes. However, the canonical boundary quantities and the natural boundary conditions may change.

Thus equivalent bulk equations do not imply that all boundary formulations are identical.

Higher-derivative actions

If the density depends on second derivatives,

$$\mathcal{L} = \mathcal{L}(\phi, \partial_\mu\phi, \partial_\mu\partial_\nu\phi),$$

then two integrations by parts are needed. The boundary contribution generally contains both $\delta\phi$ and derivatives of $\delta\phi$.

Additional boundary data are therefore required. The present chapter focuses on first-derivative densities, for which fixing ϕ on the boundary is sufficient for the standard derivation.

Diagnostic questions

Before removing a boundary term, one should ask:

1. What is the complete boundary of the integration region?
2. Are the initial and final field configurations fixed?
3. Is the field fixed on the spatial boundary?
4. If the field is free, what natural condition follows?
5. Are there external boundary interactions?
6. On an unbounded region, do the fall-off conditions make the surface integral vanish?
7. Has a total divergence changed the boundary formulation?

What has been established

The first variation of a field action contains a bulk term and a surface term. Dirichlet conditions eliminate the surface term by setting $\delta\phi = 0$ on the boundary. Free boundary values instead produce natural conditions involving the normal field momentum.

For the vibrating string, the same variation produces the interior wave equation, fixed-end conditions, or the free-end condition $u_x = 0$, depending on the admissible boundary variations.

The next section carries out the wave-equation derivation as a complete worked field-theory example.

3.6 The Wave Equation from an Action

The vibrating string provides the simplest complete example of a dynamical field equation derived from an action. Its Lagrangian density contains a time-derivative term describing inertia and a spatial-gradient term describing elastic restoring forces.

We will derive the equation, identify the boundary assumptions, verify a plane-wave solution, and interpret the coefficients.

The field and the model

Let

$$u = u(x,t)$$

be the small transverse displacement of a string whose equilibrium position is the interval

$$0 \leq x \leq L.$$

We assume:

1. the displacement is small;
2. the string has constant mass density per unit length μ;
3. the tension T is constant;
4. the motion is transverse;
5. damping and external forces are absent.

The field history is defined on

$$\Omega = [t_1, t_2] \times [0, L].$$

The Lagrangian density

The kinetic-energy density is

$$\mathcal{T} = \frac{\mu}{2} u_t^2,$$

where

$$u_t = \frac{\partial u}{\partial t}.$$

For small slopes, the elastic-energy density is

$$\mathcal{V} = \frac{T}{2} u_x^2,$$

where

$$u_x = \frac{\partial u}{\partial x}.$$

The Lagrangian density is kinetic minus potential energy density:

$$\boxed{\mathcal{L} = \frac{\mu}{2} u_t^2 - \frac{T}{2} u_x^2}.$$

The action is

$$\boxed{S[u] = \int_{t_1}^{t_2} \int_0^L \left[\frac{\mu}{2} u_t^2 - \frac{T}{2} u_x^2 \right] \, \mathrm{d}x \, \mathrm{d}t}.$$

Derivation using the Euler–Lagrange field equation

The density does not depend explicitly on u, so

$$\frac{\partial \mathcal{L}}{\partial u} = 0.$$

For the time derivative,

$$\frac{\partial \mathcal{L}}{\partial u_t} = \mu u_t.$$

Therefore

$$\frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial u_t}\right) = \mu u_{tt}.$$

For the spatial derivative,

$$\frac{\partial \mathcal{L}}{\partial u_x} = -T u_x.$$

Therefore

$$\frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial u_x}\right) = -T u_{xx}.$$

The field Euler–Lagrange equation is

$$\frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial u_t}\right) + \frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial u_x}\right) - \frac{\partial \mathcal{L}}{\partial u} = 0.$$

Substitution gives

$$\mu u_{tt} - T u_{xx} = 0.$$

Thus

$$\boxed{\mu \frac{\partial^2 u}{\partial t^2} = T \frac{\partial^2 u}{\partial x^2}}.$$

Defining

$$c^2 = \frac{T}{\mu},$$

we obtain

$$\boxed{\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}}.$$

This is the one-dimensional wave equation.

Direct variation as a check

Let

$$u_\varepsilon = u + \varepsilon \eta.$$

Then

$$\delta u = \eta, \qquad \delta u_t = \eta_t, \qquad \delta u_x = \eta_x.$$

The first variation is

$$\delta S = \int_{t_1}^{t_2} \int_0^L (\mu u_t \eta_t - T u_x \eta_x) \, \mathrm{d}x \, \mathrm{d}t.$$

After integration by parts in time and space,

$$\begin{aligned}\delta S = &\left[\int_0^L \mu u_t \eta \, \mathrm{d}x\right]_{t_1}^{t_2} \\ &- \int_{t_1}^{t_2} [T u_x \eta]_0^L \, \mathrm{d}t \\ &+ \int_{t_1}^{t_2} \int_0^L (-\mu u_{tt} + T u_{xx}) \eta \, \mathrm{d}x \, \mathrm{d}t.\end{aligned}$$

For fixed initial and final configurations and fixed string endpoints, all boundary variations vanish. Stationarity for arbitrary interior η gives

$$-\mu u_{tt} + T u_{xx} = 0,$$

which is equivalent to the equation obtained from the compact formula.

Physical interpretation

The term

$$\mu u_{tt}$$

is mass per unit length multiplied by transverse acceleration. It represents local inertia.

The term

$$T u_{xx}$$

is proportional to local curvature. A straight segment has $u_{xx} = 0$ and no net transverse restoring force. A curved segment is accelerated toward a straighter configuration.

The equation therefore expresses local force balance:

$$\text{inertial response} = \text{elastic restoring force}.$$

Dimensional check

The dimensions of tension are

$$[T] = \mathrm{N} = \mathrm{kg\,m\,s^{-2}},$$

and the dimensions of linear mass density are

$$[\mu] = \mathrm{kg\,m^{-1}}.$$

Therefore

$$\left[\frac{T}{\mu}\right] = \mathrm{m^2\,s^{-2}}.$$

Hence

$$[c] = \mathrm{m\,s^{-1}},$$

so c has the dimensions of speed.

The wave speed is

$$\boxed{c = \sqrt{\frac{T}{\mu}}}.$$

Increasing the tension increases the propagation speed, while increasing the mass density decreases it.

Plane-wave verification

Consider the complex trial solution

$$u(x,t) = Ae^{i(kx-\omega t)}.$$

Its second derivatives are

$$u_{tt} = -\omega^2 u,$$

and

$$u_{xx} = -k^2 u.$$

Substitution into

$$u_{tt} = c^2 u_{xx}$$

gives

$$-\omega^2 u = -c^2 k^2 u.$$

For a nonzero wave amplitude,

$$\boxed{\omega^2 = c^2 k^2}.$$

Thus

$$\omega = c|k|$$

for positive frequency. The phase speed is

$$\frac{\omega}{|k|} = c.$$

The real and imaginary parts of the complex exponential are real solutions because the wave equation is linear with real coefficients.

General travelling waves

The wave equation admits solutions of the form

$$u(x,t) = f(x-ct),$$

where f is twice differentiable. Indeed,

$$u_t = -cf'(x-ct), \qquad u_{tt} = c^2 f''(x-ct),$$

while

$$u_x = f'(x-ct), \qquad u_{xx} = f''(x-ct).$$

Therefore

$$u_{tt} = c^2 u_{xx}.$$

Similarly,

$$u(x,t) = g(x+ct)$$

is a left-moving solution. The general solution on an unbounded line can be written as

$$\boxed{u(x,t) = f(x-ct) + g(x+ct)}.$$

Boundary conditions restrict which combinations are allowed on a finite interval.

Boundary conditions

For fixed endpoints,

$$u(0,t) = 0, \qquad u(L,t) = 0.$$

For free endpoints, stationarity gives

$$u_x(0,t) = 0, \qquad u_x(L,t) = 0.$$

Periodic boundary conditions take the form

$$u(0,t) = u(L,t),$$

and

$$u_x(0,t) = u_x(L,t).$$

The differential equation alone does not determine a unique physical motion. Initial data and boundary data are part of the model.

Energy density

The energy density associated with the string is

$$\boxed{\mathcal{E} = \frac{\mu}{2}u_t^2 + \frac{T}{2}u_x^2}.$$

The signs differ from the Lagrangian density because energy is kinetic plus potential, whereas the Lagrangian is kinetic minus potential.

The total energy is

$$E(t) = \int_0^L \mathcal{E}(x,t)\,\mathrm{d}x.$$

Differentiating,

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \int_0^L (\mu u_t u_{tt} + T u_x u_{xt})\,\mathrm{d}x.$$

Using the field equation and integrating the second term by parts gives

$$\boxed{\frac{\mathrm{d}E}{\mathrm{d}t} = [T u_x u_t]_0^L}.$$

The change of energy inside the string is determined by energy flow through its endpoints. For fixed endpoints, $u_t = 0$ there, and the total energy is conserved.

Adding an external source

A distributed external force density $J(x,t)$ can be included through

$$\mathcal{L} = \frac{\mu}{2}u_t^2 - \frac{T}{2}u_x^2 + Ju.$$

The source is prescribed, so $\delta J = 0$. The field equation becomes

$$\boxed{\mu u_{tt} - T u_{xx} = J}.$$

The homogeneous wave equation describes free propagation. The sourced equation describes waves generated or driven by an external influence.

Common mistakes

Typical errors include:

1. using energy density instead of Lagrangian density in the action;
2. losing the minus sign in the gradient term;
3. forgetting to vary both u_t and u_x;
4. discarding time and spatial boundary terms without separate justifications;
5. assuming a plane wave is allowed without checking the boundary conditions;
6. writing $c = T/\mu$ instead of $c^2 = T/\mu$;
7. confusing the source-free and driven equations.

What has been established

The action

$$S[u] = \int\!\!\int \left[\frac{\mu}{2}u_t^2 - \frac{T}{2}u_x^2\right] \mathrm{d}x\,\mathrm{d}t$$

produces the wave equation

$$u_{tt} = c^2 u_{xx}, \qquad c^2 = \frac{T}{\mu}.$$

The derivation identifies the interior dynamics and the boundary conditions within one variational calculation. Plane waves satisfy the dispersion relation $\omega^2 = c^2k^2$, and the associated energy changes only through boundary flux.

The next section turns to static variational problems and derives the Laplace and Poisson equations from energy functionals.

3.7 Laplace and Poisson Equations

Not every variational problem describes time evolution. Static field configurations may be selected by making an energy functional stationary. The resulting Euler–Lagrange equation is then an equation in space rather than spacetime.

The Laplace and Poisson equations provide the basic examples. They appear in electrostatics, gravitation, steady heat flow, incompressible potential flow, and equilibrium elasticity.

A static energy functional

Let $\phi(\mathbf{x})$ be a real scalar field on a spatial region Σ. Consider

$$\boxed{E[\phi] = \int_\Sigma \left[\frac{\kappa}{2}|\nabla\phi|^2 - \rho(\mathbf{x})\phi\right] \mathrm{d}^3x}.$$

Here

- $\kappa > 0$ is a constant;
- $\rho(\mathbf{x})$ is a prescribed source;
- the gradient term penalizes spatial variation;
- the source term favours field values aligned with ρ.

The field ϕ is varied, while ρ is held fixed.

Variation of the gradient term

Introduce

$$\phi_\varepsilon = \phi + \varepsilon\eta.$$

Then

$$\delta\phi = \eta, \qquad \delta(\nabla\phi) = \nabla\eta.$$

The variation of the gradient contribution is

$$\delta\left(\frac{\kappa}{2}|\nabla\phi|^2\right) = \kappa\nabla\phi\cdot\nabla\eta.$$

The source contribution varies as

$$\delta(-\rho\phi) = -\rho\eta.$$

Therefore

$$\delta E = \int_\Sigma \left[\kappa\nabla\phi\cdot\nabla\eta - \rho\eta\right] \mathrm{d}^3x.$$

Spatial integration by parts

The identity

$$\nabla\cdot(\eta\nabla\phi) = \nabla\eta\cdot\nabla\phi + \eta\nabla^2\phi$$

implies

$$\nabla\phi\cdot\nabla\eta = \nabla\cdot(\eta\nabla\phi) - \eta\nabla^2\phi.$$

Thus

$$\begin{aligned}\delta E = \kappa \int_{\partial\Sigma} \eta\frac{\partial\phi}{\partial n}\,\mathrm{d}A \\ - \int_\Sigma \Big(\kappa\nabla^2\phi + ho\Big)\,\eta\,\mathrm{d}^3x.\end{aligned}$$

Here

$$\frac{\partial \phi}{\partial n} = \mathbf{n} \cdot \nabla \phi$$

is the outward normal derivative.
For fixed boundary values,

$$\eta = 0 \qquad \text{on } \partial\Sigma,$$

so the surface term vanishes. Stationarity for arbitrary interior η gives

$$\kappa \nabla^2 \phi + ho = 0.$$

Therefore

$$\boxed{\nabla^2 \phi = -\frac{\rho}{\kappa}}.$$

This is the Poisson equation for the chosen sign convention.

The Laplace equation

In a source-free region,

$$\rho = 0.$$

The field equation becomes

$$\boxed{\nabla^2 \phi = 0}.$$

This is the Laplace equation. A function satisfying it is called harmonic.
The distinction between the two equations is therefore simple:

$$\text{Poisson equation: source present,}$$

$$\text{Laplace equation: source absent.}$$

A field may satisfy the Poisson equation in a region containing sources and the Laplace equation in neighbouring source-free regions.

Electrostatic interpretation

For an electrostatic potential Φ, choose

$$\kappa = \varepsilon_0$$

and let ρ denote the electric charge density. The functional is

$$E[\Phi] = \int_\Sigma \left[\frac{\varepsilon_0}{2} |\nabla \Phi|^2 - \rho \Phi \right] \mathrm{d}^3 x.$$

Stationarity gives

$$\boxed{\nabla^2 \Phi = -\frac{\rho}{\varepsilon_0}}.$$

Using

$$\mathbf{E} = -\nabla \Phi,$$

we find

$$\nabla \cdot \mathbf{E} = -\nabla^2 \Phi = \frac{\rho}{\varepsilon_0},$$

which is Gauss's law in differential form.

This provides an early example of a familiar field equation emerging from a variational functional.

Natural boundary condition

If the boundary value of ϕ is free, then η is arbitrary on $\partial\Sigma$. Stationarity requires

$$\boxed{\kappa \frac{\partial \phi}{\partial n} = 0 \qquad \text{on the free boundary}}.$$

For nonzero κ,

$$\frac{\partial \phi}{\partial n} = 0.$$

This is the homogeneous Neumann condition. It says that the field has no normal gradient through the boundary.

A nonzero prescribed normal derivative can be generated by adding an appropriate boundary source term to the functional.

Dirichlet and Neumann data

Two common boundary-value problems are:

1. Dirichlet problem: specify ϕ on $\partial\Sigma$;
2. Neumann problem: specify $\partial\phi/\partial n$ on $\partial\Sigma$.

For the Dirichlet problem, admissible variations vanish on the boundary. For the Neumann problem, the field value may vary, but the normal derivative is constrained.

A pure Neumann problem determines ϕ only up to an additive constant, because

$$\nabla^2(\phi + C) = \nabla^2 \phi.$$

Additional information, such as the average value of ϕ, is needed to fix that constant.

Compatibility condition for a Neumann problem

Integrate the Poisson equation over Σ:

$$\int_\Sigma \nabla^2 \phi \, \mathrm{d}^3x = -\frac{1}{\kappa} \int_\Sigma \rho \, \mathrm{d}^3x.$$

By the divergence theorem,

$$\int_{\partial\Sigma} \frac{\partial \phi}{\partial n} \, \mathrm{d}A = -\frac{1}{\kappa} \int_\Sigma \rho \, \mathrm{d}^3x.$$

Thus prescribed Neumann data cannot be arbitrary. Their total flux must be compatible with the total source inside the region.

For homogeneous Neumann data,

$$\frac{\partial \phi}{\partial n} = 0,$$

the compatibility condition is

$$\int_\Sigma \rho \, \mathrm{d}^3 x = 0.$$

One-dimensional worked example

Consider the interval

$$0 \le x \le L$$

with constant source ρ_0. The Poisson equation is

$$\frac{\mathrm{d}^2 \phi}{\mathrm{d}x^2} = -\frac{\rho_0}{\kappa}.$$

Integrating once,

$$\frac{\mathrm{d}\phi}{\mathrm{d}x} = -\frac{\rho_0}{\kappa} x + C_1.$$

Integrating again,

$$\phi(x) = -\frac{\rho_0}{2\kappa} x^2 + C_1 x + C_2.$$

Impose Dirichlet conditions

$$\phi(0) = 0, \qquad \phi(L) = 0.$$

The first condition gives

$$C_2 = 0.$$

The second gives

$$0 = -\frac{\rho_0}{2\kappa} L^2 + C_1 L,$$

so

$$C_1 = \frac{\rho_0 L}{2\kappa}.$$

Therefore

$$\boxed{\phi(x) = \frac{\rho_0}{2\kappa} x (L - x)}.$$

The solution is zero at both endpoints and reaches an extremum in the middle of the interval.

Verification of the worked solution

Differentiate twice:

$$\phi'(x) = \frac{\rho_0}{2\kappa} (L - 2x),$$

and

$$\phi''(x) = -\frac{\rho_0}{\kappa}.$$

Thus the Poisson equation is satisfied. Also,

$$\phi(0) = 0, \qquad \phi(L) = 0.$$

Both the differential equation and the boundary conditions have been checked.

Uniqueness for the Dirichlet problem

Suppose ϕ_1 and ϕ_2 satisfy the same Poisson equation and the same Dirichlet boundary data. Define

$$w = \phi_1 - \phi_2.$$

Then

$$\nabla^2 w = 0$$

and

$$w = 0 \qquad \text{on } \partial\Sigma.$$

Using integration by parts,

$$\int_\Sigma |\nabla w|^2 \, \mathrm{d}^3x = \int_{\partial\Sigma} w \frac{\partial w}{\partial n} \, \mathrm{d}A - \int_\Sigma w \nabla^2 w \, \mathrm{d}^3x.$$

Both terms on the right vanish, so

$$\int_\Sigma |\nabla w|^2 \, \mathrm{d}^3x = 0.$$

Therefore

$$\nabla w = 0,$$

so w is constant. Since $w = 0$ on the boundary,

$$w = 0$$

throughout the region. Hence

$$\boxed{\phi_1 = \phi_2}.$$

The Dirichlet solution is unique under the stated regularity assumptions.

Minimum-energy interpretation

Let ϕ_* satisfy the Poisson equation with fixed boundary values. Consider

$$\phi = \phi_* + \eta, \qquad \eta|_{\partial\Sigma} = 0.$$

Expanding the functional gives

$$E[\phi] = E[\phi_*] + \delta E[\phi_*; \eta] + \frac{\kappa}{2} \int_\Sigma |\nabla \eta|^2 \, \mathrm{d}^3x.$$

The first variation vanishes because ϕ_* is stationary. Since $\kappa > 0$,

$$E[\phi] - E[\phi_*] = \frac{\kappa}{2} \int_\Sigma |\nabla \eta|^2 \, \mathrm{d}^3x \geq 0.$$

Thus the stationary solution is a minimum of the functional for fixed Dirichlet data. This is stronger than stationarity and follows from the positive quadratic gradient term.

Relation to time-dependent field theory

The static functional may be obtained from a dynamical density by restricting attention to time-independent fields. For

$$\mathcal{L} = \frac{\rho_m}{2}\phi_t^2 - \frac{\kappa}{2}|\nabla\phi|^2 + \rho\phi,$$

the field equation is

$$\rho_m\phi_{tt} - \kappa\nabla^2\phi = \rho.$$

For a static configuration,

$$\phi_{tt} = 0,$$

so

$$\nabla^2\phi = -\frac{\rho}{\kappa}.$$

The Poisson equation is therefore the equilibrium limit of a broader dynamical theory.

Common mistakes

Typical errors include:

1. confusing the energy functional with a time-dependent action;
2. losing the minus sign produced by spatial integration by parts;
3. varying the prescribed source ρ;
4. solving the differential equation without imposing boundary conditions;
5. assuming a Neumann solution is unique without fixing its additive constant;
6. ignoring the compatibility condition for prescribed flux;
7. failing to verify both the equation and the boundary data.

What has been established

The static functional

$$E[\phi] = \int_\Sigma \left[\frac{\kappa}{2}|\nabla\phi|^2 - \rho\phi\right] \mathrm{d}^3x$$

produces the Poisson equation

$$\nabla^2\phi = -\frac{\rho}{\kappa}.$$

In source-free regions this reduces to the Laplace equation. Boundary variations distinguish Dirichlet, Neumann, and natural boundary conditions. For fixed Dirichlet data and $\kappa > 0$, the stationary solution is unique and minimizes the energy functional.

The final section adds local potentials and interactions, derives nonlinear field equations, and introduces canonical field momentum and the Hamiltonian density.

3.8 Fields with Potentials and Interactions

A gradient term couples neighbouring points, but it does not determine how the field behaves locally when spatial derivatives vanish. Local behaviour is encoded by a potential $V(\phi)$. A source term describes the influence of a prescribed external distribution, while mixed potentials describe interactions among several fields.

This section derives the resulting nonlinear equations, studies equilibrium and small perturbations, and introduces canonical field momentum and the Hamiltonian density.

A general scalar-field density

Consider

$$\boxed{\mathcal{L} = \frac{\rho}{2}\phi_t^2 - \frac{\kappa}{2}|\nabla\phi|^2 - V(\phi) + J(x)\phi}.$$

We assume

$$\rho > 0, \qquad \kappa > 0,$$

and treat $J(x)$ as a prescribed source.

The Euler–Lagrange equation is

$$\boxed{\rho\phi_{tt} - \kappa\nabla^2\phi + V'(\phi) = J}.$$

Each term has a clear role:

- $\rho\phi_{tt}$ is the inertial response;
- $-\kappa\nabla^2\phi$ is the spatial restoring contribution;
- $V'(\phi)$ is the local force generated by the potential;
- J drives the field externally.

The equation is nonlinear whenever $V'(\phi)$ is nonlinear in ϕ.

Derivation term by term

The derivative with respect to the time derivative is

$$\frac{\partial\mathcal{L}}{\partial\phi_t} = \rho\phi_t.$$

Therefore

$$\frac{\partial}{\partial t}\left(\frac{\partial\mathcal{L}}{\partial\phi_t}\right) = \rho\phi_{tt}.$$

For the spatial derivatives,

$$\frac{\partial\mathcal{L}}{\partial(\partial_i\phi)} = -\kappa\partial_i\phi,$$

so

$$\sum_i \partial_i\left(\frac{\partial\mathcal{L}}{\partial(\partial_i\phi)}\right) = -\kappa\nabla^2\phi.$$

Finally,

$$\frac{\partial \mathcal{L}}{\partial \phi} = -V'(\phi) + J.$$

Substitution into the field Euler–Lagrange equation gives the stated result.

Uniform equilibrium fields

A uniform static field satisfies

$$\phi_t = 0, \qquad \nabla \phi = 0.$$

The field equation reduces to

$$\boxed{V'(\phi_0) = J}.$$

In the source-free case,

$$J = 0,$$

uniform equilibria are stationary points of the potential:

$$V'(\phi_0) = 0.$$

The second derivative distinguishes their local character:

$$V''(\phi_0) > 0$$

indicates a local minimum and usually stable small oscillations, while

$$V''(\phi_0) < 0$$

indicates a local maximum and instability.

Quadratic potential

Let

$$V(\phi) = \frac{m^2}{2}\phi^2.$$

Then

$$V'(\phi) = m^2\phi.$$

The field equation is

$$\boxed{\rho\phi_{tt} - \kappa\nabla^2\phi + m^2\phi = J}.$$

When $J = 0$, this is a linear wave equation with a local restoring term. The parameter m^2 controls the strength of the restoring force.

For a spatially uniform field,

$$\nabla \phi = 0,$$

the equation becomes

$$\rho\phi_{tt} + m^2\phi = 0.$$

Thus every spatial point behaves locally like a harmonic oscillator when gradient coupling is absent.

Plane-wave check for the quadratic theory

For $J = 0$, try

$$\phi(\mathbf{x}, t) = Ae^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}.$$

Then

$$\phi_{tt} = -\omega^2\phi, \qquad \nabla^2\phi = -|\mathbf{k}|^2\phi.$$

Substitution gives

$$-\rho\omega^2 + \kappa|\mathbf{k}|^2 + m^2 = 0.$$

Therefore

$$\boxed{\omega^2 = \frac{\kappa}{\rho}|\mathbf{k}|^2 + \frac{m^2}{\rho}}.$$

At zero wave number,

$$\omega^2(0) = \frac{m^2}{\rho}.$$

The field can therefore oscillate even when it is spatially uniform.

A quartic interaction

Consider

$$V(\phi) = \frac{m^2}{2}\phi^2 + \frac{\lambda}{4}\phi^4,$$

with $\lambda > 0$. Then

$$V'(\phi) = m^2\phi + \lambda\phi^3.$$

The source-free field equation is

$$\boxed{\rho\phi_{tt} - \kappa\nabla^2\phi + m^2\phi + \lambda\phi^3 = 0}.$$

The cubic term in the equation makes the theory nonlinear. If ϕ_1 and ϕ_2 are solutions, their sum is generally not a solution.

The quartic term also makes the potential grow positively for large $|\phi|$, so the local potential energy is bounded below when $\lambda > 0$.

Double-well potential

A useful nonlinear example is

$$\boxed{V(\phi) = \frac{\lambda}{4}(\phi^2 - v^2)^2},$$

where $\lambda > 0$ and $v > 0$.

Its derivative is

$$V'(\phi) = \lambda\phi(\phi^2 - v^2).$$

The source-free field equation is

$$\boxed{\rho\phi_{tt} - \kappa\nabla^2\phi + \lambda\phi(\phi^2 - v^2) = 0}.$$

Uniform equilibria satisfy

$$\lambda\phi_0(\phi_0^2 - v^2) = 0.$$

Therefore

$$\phi_0 = 0, \qquad \phi_0 = v, \qquad \phi_0 = -v.$$

The second derivative is

$$V''(\phi) = \lambda(3\phi^2 - v^2).$$

At $\phi_0 = 0$,

$$V''(0) = -\lambda v^2 < 0,$$

so the central equilibrium is unstable. At $\phi_0 = \pm v$,

$$V''(\pm v) = 2\lambda v^2 > 0,$$

so the two outer equilibria are stable against sufficiently small uniform perturbations.

Linearization around an equilibrium

Let ϕ_0 be a constant equilibrium satisfying

$$V'(\phi_0) = J_0.$$

Write

$$\phi(x) = \phi_0 + \psi(x)$$

and

$$J(x) = J_0 + \delta J(x),$$

where ψ and δJ are small.
Expand the potential derivative:

$$V'(\phi_0 + \psi) = V'(\phi_0) + V''(\phi_0)\psi + O(\psi^2).$$

Substitution into the field equation gives

$$\rho\psi_{tt} - \kappa\nabla^2\psi + V''(\phi_0)\psi = \delta J + O(\psi^2).$$

Keeping only first-order terms,

$$\boxed{\rho\psi_{tt} - \kappa\nabla^2\psi + V''(\phi_0)\psi = \delta J}.$$

Thus small perturbations around a nonlinear equilibrium obey an approximately linear field equation. The curvature $V''(\phi_0)$ determines the local restoring strength.

Stability from the linearized equation

For a source-free spatially uniform perturbation,

$$\rho\psi_{tt} + V''(\phi_0)\psi = 0.$$

If

$$V''(\phi_0) > 0,$$

the perturbation oscillates with frequency

$$\omega_0^2 = \frac{V''(\phi_0)}{\rho}.$$

If

$$V''(\phi_0) < 0,$$

the equation has exponentially growing solutions. The equilibrium is linearly unstable. This connects the geometry of the potential with dynamical stability.

Interactions among several fields

For two fields, a general local potential is

$$V = V(\phi, \chi).$$

The density

$$\mathcal{L} = \frac{1}{2}\phi_t^2 - \frac{c_\phi^2}{2}|\nabla\phi|^2 + \frac{1}{2}\chi_t^2 - \frac{c_\chi^2}{2}|\nabla\chi|^2 - V(\phi, \chi)$$

produces

$$\boxed{\phi_{tt} - c_\phi^2\nabla^2\phi + \frac{\partial V}{\partial \phi} = 0},$$

and

$$\boxed{\chi_{tt} - c_\chi^2\nabla^2\chi + \frac{\partial V}{\partial \chi} = 0}.$$

A mixed potential such as

$$V_{\text{int}} = g\phi^2\chi^2$$

contributes

$$\frac{\partial V_{\text{int}}}{\partial \phi} = 2g\phi\chi^2,$$

and

$$\frac{\partial V_{\text{int}}}{\partial \chi} = 2g\phi^2\chi.$$

Each field changes the effective local force acting on the other.

Energy density

For the density

$$\mathcal{L} = \frac{\rho}{2}\phi_t^2 - \frac{\kappa}{2}|\nabla\phi|^2 - V(\phi) + J\phi,$$

the corresponding energy density is

$$\boxed{\mathcal{E} = \frac{\rho}{2}\phi_t^2 + \frac{\kappa}{2}|\nabla\phi|^2 + V(\phi) - J\phi}.$$

The kinetic and gradient contributions are positive when $\rho > 0$ and $\kappa > 0$. Stability also depends on whether the effective potential

$$V_{\text{eff}}(\phi, x) = V(\phi) - J(x)\phi$$

is bounded below.

If the source depends explicitly on time, the field can exchange energy with the external system, and the field energy need not be conserved.

Canonical field momentum

In mechanics, the momentum conjugate to q is

$$p = \frac{\partial L}{\partial \dot{q}}.$$

For a field, the canonical momentum density conjugate to ϕ is defined by

$$\boxed{\pi(\mathbf{x}, t) := \frac{\partial \mathcal{L}}{\partial \phi_t}}.$$

For the general scalar density,

$$\pi = \rho\phi_t.$$

Thus the canonical momentum is itself a field. It assigns a momentum density to every spatial point.

For several fields,

$$\pi_a = \frac{\partial \mathcal{L}}{\partial(\partial_t\phi^a)}.$$

Hamiltonian density

The Hamiltonian density is defined by the field-theoretic Legendre transform

$$\boxed{\mathcal{H} = \pi\phi_t - \mathcal{L}}.$$

For

$$\pi = \rho\phi_t,$$

we obtain

$$\begin{aligned}\mathcal{H} &= \rho\phi_t^2 - \left[\frac{\rho}{2}\phi_t^2 - \frac{\kappa}{2}|\nabla\phi|^2 - V(\phi) + J\phi\right] \\ &= \frac{\rho}{2}\phi_t^2 + \frac{\kappa}{2}|\nabla\phi|^2 + V(\phi) - J\phi.\end{aligned}$$

Therefore

$$\boxed{\mathcal{H} = \frac{\pi^2}{2\rho} + \frac{\kappa}{2}|\nabla\phi|^2 + V(\phi) - J\phi}.$$

For this standard theory, the Hamiltonian density equals the energy density.
The Hamiltonian functional is

$$\boxed{H[\phi, \pi] = \int_\Sigma \mathcal{H}\, \mathrm{d}^3x}.$$

A full Hamiltonian formulation will not be developed here. The important introductory point is that both the field value and its conjugate momentum are spatially distributed variables.

When the Legendre transform fails

The definition of $\mathcal{H}$ requires that the relation

$$\pi = \frac{\partial \mathcal{L}}{\partial \phi_t}$$

can be inverted to express ϕ_t in terms of ϕ and π.
For

$$\pi = \rho\phi_t$$

with $\rho \neq 0$, inversion is immediate:

$$\phi_t = \frac{\pi}{\rho}.$$

In theories with constraints or gauge redundancy, this inversion may fail. The Hamiltonian formulation then requires a separate constraint analysis. Electromagnetism will later provide an important example.

Diagnostic checks

For a theory with a potential or interaction, one should ask:

1. Is the potential differentiated with the correct sign?
2. Is the source dynamical or prescribed?
3. Is the potential bounded below?
4. Which constant fields satisfy the equilibrium equation?
5. What is the sign of V'' at each equilibrium?
6. Does linearization reproduce the expected small-oscillation equation?
7. Is the canonical momentum definition invertible?
8. Does the Hamiltonian density have the expected energy signs?

Common mistakes

Typical errors include:

1. writing $V(\phi)$ rather than $V'(\phi)$ in the field equation;
2. changing the sign of the potential contribution incorrectly;
3. applying superposition to a nonlinear equation;
4. calling every stationary point stable;
5. linearizing without stating which terms are neglected;
6. varying an external source that should remain fixed;
7. identifying $\mathcal{L}$ with the energy density;
8. forgetting that canonical momentum is a density field.

Chapter summary

This chapter extended the action principle from particle trajectories to fields. The field action is

$$S[\phi] = \int_\Omega \mathcal{L}(\phi, \partial_\mu\phi, x)\, \mathrm{d}^4x.$$

A varied field is written as

$$\phi_\varepsilon = \phi + \varepsilon\eta,$$

and its derivative varies according to

$$\delta(\partial_\mu\phi) = \partial_\mu(\delta\phi).$$

Spacetime integration by parts separates the first variation into an interior term and a boundary term. For fixed boundary values, stationarity gives

$$\boxed{\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right) - \frac{\partial\mathcal{L}}{\partial\phi} = 0}.$$

The same method produced coupled equations for several fields, the wave equation for a vibrating string, and the Laplace and Poisson equations from a static energy functional. Local potentials produced nonlinear equations, equilibrium conditions, and linearized small-perturbation theories.

Finally, the canonical momentum density and Hamiltonian density were identified as

$$\pi = \frac{\partial\mathcal{L}}{\partial\phi_t}, \qquad \mathcal{H} = \pi\phi_t - \mathcal{L}.$$

The practical objective of the chapter has therefore been achieved: given a Lagrangian density, one can define admissible variations, control the boundary term, derive the field equation, and interpret the roles of kinetic, gradient, source, potential, and interaction terms.

The next chapter introduces Minkowski spacetime and Lorentz-covariant notation, allowing these field equations to be written in a form adapted to relativistic symmetry.

Chapter 4

Spacetime, Lorentz Symmetry, and Covariant Notation

4.1 Minkowski Spacetime and Events

Classical field theory describes quantities defined throughout space and time. In a relativistic theory, however, space and time cannot be treated as two unrelated backgrounds. Observers in relative motion disagree about spatial distances and time intervals, yet they agree on the speed of light and on the causal order of events that can influence one another.

The appropriate arena is Minkowski spacetime. Its basic elements are events, inertial coordinate systems, and worldlines. The metric structure that measures spacetime intervals will be introduced in the next section; here we first establish the geometric and physical objects that the metric will act upon.

Events as points in spacetime

An event is something that occurs at one place and one time. Examples include

- the emission of a light pulse;
- a particle passing a detector;
- two particles colliding;
- a field reaching a specified value at a specified location.

An observer using an inertial coordinate system assigns four coordinates to an event:

$$(t, x, y, z).$$

It is convenient to combine them into the spacetime coordinate array

$$\boxed{x^\mu = (x^0, x^1, x^2, x^3) = (ct, x, y, z)}.$$

The index

$$\mu = 0, 1, 2, 3$$

labels the four spacetime components. The zeroth coordinate is

$$x^0 = ct,$$

so all four components have the dimensions of length.

At this stage, the upper index on x^μ is part of the notation for a contravariant four-vector. The distinction between upper and lower indices will be developed systematically later in the chapter.

Why use ct rather than t?

Time and space have different physical units. Multiplying time by the speed of light gives

$$[ct] = \mathrm{m},$$

so the coordinates

$$ct, \qquad x, \qquad y, \qquad z$$

can be compared within one geometric expression.

This convention does not claim that time is identical to space. The distinction is encoded by the spacetime metric, which assigns a different sign to the temporal component than to the spatial components.

If natural units are later adopted, one sets

$$c = 1,$$

and then writes

$$x^\mu = (t, \mathbf{x}).$$

Until the final section of this chapter, the factor c will usually remain explicit.

Inertial reference frames

An inertial reference frame is a coordinate system in which a free particle moves with constant velocity. If no force acts, its spatial trajectory has the form

$$\mathbf{x}(t) = \mathbf{x}_0 + \mathbf{v}t.$$

Two inertial frames may move uniformly relative to one another. Their coordinates are related by a Lorentz transformation rather than by the Galilean transformation of nonrelativistic mechanics.

The two postulates underlying special relativity are:

1. the laws of physics have the same form in every inertial frame;
2. the speed of light in vacuum has the same value c in every inertial frame.

The second statement forces a modification of the classical relation between space and time. If a light pulse satisfies

$$|\Delta\mathbf{x}| = c\Delta t$$

in one inertial frame, it must satisfy the corresponding relation in every inertial frame.

Coordinate descriptions and physical events

The event itself is independent of coordinates. Different observers assign different coordinate quadruples to the same event.

Suppose observer S assigns

$$x^\mu = (ct, x, y, z)$$

to an event, while observer S' assigns

$$x'^\mu = (ct', x', y', z').$$

The coordinates differ, but the event is the same point of spacetime. A Lorentz transformation relates the two descriptions:

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}.$$

The matrix $\Lambda^{\mu}{}_{\nu}$ will be constructed explicitly in the section on Lorentz transformations.

This distinction is fundamental:

> Coordinates describe an event; they are not the event itself.

The same principle applies to fields. A scalar field assigns one physical value to an event, while its coordinate formula changes when the coordinate system changes.

Worldlines

The history of a point particle is represented by a curve in spacetime called its worldline. In a chosen inertial frame, the worldline may be written as

$$x^{\mu}(t) = \big(ct, \mathbf{x}(t)\big).$$

A particle at rest at spatial position $\mathbf{x}_0$ has

$$\mathbf{x}(t) = \mathbf{x}_0,$$

so its worldline is parallel to the time axis.

A particle moving uniformly with velocity $\mathbf{v}$ has

$$\mathbf{x}(t) = \mathbf{x}_0 + \mathbf{v}t.$$

In a one-dimensional spacetime diagram with horizontal axis x and vertical axis ct, the worldline is a straight line. Its slope satisfies

$$\frac{\Delta(ct)}{\Delta x} = \frac{c}{v}.$$

For a material particle,

$$|v| < c,$$

so its worldline is steeper than a light ray in such a diagram.

Light rays and the light cone

A light signal moving in one spatial dimension obeys

$$\Delta x = \pm c \Delta t.$$

Since

$$\Delta x^0 = c\Delta t,$$

this becomes

$$\Delta x = \pm \Delta x^0.$$

In a spacetime diagram, light rays form lines at 45° when the axes ct and x are drawn with the same scale.

In three spatial dimensions, the set of all light signals emitted from one event forms a light cone. The cone divides spacetime into four regions:

- the future light cone;
- the past light cone;
- the region outside the cone in spatial directions;
- the cone itself.

An event inside the future light cone can be influenced by a signal emitted at the original event without exceeding the speed of light. An event inside the past light cone can influence the original event. Events outside the cone cannot be connected to the original event by a material particle or signal moving at or below c.

Causal classification without the metric notation

Let two events have coordinate differences

$$\Delta t = t_2 - t_1, \qquad \Delta \mathbf{x} = \mathbf{x}_2 - \mathbf{x}_1.$$

Compare

$$c^2(\Delta t)^2$$

with

$$|\Delta \mathbf{x}|^2.$$

Three cases occur.

1. If
$$c^2(\Delta t)^2 > |\Delta \mathbf{x}|^2,$$
the separation is timelike. A massive particle moving slower than light can travel from one event to the other.

2. If
$$c^2(\Delta t)^2 = |\Delta \mathbf{x}|^2,$$
the separation is null or lightlike. Only a signal moving at speed c can connect the events.

3. If
$$c^2(\Delta t)^2 < |\Delta \mathbf{x}|^2,$$
the separation is spacelike. Connecting the events would require speed greater than c.

The next section will combine these three comparisons into one invariant quantity called the spacetime interval.

Worked example: can one event influence another?

Event A occurs at

$$t_A = 0, \qquad x_A = 0.$$

Event B occurs at

$$t_B = 2.0\,\mu\text{s}, \qquad x_B = 400\,\text{m}.$$

The coordinate differences are

$$\Delta t = 2.0 \times 10^{-6}\,\mathrm{s},$$

and

$$\Delta x = 400\,\mathrm{m}.$$

A light signal travels during this time the distance

$$c\Delta t = (3.00 \times 10^{8}\,\mathrm{m\,s^{-1}})(2.0 \times 10^{-6}\,\mathrm{s}) = 600\,\mathrm{m}.$$

Since

$$|\Delta x| = 400\,\mathrm{m} < 600\,\mathrm{m} = c\Delta t,$$

the separation is timelike. A signal moving at speed

$$v = \frac{\Delta x}{\Delta t} = 2.0 \times 10^{8}\,\mathrm{m\,s^{-1}}$$

can connect the events, and this speed satisfies

$$v < c.$$

Therefore event A can causally influence event B.

Worked example: spacelike separation

Suppose two events satisfy

$$\Delta t = 1.0\,\mu\mathrm{s}, \qquad |\Delta x| = 500\,\mathrm{m}.$$

Light travels only

$$c\Delta t = 300\,\mathrm{m}$$

during this time. The speed required to connect the events would be

$$v = \frac{500\,\mathrm{m}}{1.0 \times 10^{-6}\,\mathrm{s}} = 5.0 \times 10^{8}\,\mathrm{m\,s^{-1}},$$

which is greater than c. The separation is spacelike, so no causal influence can travel from one event to the other within special relativity.

Simultaneity is frame dependent

Two events are simultaneous in a frame S if

$$\Delta t = 0.$$

For events separated in space, a Lorentz transformation generally gives

$$\Delta t' \neq 0.$$

Thus distant simultaneity is not an observer-independent relation. This does not create a causal contradiction because spacelike-separated events cannot influence one another.

Events occurring at the same spacetime point are different. If

$$\Delta \mathbf{x} = 0$$

and

$$\Delta t = 0,$$

then the two descriptions refer to one event, and every inertial observer agrees that the events coincide.

Fields on Minkowski spacetime

A scalar field is a function

$$\phi : \mathbb{M}^{1,3} \longrightarrow \mathbb{R},$$

where $\mathbb{M}^{1,3}$ denotes four-dimensional Minkowski spacetime.

In coordinates, one writes

$$\phi = \phi(x^\mu) = \phi(ct, x, y, z).$$

A vector field assigns a four-vector to every event:

$$A^\mu = A^\mu(x).$$

The central requirement of relativistic field theory is that physical laws should be expressible in a form that is valid in every inertial frame. Covariant notation is the language used to make this requirement visible.

Structural checks

A spacetime description should pass several immediate checks.

1. Every event must be assigned four coordinates in a chosen frame.
2. The zeroth coordinate $x^0 = ct$ must have the same dimensions as the spatial coordinates.
3. A material worldline must remain inside the light cone.
4. A light ray must lie on the light cone.
5. A causal conclusion must not depend on one particular inertial coordinate system.

Common mistakes

Typical errors include:

1. identifying an event with its coordinates rather than treating coordinates as one description of the event;
2. using t as the zeroth component while treating all components as having units of length;
3. drawing a light ray at 45° when the time and space axes use different scales;
4. assuming that all observers agree on distant simultaneity;
5. confusing a worldline with the spatial path seen at one time;
6. treating spacelike separation as merely a large distance rather than a comparison between distance and elapsed time.

What has been established

Minkowski spacetime is the arena of special-relativistic field theory. Its points are events, and an inertial observer assigns coordinates

$$x^\mu = (ct, \mathbf{x}).$$

A particle traces a worldline, light signals generate the light cone, and pairs of events are classified as timelike, null, or spacelike according to whether their spatial separation is smaller than, equal to, or greater than the distance light can travel during the elapsed time.

The next section introduces the Minkowski metric, combines this causal classification into the invariant spacetime interval, and defines proper time.

4.2 Spacetime Interval and Metric

Different inertial observers assign different coordinate differences to the same pair of events. A relativistic geometry therefore requires a quantity that combines temporal and spatial separations and has the same value in every inertial frame.

That quantity is the spacetime interval. It is defined using the Minkowski metric, which also establishes the sign convention used throughout the remainder of these notes.

The metric convention

We adopt the mostly-minus signature

$$\boxed{\eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)}.$$

In matrix form,

$$\eta_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

The symbol $\eta_{\mu\nu}$ denotes the metric of flat Minkowski spacetime. Greek indices run over

$$\mu, \nu = 0, 1, 2, 3.$$

The temporal component carries a positive sign, while each spatial component carries a negative sign.

This convention must be used consistently. A text using the opposite signature

$$(-, +, +, +)$$

will display sign changes in norms, raised derivatives, the d'Alembert operator, and some Lagrangian densities, even though the physical predictions are unchanged.

The spacetime interval

For an infinitesimal displacement

$$\mathrm{d}x^\mu = (c\,\mathrm{d}t, \mathrm{d}x, \mathrm{d}y, \mathrm{d}z),$$

the spacetime interval is

$$\boxed{\mathrm{d}s^2 = \eta_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu}.$$

Expanding the components gives

$$\boxed{\mathrm{d}s^2 = c^2\mathrm{d}t^2 - \mathrm{d}x^2 - \mathrm{d}y^2 - \mathrm{d}z^2}.$$

Using

$$\mathrm{d}\mathbf{x}^2 = \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2,$$

we may write

$$\mathrm{d}s^2 = c^2\mathrm{d}t^2 - \mathrm{d}\mathbf{x}^2.$$

For two finite events, define

$$\Delta x^\mu = x_2^\mu - x_1^\mu.$$

Their squared separation is

$$\boxed{\Delta s^2 = \eta_{\mu\nu}\Delta x^\mu \Delta x^\nu = c^2(\Delta t)^2 - |\Delta\mathbf{x}|^2}.$$

Why the interval is not an ordinary Euclidean distance

In Euclidean space, the squared distance between nearby points is

$$\mathrm{d}\ell^2 = \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2.$$

Every nonzero Euclidean squared distance is positive. In Minkowski spacetime, by contrast,

$$\mathrm{d}s^2 = c^2\mathrm{d}t^2 - \mathrm{d}\mathbf{x}^2$$

may be positive, zero, or negative.

This indefinite sign is not a technical inconvenience. It encodes causal structure. Temporal and spatial directions are geometrically different, and the sign of Δs^2 determines which events can be causally connected.

Timelike, null, and spacelike separations

With the chosen signature, the three cases are:

1. Timelike separation:
$$\boxed{\Delta s^2 > 0}.$$
   Then
$$c^2(\Delta t)^2 > |\Delta\mathbf{x}|^2.$$
   A massive particle moving slower than light can connect the events.

2. Null or lightlike separation:
$$\boxed{\Delta s^2 = 0}.$$
   Then
$$|\Delta\mathbf{x}| = c|\Delta t|.$$
   A light signal can connect the events.

3. Spacelike separation:

$$\boxed{\Delta s^2 < 0}.$$

   Then

$$|\Delta \mathbf{x}| > c|\Delta t|.$$

   Connecting the events would require a signal faster than light.

The sign assignments reverse when the opposite metric signature is used. The causal classification itself does not change.

Lorentz invariance of the interval

A Lorentz transformation is a linear coordinate transformation

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}$$

that preserves the Minkowski metric:

$$\boxed{\Lambda^{\mathsf{T}} \eta \Lambda = \eta}.$$

Therefore

$$\begin{aligned}
\Delta s'^2 &= \eta_{\mu\nu} \Delta x'^{\mu} \Delta x'^{\nu} \\
&= \eta_{\mu\nu} \Lambda^{\mu}{}_{\alpha} \Lambda^{\nu}{}_{\beta} \Delta x^{\alpha} \Delta x^{\beta} \\
&= \eta_{\alpha\beta} \Delta x^{\alpha} \Delta x^{\beta} \\
&= \Delta s^2.
\end{aligned}$$

Thus

$$\boxed{\Delta s'^2 = \Delta s^2}.$$

The explicit form of $\Lambda^{\mu}{}_{\nu}$ and a direct component verification will be given later. Here the defining geometric property is already visible: Lorentz transformations preserve the interval.

Lowering the coordinate index

The metric converts a contravariant coordinate into a covariant coordinate:

$$x_{\mu} = \eta_{\mu\nu} x^{\nu}.$$

For

$$x^{\mu} = (ct, x, y, z),$$

we obtain

$$\boxed{x_{\mu} = (ct, -x, -y, -z)}.$$

The interval may then be written compactly as

$$\Delta s^2 = \Delta x_{\mu} \Delta x^{\mu}.$$

This is a contraction: the repeated index appears once upstairs and once downstairs and is summed over all four values.

The procedure of raising and lowering general vector indices will be developed fully in the next section.

Proper time

For a timelike worldline, the proper time τ is defined by

$$\boxed{c^2 \mathrm{d}\tau^2 = \mathrm{d}s^2}.$$

Using

$$\mathrm{d}\mathbf{x} = \mathbf{v}\,\mathrm{d}t,$$

we find

$$\begin{aligned} c^2 \mathrm{d}\tau^2 &= c^2 \mathrm{d}t^2 - |\mathbf{v}|^2 \mathrm{d}t^2 \\ &= c^2 \mathrm{d}t^2 \left(1 - \frac{v^2}{c^2}\right). \end{aligned}$$

Therefore

$$\boxed{\mathrm{d}\tau = \mathrm{d}t \sqrt{1 - \frac{v^2}{c^2}}}.$$

Define the Lorentz factor

$$\boxed{\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}}.$$

Then

$$\boxed{\mathrm{d}t = \gamma\,\mathrm{d}\tau}.$$

Proper time is the time measured by a clock moving along the worldline. It is invariant because it is constructed from the invariant interval.

Coordinate time and proper time

If a clock is at rest in an inertial frame, then

$$\mathrm{d}\mathbf{x} = 0,$$

so

$$\mathrm{d}\tau = \mathrm{d}t.$$

If the clock moves with speed v, then

$$\mathrm{d}\tau < \mathrm{d}t$$

for $v \neq 0$. The moving clock accumulates less proper time between the same two events than the coordinate time measured in that frame.

This is time dilation expressed geometrically. The statement follows directly from the interval rather than from a separate dynamical assumption.

Worked example: proper time of a moving particle

A particle moves at constant speed

$$v = 0.80c$$

for a coordinate time interval

$$\Delta t = 10.0\,\mathrm{s}.$$

The Lorentz factor is

$$\gamma = \frac{1}{\sqrt{1-0.80^2}} = \frac{1}{0.60} = \frac{5}{3}.$$

The elapsed proper time is

$$\Delta\tau = \frac{\Delta t}{\gamma} = \frac{10.0\,\mathrm{s}}{5/3} = 6.0\,\mathrm{s}.$$

Thus the particle's comoving clock records

$$\boxed{\Delta\tau = 6.0\,\mathrm{s}}$$

between the departure and arrival events.
A direct interval calculation gives the same result. Since

$$\Delta x = v\Delta t = 0.80c\Delta t,$$

we have

$$\begin{aligned}\Delta s^2 &= c^2(\Delta t)^2 - (0.80c\Delta t)^2\\ &= 0.36c^2(\Delta t)^2\\ &= c^2(6.0\,\mathrm{s})^2.\end{aligned}$$

Hence

$$\Delta\tau = \frac{\sqrt{\Delta s^2}}{c} = 6.0\,\mathrm{s}.$$

Worked example: classifying a separation

Suppose

$$\Delta t = 3.0\,\mu\mathrm{s}, \qquad |\Delta\mathbf{x}| = 1.20\,\mathrm{km}.$$

The temporal contribution is

$$c\Delta t = (3.00\times10^8\,\mathrm{m\,s^{-1}})(3.0\times10^{-6}\,\mathrm{s}) = 900\,\mathrm{m}.$$

Therefore

$$\Delta s^2 = (900\,\mathrm{m})^2 - (1200\,\mathrm{m})^2.$$

Calculating,

$$\Delta s^2 = 810000\,\mathrm{m}^2 - 1440000\,\mathrm{m}^2 = -630000\,\mathrm{m}^2.$$

Since

$$\Delta s^2 < 0,$$

the separation is spacelike.

Proper distance for spacelike separation

For a spacelike separation, one may define the invariant proper distance

$$\boxed{\ell_{\text{proper}} = \sqrt{-\Delta s^2}}.$$

The minus sign is needed because $\Delta s^2 < 0$ with our signature.

There exists an inertial frame in which the two spacelike-separated events are simultaneous:

$$\Delta t' = 0.$$

In that frame,

$$\Delta s^2 = -|\Delta \mathbf{x}'|^2,$$

so

$$\ell_{\text{proper}} = |\Delta \mathbf{x}'|.$$

This is the invariant spatial separation measured in the frame where the events are simultaneous.

Null intervals

For light,

$$\Delta s^2 = 0.$$

Therefore the proper time along a light ray is

$$\Delta \tau = 0.$$

This does not mean that a physical rest frame can be attached to light. A massive clock cannot move at speed c, and no inertial frame exists in which a photon is at rest.

The vanishing interval means only that the lightlike path has zero Minkowski length according to the metric.

The inverse metric

The inverse metric $\eta^{\mu\nu}$ is defined by

$$\eta^{\mu\alpha}\eta_{\alpha\nu} = \delta^{\mu}{}_{\nu},$$

where

$$\delta^{\mu}{}_{\nu} = \begin{cases} 1, & \mu = \nu, \\ 0, & \mu \neq \nu. \end{cases}$$

For the Minkowski metric in Cartesian inertial coordinates,

$$\boxed{\eta^{\mu\nu} = \eta_{\mu\nu} = \operatorname{diag}(1, -1, -1, -1)}.$$

The equality of the numerical matrices is special to this diagonal convention. The positions of the indices still carry mathematical meaning.

Structural checks

A calculation involving the interval should satisfy:

1. every term in Δs^2 has dimensions of length squared;
2. the temporal term is positive and spatial terms are negative in the chosen convention;
3. timelike, null, and spacelike classifications agree with the light-cone picture;
4. proper time is real only for timelike motion;
5. the interval remains unchanged after a Lorentz transformation.

Common mistakes

Typical errors include:

1. mixing the signatures $(+,-,-,-)$ and $(-,+,+,+)$ in one derivation;
2. writing the spatial terms with positive signs after choosing the mostly-minus metric;
3. confusing Δs with Δs^2;
4. taking $\sqrt{\Delta s^2}$ for a spacelike interval without accounting for its negative sign;
5. treating proper time as a coordinate attached to every observer rather than the invariant time along a timelike worldline;
6. claiming that light has a rest frame because its proper time vanishes;
7. forgetting the factor c when comparing time and distance.

What has been established

The Minkowski metric is fixed throughout these notes as

$$\eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1).$$

It defines the invariant interval

$$\Delta s^2 = c^2(\Delta t)^2 - |\Delta \mathbf{x}|^2,$$

whose sign classifies separations as timelike, null, or spacelike. For a timelike worldline, the interval defines proper time through

$$c^2 \mathrm{d}\tau^2 = \mathrm{d}s^2,$$

leading to

$$\mathrm{d}t = \gamma \mathrm{d}\tau.$$

The next section generalizes the coordinate displacement to arbitrary four-vectors and introduces covectors, scalar products, and systematic index raising and lowering.

4.3 Four-Vectors and Covectors

A relativistic law should retain its form when one inertial observer is replaced by another. This requirement is easiest to express using objects with known Lorentz-transformation rules.

Four-vectors generalize ordinary spatial vectors to Minkowski spacetime. Covectors are the corresponding dual objects. The metric connects the two, and their contraction produces Lorentz scalars.

Contravariant four-vectors

A contravariant four-vector is an ordered set of four components

$$A^\mu = (A^0, A^1, A^2, A^3)$$

that transforms under a Lorentz transformation according to

$$\boxed{A'^\mu = \Lambda^\mu{}_\nu A^\nu}.$$

The superscript is not an exponent. It identifies the component and indicates the transformation type.

A collection of four numbers is not automatically a four-vector. The defining condition is the transformation law. Four quantities assembled into one column become a four-vector only if they transform together through the Lorentz matrix.

Examples of contravariant four-vectors

Important examples include:

- the position four-vector
$$x^\mu = (ct, \mathbf{x});$$
- the displacement four-vector
$$\Delta x^\mu = x_2^\mu - x_1^\mu;$$
- the four-velocity
$$U^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau};$$
- the four-momentum
$$p^\mu = mU^\mu;$$
- a four-current
$$J^\mu;$$
- the electromagnetic four-potential
$$A^\mu.$$

The last two examples will be developed in later chapters.

Covectors

A covector has components

$$A_\mu = (A_0, A_1, A_2, A_3)$$

and transforms with the inverse Lorentz transformation:

$$\boxed{A'_\mu = (\Lambda^{-1})^\nu{}_\mu A_\nu}.$$

Covectors are linear maps from vectors to scalars. Given a vector V^μ, the covector A_μ produces the contraction

$$A_\mu V^\mu.$$

This quantity is invariant under Lorentz transformations.

In Euclidean Cartesian coordinates, vectors and covectors often have the same numerical components, so their distinction can remain hidden. In Minkowski spacetime, lowering a spatial index changes its sign, making the distinction immediately visible.

Lowering an index

The Minkowski metric converts a contravariant vector into a covector:

$$\boxed{A_\mu = \eta_{\mu\nu} A^\nu}.$$

For

$$A^\mu = (A^0, A^1, A^2, A^3),$$

and

$$\eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1),$$

we obtain

$$\boxed{A_\mu = (A^0, -A^1, -A^2, -A^3)}.$$

Thus

$$A_0 = A^0, \qquad A_i = -A^i$$

for spatial indices $i = 1, 2, 3$.

Raising an index

The inverse metric raises a covariant index:

$$\boxed{A^\mu = \eta^{\mu\nu} A_\nu}.$$

Since

$$\eta^{\mu\nu} = \mathrm{diag}(1, -1, -1, -1),$$

raising a spatial index changes its sign again:

$$A^0 = A_0, \qquad A^i = -A_i.$$

Applying lowering followed by raising returns the original vector:

$$\eta^{\mu\alpha}\eta_{\alpha\nu} A^\nu = \delta^\mu{}_\nu A^\nu = A^\mu.$$

The Minkowski scalar product

The scalar product of four-vectors A^μ and B^μ is

$$\boxed{A \cdot B = \eta_{\mu\nu} A^\mu B^\nu = A_\mu B^\mu}.$$

Expanding the components,

$$\boxed{A \cdot B = A^0 B^0 - A^1 B^1 - A^2 B^2 - A^3 B^3}.$$

Using spatial vector notation,

$$A \cdot B = A^0 B^0 - \mathbf{A} \cdot \mathbf{B}.$$

The contraction is a Lorentz scalar. Its numerical value is the same in every inertial frame.

The norm of a four-vector

The squared norm is

$$\boxed{A^2 := A \cdot A = A_\mu A^\mu}.$$

Explicitly,

$$A^2 = (A^0)^2 - |\mathbf{A}|^2.$$

Unlike a Euclidean norm, this quantity can be positive, zero, or negative.
A four-vector is called

- timelike if $A^2 > 0$;
- null if $A^2 = 0$;
- spacelike if $A^2 < 0$.

For the displacement four-vector, this classification reproduces the causal classification of spacetime separations.

Worked example: lowering a vector index

Let

$$A^\mu = (5, 2, -1, 3).$$

Then

$$A_\mu = \eta_{\mu\nu} A^\nu = (5, -2, 1, -3).$$

The squared norm is

$$\begin{aligned} A_\mu A^\mu &= 5(5) + (-2)(2) + (1)(-1) + (-3)(3) \\ &= 25 - 4 - 1 - 9 \\ &= 11. \end{aligned}$$

Equivalently,

$$A^2 = 5^2 - 2^2 - (-1)^2 - 3^2 = 11.$$

Since

$$A^2 > 0,$$

the vector is timelike.

Worked example: scalar product

Let

$$A^\mu = (4, 1, 2, 0), \qquad B^\mu = (3, -2, 1, 5).$$

Then

$$\begin{aligned} A \cdot B &= A^0 B^0 - A^1 B^1 - A^2 B^2 - A^3 B^3 \\ &= 4(3) - 1(-2) - 2(1) - 0(5) \\ &= 12 + 2 - 2 \\ &= 12. \end{aligned}$$

Using the lowered vector

$$A_\mu = (4, -1, -2, 0),$$

we obtain the same result:

$$A_\mu B^\mu = 4(3) + (-1)(-2) + (-2)(1) + 0(5) = 12.$$

The position four-vector

The position four-vector is

$$\boxed{x^\mu = (ct, \mathbf{x})}.$$

Its covariant components are

$$\boxed{x_\mu = (ct, -\mathbf{x})}.$$

The contraction

$$x_\mu x^\mu = c^2 t^2 - |\mathbf{x}|^2$$

is the squared interval from the coordinate origin to the event.

The coordinate origin itself has no physical privilege. Translating the origin changes x^μ, but differences

$$\Delta x^\mu$$

between events transform as four-vectors and carry direct geometric meaning.

The four-velocity

For a massive particle, the four-velocity is defined using proper time:

$$\boxed{U^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}}.$$

Since

$$\frac{\mathrm{d}t}{\mathrm{d}\tau} = \gamma,$$

we obtain

$$U^0 = \frac{\mathrm{d}(ct)}{\mathrm{d}\tau} = \gamma c,$$
$$\mathbf{U} = \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}\tau} = \gamma \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \gamma \mathbf{v}.$$

Therefore

$$\boxed{U^\mu = \gamma(c, \mathbf{v})}.$$

The four-velocity is not simply $(c, \mathbf{v})$. The factor γ is required for the object to transform as a four-vector.

Normalization of the four-velocity

The squared norm is

$$\begin{aligned} U_\mu U^\mu &= \gamma^2(c^2 - v^2) \\ &= \gamma^2 c^2 \left(1 - \frac{v^2}{c^2}\right) \\ &= c^2. \end{aligned}$$

Thus

$$\boxed{U_\mu U^\mu = c^2}.$$

Every massive particle has the same four-velocity norm, even though different observers assign different three-velocities to it.

This provides an important structural check. Any proposed formula for a four-velocity should satisfy this normalization.

The four-momentum

For a particle of invariant mass m, define

$$\boxed{p^\mu = mU^\mu}.$$

Using the four-velocity components,

$$p^\mu = (\gamma mc, \gamma m\mathbf{v}).$$

The relativistic three-momentum is

$$\mathbf{p} = \gamma m \mathbf{v},$$

and the relativistic energy is

$$E = \gamma mc^2.$$

Therefore

$$\boxed{p^\mu = \left(\frac{E}{c}, \mathbf{p}\right)}.$$

Lowering the index gives

$$\boxed{p_\mu = \left(\frac{E}{c}, -\mathbf{p}\right)}.$$

Norm of the four-momentum

Since

$$p^\mu = mU^\mu,$$

we have

$$\begin{aligned} p_\mu p^\mu &= m^2 U_\mu U^\mu \\ &= m^2 c^2. \end{aligned}$$

Thus

$$\boxed{p_\mu p^\mu = m^2 c^2}.$$

Using

$$p^\mu = (E/c, \mathbf{p}),$$

this becomes

$$\frac{E^2}{c^2} - |\mathbf{p}|^2 = m^2 c^2.$$

Multiplying by c^2,

$$\boxed{E^2 = |\mathbf{p}|^2 c^2 + m^2 c^4}.$$

This relation will be revisited in the section on relativistic invariants.

Massless four-momentum

For a massless particle,

$$m = 0,$$

so

$$p_\mu p^\mu = 0.$$

The energy-momentum relation reduces to

$$E^2 = |\mathbf{p}|^2 c^2,$$

or, for positive energy,

$$\boxed{E = c|\mathbf{p}|}.$$

A massless particle has null four-momentum. Its four-momentum is well defined even though proper time and four-velocity are not defined along a null worldline in the same way as for a massive particle.

Basis-vector interpretation

A four-vector is a geometric object independent of coordinates. In a basis $\{\mathbf{e}_\mu\}$, it may be written as

$$\mathbf{A} = A^\mu \mathbf{e}_\mu.$$

When the inertial frame changes, both the components and the basis change in compensating ways, leaving the geometric vector unchanged.

A covector belongs to the dual basis $\{\boldsymbol{\theta}^\mu\}$:

$$\boldsymbol{\alpha} = \alpha_\mu \boldsymbol{\theta}^\mu.$$

The dual pairing satisfies

$$\boldsymbol{\theta}^\mu(\mathbf{e}_\nu) = \delta^\mu{}_\nu.$$

This geometric viewpoint explains why upper and lower indices transform differently.

Structural checks

A proposed four-vector calculation should satisfy:

1. the object has four components;
2. its components transform with the Lorentz matrix;
3. lowering or raising an index uses the metric exactly once;
4. scalar products contain one upper and one lower occurrence of the contracted index;
5. the norm has the correct sign for its causal type;
6. the four-velocity satisfies $U^2 = c^2$;
7. the four-momentum satisfies $p^2 = m^2c^2$.

Common mistakes

Typical errors include:

1. treating the index as an exponent;
2. assuming that every list of four numbers is a four-vector;
3. lowering an index without changing the spatial signs;
4. using the Euclidean scalar product instead of the Minkowski scalar product;
5. writing $U^\mu = (c, \mathbf{v})$ and omitting γ;
6. confusing the invariant mass m with the frame-dependent energy E;
7. defining a four-velocity for a light ray using proper time, even though $d\tau = 0$ on a null worldline.

What has been established

A contravariant four-vector transforms as

$$A'^{\mu} = \Lambda^{\mu}{}_{\nu} A^{\nu},$$

while the metric produces the corresponding covector

$$A_{\mu} = \eta_{\mu\nu} A^{\nu}.$$

Their contraction defines the Lorentz-invariant scalar product. The position, four-velocity, and four-momentum are central examples:

$$x^{\mu} = (ct, \mathbf{x}),$$

$$U^{\mu} = \gamma(c, \mathbf{v}),$$

and

$$p^{\mu} = (E/c, \mathbf{p}).$$

The next section develops the index notation needed to manipulate these objects reliably, including free indices, dummy indices, contractions, and tensor equations.

4.4 Index Notation and Contractions

Tensor notation compresses many component equations into one expression. Its usefulness depends on strict bookkeeping. A misplaced, repeated, or missing index changes the mathematical meaning of a formula and may make the expression invalid.

This section develops a practical syntax for relativistic calculations: the Einstein summation convention, free and dummy indices, contractions, traces, symmetry, and diagnostic checks.

The Einstein summation convention

When an index appears exactly twice in one term, once upstairs and once downstairs, summation over that index is understood:

$$A_{\mu} B^{\mu} := \sum_{\mu=0}^{3} A_{\mu} B^{\mu}.$$

Thus

$$A_{\mu} B^{\mu} = A_0 B^0 + A_1 B^1 + A_2 B^2 + A_3 B^3.$$

With the mostly-minus metric, if A_{μ} is obtained by lowering A^{μ}, this becomes

$$A_{\mu} B^{\mu} = A^0 B^0 - A^1 B^1 - A^2 B^2 - A^3 B^3.$$

The summation convention removes explicit summation symbols, but it does not remove the need to know which indices are being summed.

Dummy indices

A repeated index that is summed is called a dummy index. Its name has no intrinsic meaning. For example,

$$A_\mu B^\mu = A_\nu B^\nu = A_\alpha B^\alpha.$$

All three expressions represent the same scalar.

Renaming a dummy index is allowed provided the new symbol does not conflict with another index already present in the term.

For example,

$$T^\mu{}_\nu V^\nu = T^\mu{}_\alpha V^\alpha$$

is valid. The index μ must not be renamed independently because it is not summed.

Free indices

An index that appears once in a term is called a free index. It labels the components of the resulting tensor.

In

$$W^\mu = T^\mu{}_\nu V^\nu,$$

the index ν is dummy, while μ is free. The equation represents four component equations, one for each value of μ.

Every term in a valid tensor equation must have the same free indices in the same positions. Thus

$$A^\mu + B^\mu = C^\mu$$

is structurally valid, while

$$A^\mu + B^\nu = C^\mu$$

is invalid because the second term has a different free index.

One term, one index pattern

Consider

$$X^\mu = A^\mu + T^\mu{}_\nu B^\nu.$$

Both terms on the right have one free upper index μ. The second term also contains the summed index ν, but after contraction its result is a vector with the same index structure as A^μ.

By contrast,

$$A^\mu + T^{\mu\nu}$$

cannot be added because the first object is a vector and the second is a rank-two tensor.

The index pattern reveals the tensor type before any components are calculated.

An index may not appear three times

Within a single monomial term, an index should not appear more than twice. The expression

$$A_\mu B^\mu C^\mu$$

is invalid because μ appears three times.

A correct expression would need distinct contractions, for example

$$A_\mu B^\mu C^\nu,$$

which is a vector with free index ν, or

$$A_\mu B^\mu C_\nu D^\nu,$$

which is a scalar.

Contraction

A contraction pairs one upper and one lower index and sums over them. It reduces the tensor rank by two.

For a rank-two tensor,

$$T^\mu{}_\nu,$$

the contraction

$$\boxed{T^\mu{}_\mu}$$

is a scalar called the trace.

For a rank-three tensor

$$R^{\mu\nu}{}_\alpha,$$

contracting ν with α gives

$$R^{\mu\nu}{}_\nu,$$

which is a vector with free index μ.

Contraction is the tensorial analogue of taking a dot product or matrix trace.

The Kronecker delta

The mixed tensor

$$\delta^\mu{}_\nu$$

acts as the identity:

$$\boxed{\delta^\mu{}_\nu A^\nu = A^\mu}.$$

Its components are

$$\delta^\mu{}_\nu = \begin{cases} 1, & \mu = \nu, \\ 0, & \mu \neq \nu. \end{cases}$$

The trace in four-dimensional spacetime is

$$\delta^\mu{}_\mu = 4.$$

The metric and its inverse satisfy

$$\eta^{\mu\alpha}\eta_{\alpha\nu} = \delta^{\mu}{}_{\nu}.$$

Raising and lowering tensor indices

The metric acts on each index separately. For a covariant rank-two tensor,

$$T_{\mu\nu},$$

raising the first index gives

$$\boxed{T^{\mu}{}_{\nu} = \eta^{\mu\alpha}T_{\alpha\nu}}.$$

Raising both indices gives

$$\boxed{T^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}T_{\alpha\beta}}.$$

Likewise,

$$T_{\mu\nu} = \eta_{\mu\alpha}\eta_{\nu\beta}T^{\alpha\beta}.$$

Each spatial index that is raised or lowered contributes one sign change in Cartesian Minkowski coordinates.

Worked component example

Let $T^{\mu\nu}$ have the component

$$T^{01} = a.$$

Lowering both indices gives

$$T_{01} = \eta_{0\alpha}\eta_{1\beta}T^{\alpha\beta}.$$

Because the metric is diagonal, only

$$\alpha = 0, \qquad \beta = 1$$

contribute. Therefore

$$T_{01} = \eta_{00}\eta_{11}T^{01} = (1)(-1)a = -a.$$

For a purely spatial component,

$$T^{12} = b,$$

we find

$$T_{12} = \eta_{11}\eta_{22}T^{12} = (-1)(-1)b = b.$$

One lowered spatial index changes the sign; two lowered spatial indices restore it.

Matrix multiplication in index notation

If

$$C^{\mu}{}_{\nu} = A^{\mu}{}_{\alpha} B^{\alpha}{}_{\nu},$$

then the dummy index α represents matrix multiplication. The free indices μ and ν identify the row and column of the resulting matrix.

Writing the sum explicitly,

$$C^{\mu}{}_{\nu} = \sum_{\alpha=0}^{3} A^{\mu}{}_{\alpha} B^{\alpha}{}_{\nu}.$$

The order matters:

$$A^{\mu}{}_{\alpha} B^{\alpha}{}_{\nu}$$

need not equal

$$B^{\mu}{}_{\alpha} A^{\alpha}{}_{\nu}.$$

Symmetric tensors

A rank-two tensor is symmetric if

$$\boxed{S_{\mu\nu} = S_{\nu\mu}}.$$

The Minkowski metric is symmetric:

$$\eta_{\mu\nu} = \eta_{\nu\mu}.$$

For a symmetric tensor, exchanging its two indices does not change the component.

Any rank-two tensor can be decomposed into symmetric and antisymmetric parts:

$$T_{\mu\nu} = T_{(\mu\nu)} + T_{[\mu\nu]},$$

where

$$\boxed{T_{(\mu\nu)} = \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu})}$$

and

$$\boxed{T_{[\mu\nu]} = \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu})}.$$

Antisymmetric tensors

A tensor is antisymmetric if

$$\boxed{F_{\mu\nu} = -F_{\nu\mu}}.$$

Setting $\mu = \nu$ gives

$$F_{\mu\mu} = -F_{\mu\mu},$$

so

$$\boxed{F_{\mu\mu} = 0}$$

with no summation intended in this statement.

The electromagnetic field tensor will be an important antisymmetric tensor.

Contraction of symmetric and antisymmetric tensors

Let $S^{\mu\nu}$ be symmetric and $F_{\mu\nu}$ antisymmetric. Consider

$$S^{\mu\nu} F_{\mu\nu}.$$

Rename the dummy indices $\mu \leftrightarrow \nu$:

$$S^{\mu\nu} F_{\mu\nu} = S^{\nu\mu} F_{\nu\mu}.$$

Using symmetry and antisymmetry,

$$S^{\nu\mu} F_{\nu\mu} = S^{\mu\nu}(-F_{\mu\nu}).$$

Therefore

$$S^{\mu\nu} F_{\mu\nu} = -S^{\mu\nu} F_{\mu\nu},$$

which implies

$$\boxed{S^{\mu\nu} F_{\mu\nu} = 0}.$$

This is a typical example in which dummy-index renaming makes a structural cancellation visible.

Checking a proposed tensor equation

Consider

$$\partial_\mu T^{\mu\nu} = J^\nu.$$

The index μ is summed, while ν is free. Both sides therefore have one free upper index ν, so the equation is structurally possible.

By contrast,

$$\partial_\mu T^{\mu\nu} = J^\mu$$

is invalid as written. On the left, μ is dummy and ν is free. On the right, μ is free. The free-index patterns do not match.

Worked example: expanding a contraction

Let

$$J^\mu = (c\rho, J^1, J^2, J^3)$$

and

$$A_\mu = (A_0, A_1, A_2, A_3).$$

Then

$$A_\mu J^\mu = A_0 c\rho + A_1 J^1 + A_2 J^2 + A_3 J^3.$$

If A_μ was obtained from

$$A^\mu = (A^0, \mathbf{A}),$$

then

$$A_\mu = (A^0, -\mathbf{A}),$$

and

$$A_\mu J^\mu = A^0 c\rho - \mathbf{A} \cdot \mathbf{J}.$$

The compact contraction contains both temporal and spatial terms with the metric signs built in.

Changing dummy indices safely

Suppose

$$X^\mu = T^\mu{}_\nu A^\nu + S^\mu{}_\alpha B^\alpha.$$

The two dummy indices may be given the same name because they occur in separate terms:

$$X^\mu = T^\mu{}_\nu A^\nu + S^\mu{}_\nu B^\nu.$$

However, in a product such as

$$T^\mu{}_\nu S^\nu{}_\alpha A^\alpha,$$

the indices ν and α represent different contractions and cannot both be renamed to the same symbol.

Free-index equations represent component families

The equation

$$A^\mu = B^\mu$$

stands for four equations:

$$A^0 = B^0, \qquad A^1 = B^1, \qquad A^2 = B^2, \qquad A^3 = B^3.$$

Similarly,

$$T^{\mu\nu} = S^{\mu\nu}$$

stands for sixteen component equations before symmetries are taken into account.

A fully contracted equation such as

$$A_\mu B^\mu = C$$

contains no free indices and is one scalar equation.

A practical index-checking algorithm

For every tensor expression:

1. inspect each term separately;
2. identify every repeated index;
3. confirm that each dummy index appears exactly twice;
4. confirm that a contracted pair consists of one upper and one lower index;
5. identify the free indices;
6. check that every term has the same free-index pattern;
7. only then perform algebra or rename dummy indices.

This procedure catches many errors before any component calculation begins.

Common mistakes

Typical errors include:

1. using the same index three times in one term;
2. changing a free index in only one term;
3. renaming a dummy index to a symbol already used elsewhere in the term;
4. adding tensors with different free-index structures;
5. contracting two upper indices without first using the metric;
6. assuming that exchanging indices is allowed without a symmetry property;
7. confusing a tensor component equation with a scalar equation.

What has been established

The Einstein convention sums an index appearing once upstairs and once downstairs. Dummy indices may be renamed, while free indices determine the tensor type and must agree across an equation. Contraction reduces tensor rank, the Kronecker delta acts as the identity, and the metric raises or lowers each index separately.

Symmetry and antisymmetry provide powerful structural simplifications. These rules will now be applied to Lorentz transformations, where the transformation matrix itself carries one upper and one lower index.

4.5 Lorentz Transformations

Lorentz transformations relate the coordinates assigned to the same event by different inertial observers. They replace Galilean transformations when the invariant speed c must be preserved.

The basic example is a boost: two frames whose spatial axes are parallel and whose origins move uniformly relative to one another. Rotations are also Lorentz transformations, but boosts reveal the specifically relativistic mixing of space and time.

Standard configuration

Let frames S and S' satisfy the following conditions:

- their spatial axes are parallel;
- the frame S' moves with constant speed v in the positive x-direction relative to S;
- the origins coincide at
$$t = t' = 0.$$

The transformation affects only the coordinates ct and x. The transverse coordinates remain unchanged:

$$y' = y, \qquad z' = z.$$

Define

$$\boxed{\beta = \frac{v}{c}}$$

and

$$\boxed{\gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-v^2/c^2}}}.$$

For a physical inertial frame,

$$|v| < c,$$

so γ is real and satisfies

$$\gamma \geq 1.$$

The standard Lorentz boost

The coordinate transformation is

$$\boxed{ct' = \gamma(ct - \beta x)}$$

and

$$\boxed{x' = \gamma(x - \beta ct)}.$$

Equivalently, using time rather than ct,

$$\boxed{t' = \gamma\left(t - \frac{vx}{c^2}\right)}$$

and

$$\boxed{x' = \gamma(x - vt)}.$$

The transverse coordinates satisfy

$$y' = y, \qquad z' = z.$$

Space and time are mixed. The transformed time depends on both t and x, while the transformed position depends on both x and t.

Matrix form

For

$$x^\mu = (ct, x, y, z),$$

the boost is

$$\boxed{x'^\mu = \Lambda^\mu{}_\nu x^\nu}$$

with

$$\boxed{\Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}.$$

The first row gives the transformed time coordinate, and the second row gives the transformed longitudinal spatial coordinate.

Why this describes a moving frame

The origin of S' is defined by

$$x' = 0.$$

Using

$$x' = \gamma(x - vt),$$

we find

$$x = vt.$$

Thus the origin of S' moves with speed v in the positive x-direction as measured in S. Similarly, the origin of S, defined by $x = 0$, moves in S' according to

$$x' = -vt'.$$

The relative speed has the same magnitude and opposite direction.

The inverse transformation

The inverse boost is obtained by replacing

$$v \longrightarrow -v, \qquad \beta \longrightarrow -\beta.$$

Therefore

$$\boxed{ct = \gamma(ct' + \beta x')}$$

and

$$\boxed{x = \gamma(x' + \beta ct')}.$$

Equivalently,

$$t = \gamma\left(t' + \frac{vx'}{c^2}\right),$$

and

$$x = \gamma(x' + vt').$$

This symmetry reflects the absence of a preferred inertial frame.

Direct verification of interval invariance

Consider the transformed temporal and longitudinal coordinates:

$$ct' = \gamma(ct - \beta x),$$

$$x' = \gamma(x - \beta ct).$$

Calculate

$$\begin{aligned}(ct')^2 - (x')^2 &= \gamma^2 \left[(ct - \beta x)^2 - (x - \beta ct)^2\right] \\ &= \gamma^2 \left[(c^2t^2 - 2\beta ctx + \beta^2 x^2) - (x^2 - 2\beta ctx + \beta^2 c^2 t^2)\right] \\ &= \gamma^2(1 - \beta^2)(c^2t^2 - x^2).\end{aligned}$$

Since

$$\gamma^2(1 - \beta^2) = 1,$$

we obtain

$$(ct')^2 - (x')^2 = c^2t^2 - x^2.$$

Because

$$y' = y, \qquad z' = z,$$

it follows that

$$\boxed{c^2t'^2 - x'^2 - y'^2 - z'^2 = c^2t^2 - x^2 - y^2 - z^2}.$$

The spacetime interval is invariant.

The metric-preservation condition

In matrix notation, interval invariance is

$$(\Lambda x)^{\mathsf{T}} \eta (\Lambda x) = x^{\mathsf{T}} \eta x.$$

Since this must hold for every x,

$$\boxed{\Lambda^{\mathsf{T}} \eta \Lambda = \eta}.$$

This equation defines the Lorentz group. It is the Minkowski analogue of the Euclidean condition

$$R^{\mathsf{T}} R = I$$

for rotation matrices.

Transformation of a four-vector

Every contravariant four-vector transforms with the same Lorentz matrix:

$$A'^{\mu} = \Lambda^{\mu}{}_{\nu} A^{\nu}.$$

For a boost in the x-direction,

$$\boxed{A'^0 = \gamma(A^0 - \beta A^1)}$$

and

$$\boxed{A'^1 = \gamma(A^1 - \beta A^0)}.$$

The transverse components are unchanged:

$$A'^2 = A^2, \qquad A'^3 = A^3.$$

This applies to displacement, four-velocity, four-momentum, four-current, and every other contravariant four-vector.

Worked example: transforming an event

An event has coordinates in S

$$t = 5.0\,\mu\text{s}, \qquad x = 900\,\text{m}.$$

Let S' move at

$$v = 0.60c.$$

Then

$$\beta = 0.60, \qquad \gamma = \frac{1}{\sqrt{1 - 0.60^2}} = 1.25.$$

First compute

$$ct = (3.00 \times 10^8\,\text{m}\,\text{s}^{-1})(5.0 \times 10^{-6}\,\text{s}) = 1500\,\text{m}.$$

The transformed coordinates are

$$\begin{aligned} ct' &= 1.25(1500\,\text{m} - 0.60 \times 900\,\text{m}) \\ &= 1.25(960\,\text{m}) \\ &= 1200\,\text{m}, \end{aligned}$$

and

$$\begin{aligned} x' &= 1.25(900\,\text{m} - 0.60 \times 1500\,\text{m}) \\ &= 0. \end{aligned}$$

Thus

$$t' = \frac{1200\,\text{m}}{3.00 \times 10^8\,\text{m}\,\text{s}^{-1}} = 4.0\,\mu\text{s}.$$

The event lies on the worldline of the origin of S', as confirmed by $x' = 0$.

Time dilation from the Lorentz transformation

Consider two events occurring at the same place in S':

$$\Delta x' = 0.$$

Their time separation in S' is the proper time

$$\Delta t' = \Delta\tau.$$

Using the inverse transformation,

$$\Delta t = \gamma\left(\Delta t' + \frac{v\Delta x'}{c^2}\right),$$

we obtain

$$\boxed{\Delta t = \gamma\Delta\tau}.$$

The coordinate time between the events in a frame where the clock moves is larger than the proper time recorded by the clock.

Length contraction

Let a rod be at rest in S'. Its proper length is

$$L_0 = \Delta x',$$

measured between its endpoints at the same time in S':

$$\Delta t' = 0.$$

To measure the rod's length in S, the endpoint positions must be recorded simultaneously in S:

$$\Delta t = 0.$$

Using

$$\Delta x' = \gamma(\Delta x - v\Delta t),$$

we find

$$L_0 = \gamma L.$$

Therefore

$$\boxed{L = \frac{L_0}{\gamma}}.$$

The moving rod is shorter along the direction of motion. The simultaneity condition is essential; measuring the two endpoints at different times does not define the rod's length in one frame.

Relativity of simultaneity

Suppose two events are simultaneous in S:

$$\Delta t = 0.$$

Their time difference in S' is

$$\begin{aligned}\Delta t' &= \gamma\left(\Delta t - \frac{v\Delta x}{c^2}\right)\\ &= -\gamma\frac{v\Delta x}{c^2}.\end{aligned}$$

If

$$\Delta x \neq 0,$$

then generally

$$\boxed{\Delta t' \neq 0}.$$

Distant simultaneity is frame dependent. Events at the same spacetime point remain coincident in every frame.

Transformation of velocities

Let a particle have longitudinal velocity

$$u_x = \frac{\mathrm{d}x}{\mathrm{d}t}$$

in S. Differentiate the Lorentz transformation:

$$\mathrm{d}x' = \gamma(\mathrm{d}x - v\,\mathrm{d}t),$$

and

$$\mathrm{d}t' = \gamma\left(\mathrm{d}t - \frac{v\,\mathrm{d}x}{c^2}\right).$$

Therefore

$$\boxed{u'_x = \frac{u_x - v}{1 - u_x v/c^2}}.$$

For the transverse components,

$$\boxed{u'_y = \frac{u_y}{\gamma(1 - u_x v/c^2)}}$$

and

$$\boxed{u'_z = \frac{u_z}{\gamma(1 - u_x v/c^2)}}.$$

These formulas replace Galilean velocity addition.

Light-speed check

If a light ray moves in the positive x-direction in S, then

$$u_x = c.$$

The transformed speed is

$$\begin{aligned} u'_x &= \frac{c - v}{1 - v/c} \\ &= c. \end{aligned}$$

Thus every inertial frame measures the same light speed.

Nonrelativistic limit

When

$$|v| \ll c,$$

we have

$$\beta \ll 1, \qquad \gamma \approx 1.$$

The Lorentz transformation reduces approximately to

$$x' \approx x - vt,$$

and

$$t' \approx t.$$

These are the Galilean transformation formulas. Relativistic corrections become important when speeds are not small compared with c.

Composition of collinear boosts

Suppose S' moves at speed v relative to S, and S'' moves at speed u' relative to S', all along the same axis. The speed of S'' relative to S is

$$\boxed{u = \frac{u' + v}{1 + u'v/c^2}}.$$

If

$$|u'| < c, \qquad |v| < c,$$

then

$$|u| < c.$$

No composition of subluminal boosts produces a superluminal relative speed.

Rapidity

A useful parameter for a boost is the rapidity χ, defined by

$$\tanh \chi = \beta.$$

Then

$$\cosh \chi = \gamma, \qquad \sinh \chi = \beta\gamma.$$

The boost matrix in the (ct, x)-plane becomes

$$\Lambda(\chi) = \begin{pmatrix} \cosh \chi & -\sinh \chi \\ -\sinh \chi & \cosh \chi \end{pmatrix}.$$

Collinear boosts add rapidities:

$$\chi_{\text{total}} = \chi_1 + \chi_2.$$

This makes a boost resemble a hyperbolic rotation in Minkowski spacetime.

Structural checks

A Lorentz-transformation calculation should satisfy:

1. setting $v = 0$ gives the identity transformation;
2. replacing v by $-v$ gives the inverse;
3. the interval is preserved;
4. a light ray remains a light ray;
5. the origin of S' follows $x = vt$ in S;
6. the nonrelativistic limit reproduces Galilean kinematics.

Common mistakes

Typical errors include:

1. using the same sign for the forward and inverse boosts;
2. forgetting the factor c^2 in the transformation of time;
3. applying length contraction without imposing simultaneity in the measuring frame;
4. treating time dilation as a statement about one isolated event rather than two events on one clock's worldline;
5. using Galilean velocity addition at relativistic speeds;
6. assuming that simultaneity is invariant;
7. confusing the Lorentz matrix with the Minkowski metric.

What has been established

A standard boost along the x-axis is

$$ct' = \gamma(ct - \beta x), \qquad x' = \gamma(x - \beta ct).$$

It preserves the Minkowski interval and transforms every contravariant four-vector through the same matrix $\Lambda^\mu{}_\nu$. Time dilation, length contraction, relativity of simultaneity, and relativistic velocity addition follow from this transformation.

The next section develops four-dimensional differentiation and constructs the d'Alembert operator used in covariant wave and field equations.

4.6 Four-Derivatives and the d'Alembert Operator

Relativistic field equations contain derivatives with respect to both time and space. To write them covariantly, these derivatives must be assembled into an object with a definite Lorentz-transformation rule.

The resulting object is the four-gradient. Contracting the four-gradient with itself produces the d'Alembert operator, the relativistic analogue of the Laplacian.

Coordinates and partial derivatives

The spacetime coordinates are

$$x^\mu = (ct, x, y, z).$$

The derivative with respect to the zeroth coordinate is

$$\frac{\partial}{\partial x^0} = \frac{\partial}{\partial(ct)} = \frac{1}{c}\frac{\partial}{\partial t}.$$

The spatial derivatives are

$$\frac{\partial}{\partial x^1} = \frac{\partial}{\partial x}, \qquad \frac{\partial}{\partial x^2} = \frac{\partial}{\partial y}, \qquad \frac{\partial}{\partial x^3} = \frac{\partial}{\partial z}.$$

The covariant four-gradient

Define

$$\boxed{\partial_\mu := \frac{\partial}{\partial x^\mu}}.$$

Its components are

$$\boxed{\partial_\mu = \left(\frac{1}{c}\frac{\partial}{\partial t}, \nabla\right)}.$$

Here

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right).$$

The lower index is essential. By the chain rule, derivatives with respect to contravariant coordinates transform as covectors.

Transformation law from the chain rule

Let

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu.$$

The inverse relation is

$$x^\nu = (\Lambda^{-1})^\nu{}_\mu x'^\mu.$$

Using the chain rule,

$$\begin{aligned}\partial'_\mu &= \frac{\partial}{\partial x'^\mu} \\ &= \frac{\partial x^\nu}{\partial x'^\mu}\frac{\partial}{\partial x^\nu} \\ &= (\Lambda^{-1})^\nu{}_\mu \partial_\nu.\end{aligned}$$

Thus

$$\boxed{\partial'_\mu = (\Lambda^{-1})^\nu{}_\mu \partial_\nu}.$$

The four-gradient is a covector operator.

The contravariant four-gradient

Raise the index with the inverse metric:

$$\boxed{\partial^\mu = \eta^{\mu\nu}\partial_\nu}.$$

With

$$\eta^{\mu\nu} = \mathrm{diag}(1, -1, -1, -1),$$

we obtain

$$\boxed{\partial^\mu = \left(\frac{1}{c}\frac{\partial}{\partial t}, -\nabla\right)}.$$

Thus

$$\partial^0 = \partial_0, \qquad \partial^i = -\partial_i.$$

The sign difference between ∂_μ and ∂^μ is a direct consequence of the metric convention.

Derivative of a scalar field

Let $\phi(x)$ be a Lorentz scalar:

$$\phi'(x') = \phi(x).$$

Its gradient

$$\partial_\mu \phi$$

is a covector. The raised gradient

$$\partial^\mu \phi$$

is a contravariant four-vector.

In components,

$$\boxed{\partial_\mu \phi = \left(\frac{1}{c}\phi_t, \nabla\phi\right)}$$

and

$$\boxed{\partial^\mu \phi = \left(\frac{1}{c}\phi_t, -\nabla\phi\right)}.$$

The quantity

$$\partial_\mu \phi\, \partial^\mu \phi$$

is a Lorentz scalar.
Expanding,

$$\boxed{\partial_\mu \phi\, \partial^\mu \phi = \frac{1}{c^2}\left(\frac{\partial \phi}{\partial t}\right)^2 - |\nabla\phi|^2}.$$

This combination will become the kinetic term of the relativistic scalar-field Lagrangian.

Four-divergence

For a contravariant four-vector field $A^\mu(x)$, its divergence is

$$\boxed{\partial_\mu A^\mu}.$$

The derivative index is contracted with the vector index, so the result is a Lorentz scalar.
If

$$A^\mu = (A^0, \mathbf{A}),$$

then

$$\boxed{\partial_\mu A^\mu = \frac{1}{c}\frac{\partial A^0}{\partial t} + \nabla\cdot\mathbf{A}}.$$

For the four-current

$$J^\mu = (c\rho, \mathbf{J}),$$

we find

$$\partial_\mu J^\mu = \frac{1}{c}\frac{\partial (c\rho)}{\partial t} + \nabla\cdot\mathbf{J}.$$

Therefore

$$\boxed{\partial_\mu J^\mu = 0}$$

is equivalent to the continuity equation

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla\cdot\mathbf{J} = 0}.$$

This is an example of a familiar three-dimensional conservation law written as one covariant scalar equation.

The d'Alembert operator

Contract the two four-derivatives:

$$\boxed{\Box := \partial_\mu \partial^\mu = \eta^{\mu\nu}\partial_\mu\partial_\nu}.$$

The symbol $\Box$ is read as d'Alembertian or wave operator.

Expanding the components,

$$\begin{aligned}\Box &= \partial_0\partial^0 + \partial_1\partial^1 + \partial_2\partial^2 + \partial_3\partial^3 \\ &= \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2}.\end{aligned}$$

Thus

$$\boxed{\Box = \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2}.$$

With the opposite metric signature, the overall sign of $\Box$ is often defined oppositely. The convention must always be checked.

Covariant wave equation

The ordinary wave equation is

$$\frac{\partial^2 \phi}{\partial t^2} = c^2 \nabla^2 \phi.$$

Move all terms to one side and divide by c^2:

$$\frac{1}{c^2}\frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi = 0.$$

Therefore

$$\boxed{\Box \phi = 0}.$$

The compact equation is Lorentz covariant because $\Box\phi$ is a scalar when ϕ is a scalar field.

Why $\Box$ is Lorentz invariant

The operator is a contraction:

$$\Box = \partial_\mu \partial^\mu.$$

The first derivative transforms covariantly and the second contravariantly. Their contracted index disappears, leaving a scalar operator.

More explicitly,

$$\partial'_\mu \partial'^\mu = \partial_\mu \partial^\mu.$$

Therefore if

$$\Box\phi = 0$$

holds in one inertial frame, the same covariant equation holds in every inertial frame.

Worked example: applying $\Box$

Let

$$\phi(x,t) = A\cos(kx - \omega t)$$

in one spatial dimension. Then

$$\frac{\partial^2 \phi}{\partial t^2} = -\omega^2 \phi,$$

and

$$\frac{\partial^2 \phi}{\partial x^2} = -k^2 \phi.$$

Therefore

$$\begin{aligned}\Box\phi &= \frac{1}{c^2}(-\omega^2\phi) - (-k^2\phi) \\ &= \left(-\frac{\omega^2}{c^2} + k^2\right)\phi.\end{aligned}$$

The equation

$$\Box\phi = 0$$

requires

$$\boxed{\omega^2 = c^2 k^2}.$$

The covariant equation therefore contains the familiar dispersion relation of a massless wave.

The wave four-vector

A plane-wave phase may be written as

$$\mathbf{k}\cdot\mathbf{x} - \omega t.$$

Define the contravariant wave four-vector

$$\boxed{k^\mu = \left(\frac{\omega}{c}, \mathbf{k}\right)}.$$

Its covariant components are

$$\boxed{k_\mu = \left(\frac{\omega}{c}, -\mathbf{k}\right)}.$$

Then

$$k_\mu x^\mu = \omega t - \mathbf{k}\cdot\mathbf{x}.$$

Therefore

$$\mathbf{k}\cdot\mathbf{x} - \omega t = -k_\mu x^\mu.$$

A plane wave may be written as

$$\phi(x) = Ae^{-ik_\mu x^\mu}.$$

The phase is a Lorentz scalar.

Derivative of a plane wave

For

$$\phi(x) = Ae^{-ik_\nu x^\nu},$$

we have

$$\partial_\mu \phi = -ik_\mu \phi.$$

Applying another derivative,

$$\begin{aligned}\Box\phi &= \partial_\mu \partial^\mu \phi \\ &= -k_\mu k^\mu \phi.\end{aligned}$$

Thus the massless wave equation implies

$$\boxed{k_\mu k^\mu = 0}.$$

The wave four-vector is null for a wave with dispersion relation

$$\omega^2 = c^2 |\mathbf{k}|^2.$$

Product rule

The four-derivative obeys the ordinary product rule:

$$\boxed{\partial_\mu (fg) = (\partial_\mu f)g + f(\partial_\mu g)}.$$

For a scalar f and vector A^μ,

$$\partial_\mu (fA^\mu) = (\partial_\mu f)A^\mu + f\partial_\mu A^\mu.$$

This identity is the covariant form used in spacetime integration by parts and in derivations of Noether currents.

Commutation of partial derivatives

For a sufficiently smooth field in Cartesian Minkowski coordinates,

$$\boxed{\partial_\mu \partial_\nu \phi = \partial_\nu \partial_\mu \phi}.$$

The order of ordinary partial derivatives may therefore be exchanged.

This property will later imply that the antisymmetrized second derivative of a scalar vanishes:

$$\partial_\mu \partial_\nu \phi - \partial_\nu \partial_\mu \phi = 0.$$

In curved spacetime or in gauge theory, generalized covariant derivatives need not commute. Their noncommutativity carries geometric or physical information.

Dimensions

Since

$$[x^\mu] = \mathrm{m},$$

all components of ∂_μ have dimensions

$$[\partial_\mu] = \mathrm{m}^{-1}.$$

The d'Alembertian has dimensions

$$[\Box] = \mathrm{m}^{-2}.$$

Thus every term added to $\Box\phi$ in a field equation must have compatible dimensions.

Structural checks

A covariant derivative calculation should satisfy:

1. ∂_μ has a lower index;
2. raising the index changes the signs of the spatial components;
3. the contraction $\partial_\mu A^\mu$ is a scalar;
4. $\Box$ expands to $c^{-2}\partial_t^2 - \nabla^2$ in the chosen convention;
5. a plane wave produces the expected dispersion relation;
6. all terms have compatible dimensions.

Common mistakes

Typical errors include:

1. writing $\partial_0 = \partial/\partial t$ instead of $c^{-1}\partial/\partial t$ when $x^0 = ct$;
2. treating ∂^μ and ∂_μ as numerically identical;
3. using the wrong sign in front of the Laplacian;
4. calling $\nabla\phi$ by itself a four-vector;
5. forgetting that the wave phase must be a Lorentz scalar;
6. confusing the d'Alembertian with the spatial Laplacian;
7. applying commutation of derivatives without sufficient smoothness or in a context involving covariant derivatives.

What has been established

The covariant four-gradient is

$$\partial_\mu = \left(\frac{1}{c}\partial_t, \nabla\right),$$

while

$$\partial^\mu = \left(\frac{1}{c}\partial_t, -\nabla\right).$$

Their contraction defines the d'Alembert operator

$$\Box = \partial_\mu \partial^\mu = \frac{1}{c^2}\partial_t^2 - \nabla^2.$$

The ordinary wave equation becomes the covariant scalar equation $\Box\phi = 0$. The next section develops a broader collection of relativistic invariants and shows how scalar contractions encode observer-independent physical statements.

4.7 Relativistic Invariants

4.8 Sign Conventions and Units

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Symmetries, Currents, and Noether's Theorem

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Electromagnetic Potentials and Gauge Freedom

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